

Virtuoso[®] Layout Pro: T4 Advanced Commands

Course Version IC 25.1

Lab Manual

Revision 1.0

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Module 1: About This Course

There are no labs in this module.

Module 2: Generating Clones

Lab 2-1 Exploring the Types of Clones

Objective: To examine the cloning and the synchronous copy features.

Cloning has been available in Virtuoso® Layout Suite XL for several releases and is widely used. You have the capability to create clones as *Free Objects*, *Grouped Objects* and *Synchronized Family*. However, some users find its interface unfamiliar and with some use model restrictions that experienced XL users prefer removed. The new *Synchronous copy* functionality addresses both issues.

This new functionality is presented as options in the Copy and Repeat Copy commands. It extends the available functionality so that as you make the transition to *EXL*, you do not have to learn a new interface. When *Synchronous copy* is enabled, both the source of the copy and the target(s) will become synchronous groups. This means that subsequent edits of one of the groups will be replicated in all, giving you a productivity boost.

Before IC 6.1.5, it was only possible to clone objects that were in the schematic. The new *Synchronous copy* functionality removes this restriction and lets you create synchronous groups of objects regardless of whether they are in the schematic or not.

In this lab, you create the different types of clones.

Lab Setup

Follow the steps below to set up the environment and perform the lab:

1. Untar the lab database file, using the following command.

```
tar -xzf <filename.tar.gz>
```

2. Update the following line in the /VLPT4_IC_25_1/.cshrc_VLPT4 file with the appropriate valid local tool paths for Virtuoso.

```
setenv CDSHOME /ccstools/cdsind1/Software/IC251ISR_lnx86
```

3. Save and close the file.

4. Source the updated class setup file.

```
source .cshrc_VLPT4
```

Invoking the virtuoso Executable

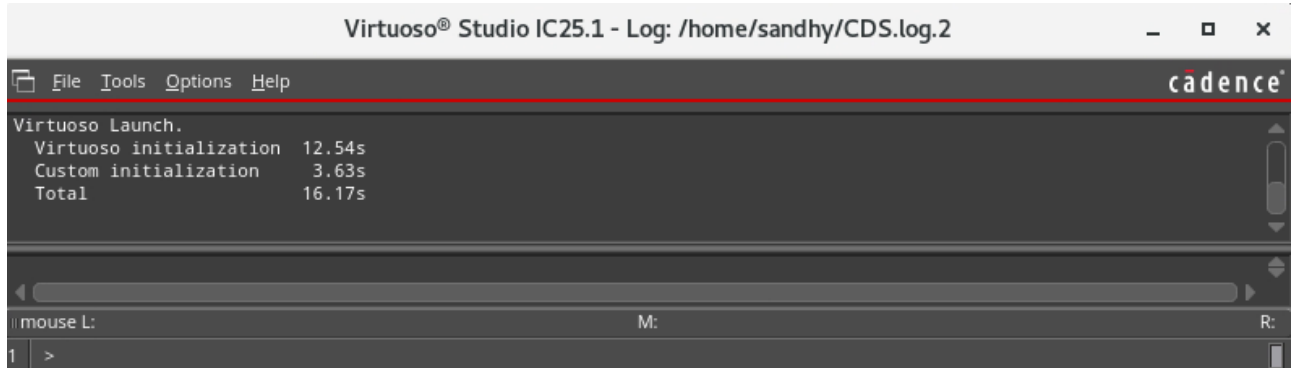
1. In a terminal window, enter the following command from your home (login) directory to get to the lab database directory when you initially logged in:

```
cd ./VLPT4_IC_25_1/VLPT4_M02_GenC
```

- To start the Cadence® Virtuoso software, enter this command in the same terminal window:

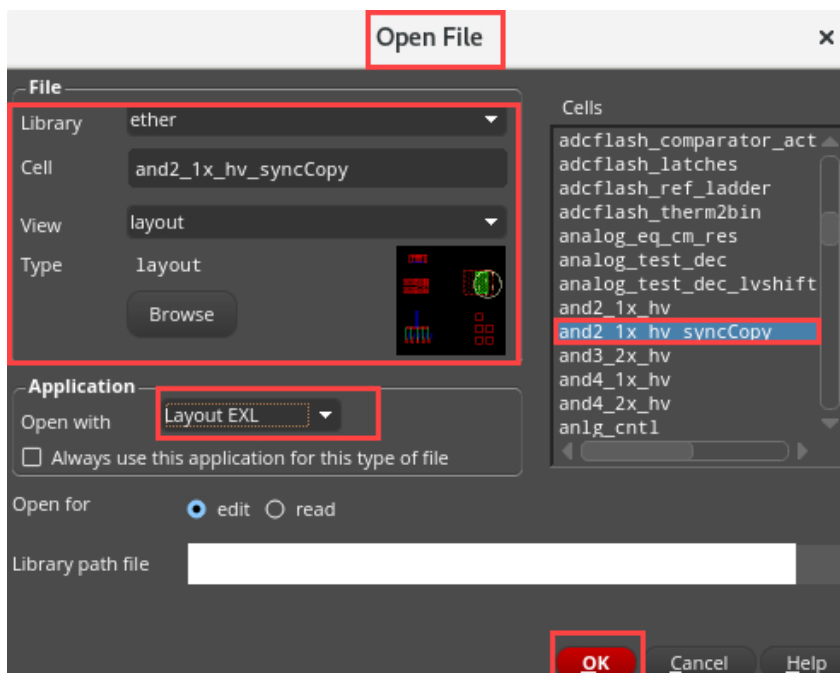
```
virtuoso &
```

The Command Interpreter Window (CIW) appears.



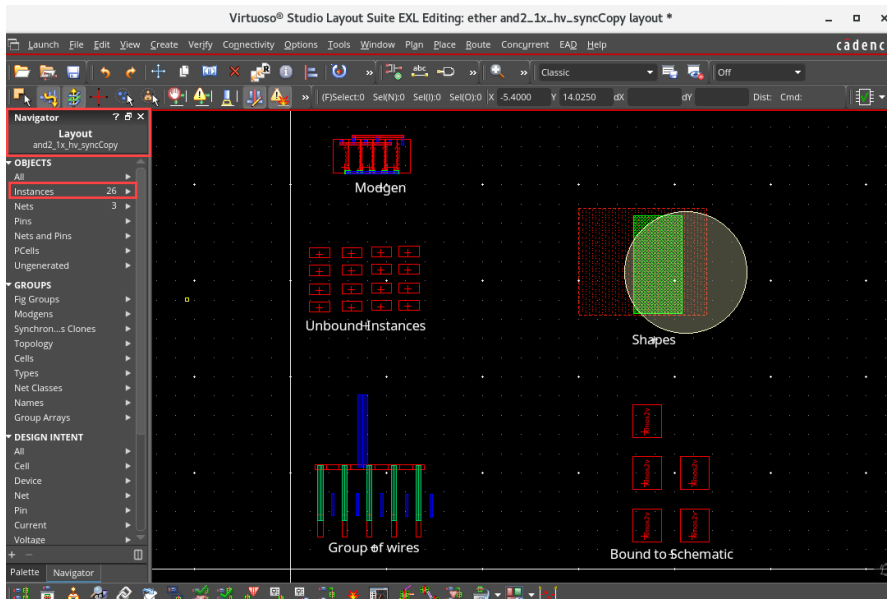
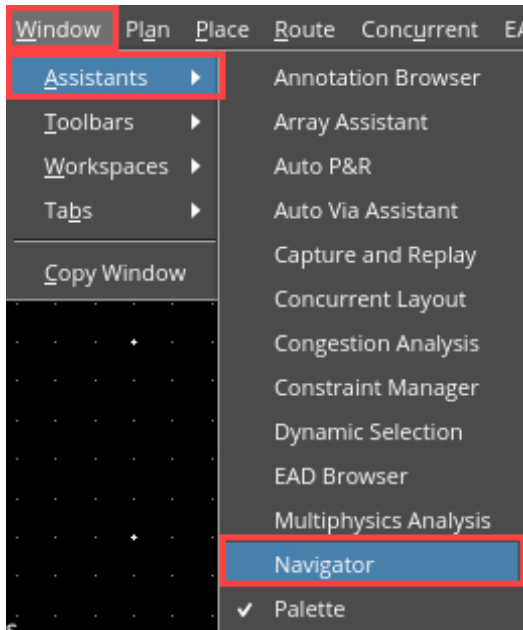
- In the CIW, choose **File – Open** menu command and enter these values in the form.

Library	cell	view
ether	and2_lx_hv_syncCopy	layout

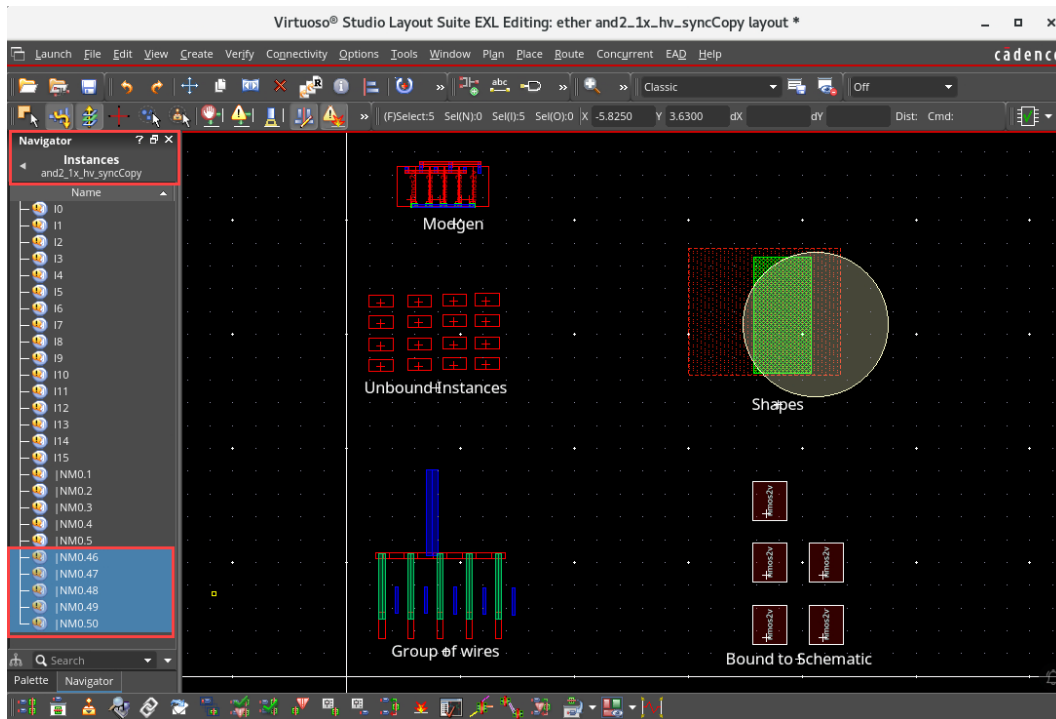


The schematic view appears in XL mode and layout views appear in EXL mode.

- If Navigator is not already visible, in the layout canvas, activate the Navigator assistant by choosing the **Window – Assistants – Navigator** menu and activating **Instances** in the *OBJECTS* category.

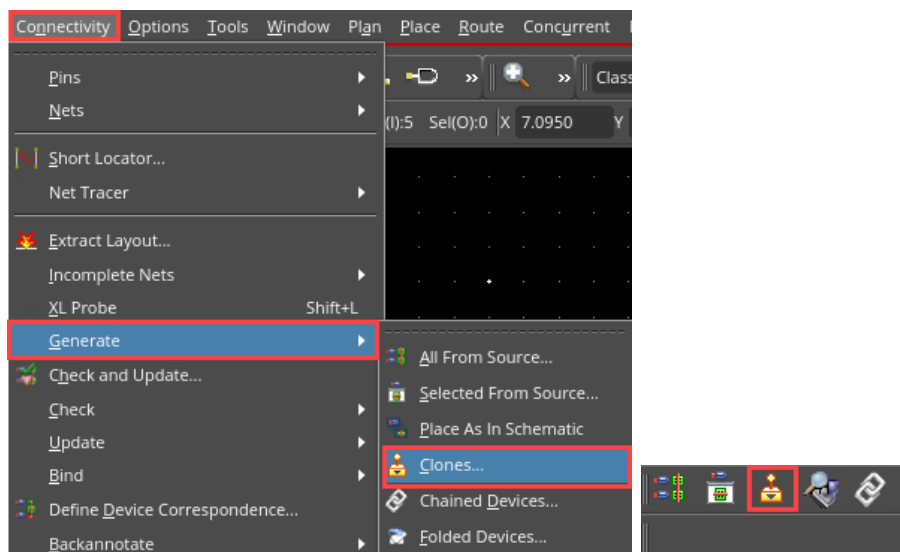


5. In the layout canvas, in the Navigator assistant, click-drag to select the instances **NM0.46** through **NM0.50**, as shown here.



Note: Since you are using **EXL**, you can also select these five devices in the layout, and they are highlighted in the Navigator assistant as seen in the graphic.

6. In the layout canvas, choose the **Connectivity – Generate – Clones** menu command or click the **Generate Clones** icon in the Layout EXL toolbar to start the Clone command, as shown here.



The Generate Clones form appears, as shown here.

Generate Clones <ether_and2_1x_hv_syncCopy layout> x

Clone Source

Layout Cellview: ether_and2_1x_hv_syncCopy_layout

Schematic Instances: /NM0.46 /NM0.47 /NM0.48 /NM0.49 /NM0.50

Layout Instances: |NM0.46 |NM0.47 |NM0.48 |NM0.49 |NM0.50

Update From Selected

Search

In: Schematic ☐ Selected Set Only ☐ Constraint Only

With ☒ Allowed Permutation ☐ Allowed Different Abutment

☐ Exact Parameters ☒ Exact Connectivity

More Options...

Search

Create Options

Create Clones as: Synchronized Family ☐ Update Constraint

☒ Update Layout Parameters to Match Schematic

☒ Display Draglines

☐ Place Clone as Free Array ...

Clones Found

Create Group Array

Edit...

Select All

Justification

Center

Transform From

Source

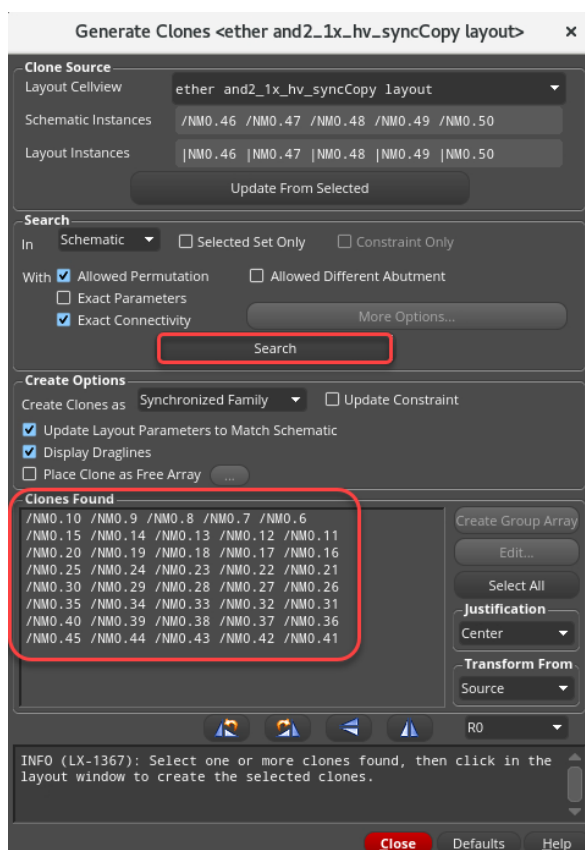
INFO (LX-1366): Click the "Search" button to find clones of the selected source.

Close Defaults Help

Notice that in the Clone Source section, *Layout Cellview* is populated with *ether/and2_1x_hv_syncCopy/layout* and the entries in the *Schematic* and *Layout Instances* fields are already populated due to your instance selection in the Navigator.

7. In the Generate Clones form, click the **Search** button to find the matches in the schematic.

The matches are returned in the *Clones Found* pane, from where they can be selected and placed in the layout canvas.

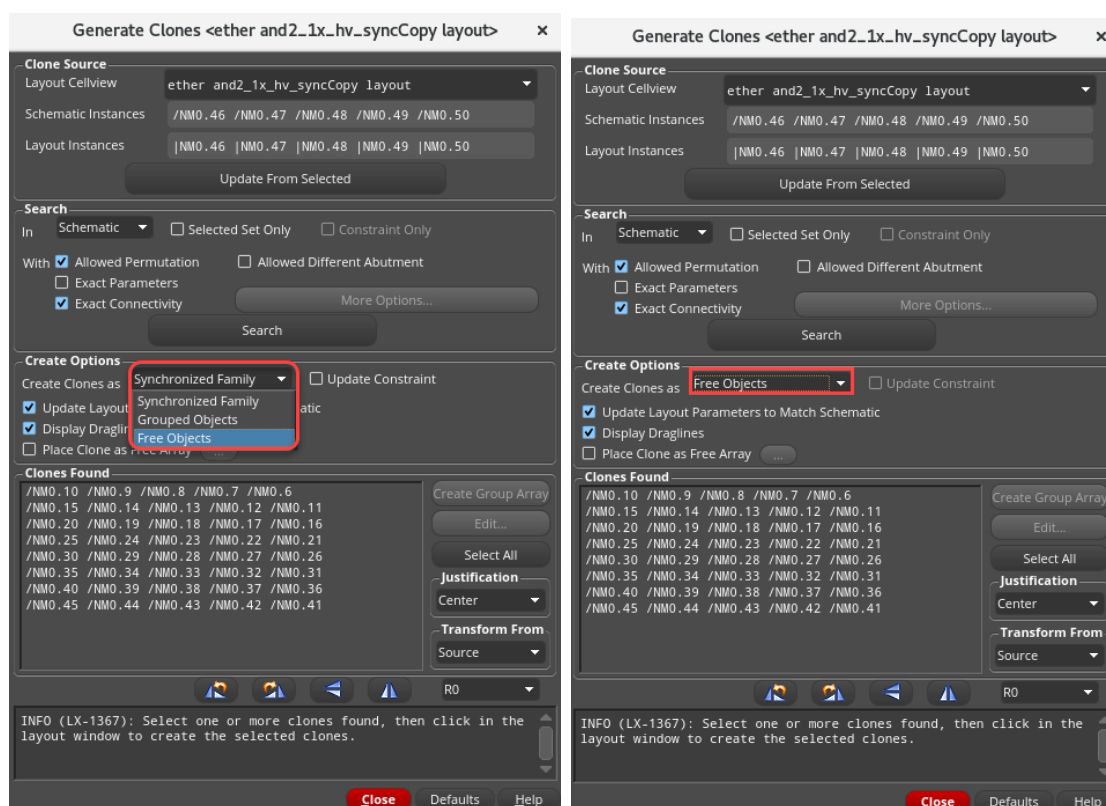


In the Generate Clones form, in the *Create Options* section, in the *Create Clones as* group box, you can control how the target structures need to be created. The three types are **Free Objects**, **Grouped Objects**, and **Synchronized Family**.

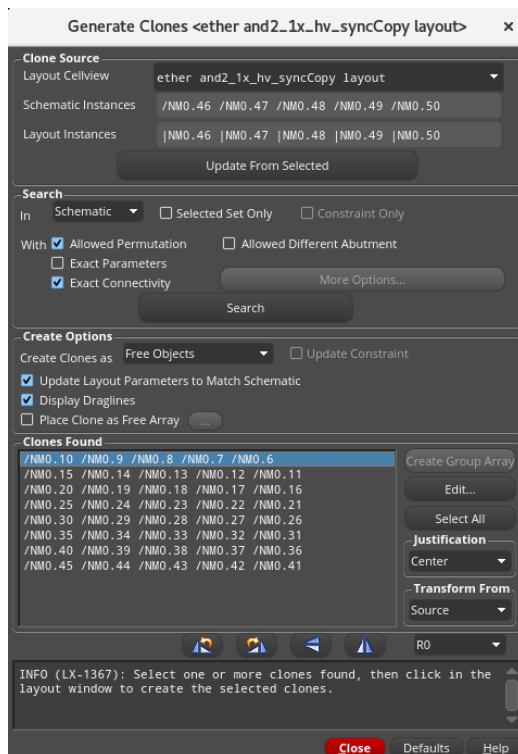
Generating Clones as Free Objects

Now, you create clones as **Free Objects**.

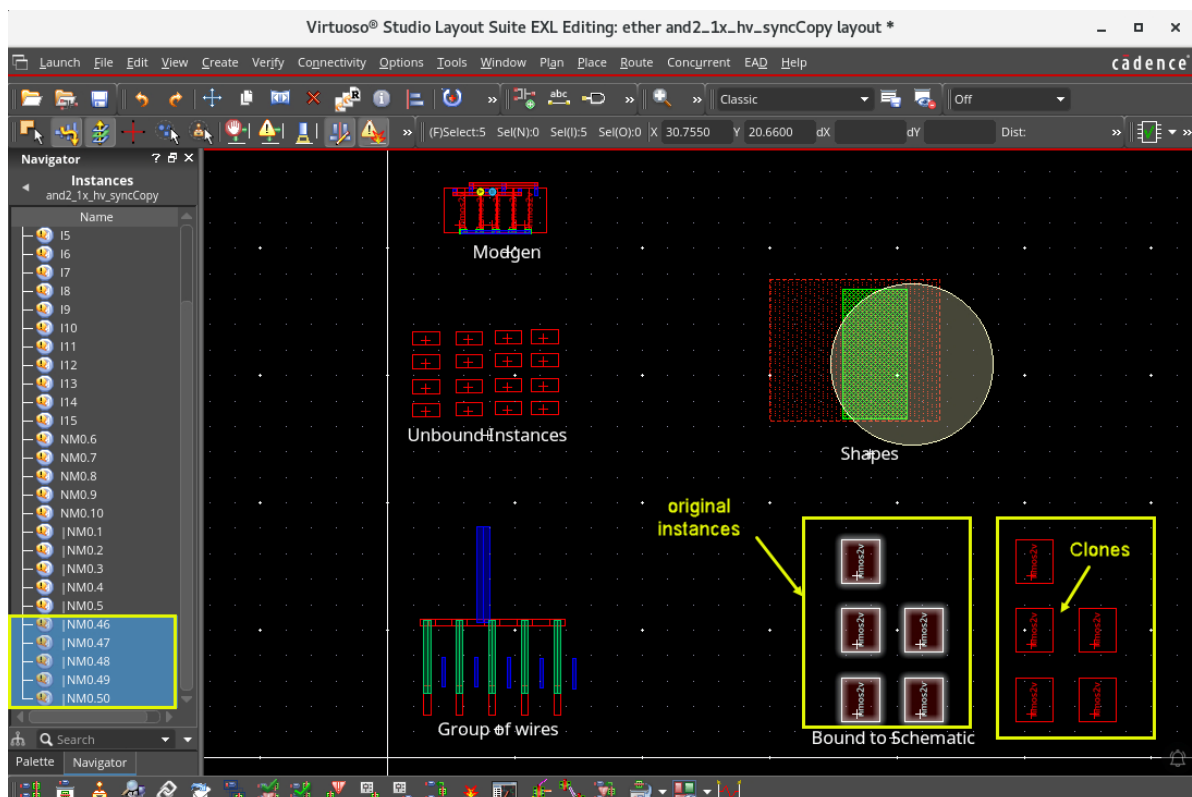
1. Select **Free Objects** in the *Create Clones as* group box in the *Create Options* section in the Generate Clones form to create clones as Free Objects.



- In the Generate Clones form, in the *Clones Found* pane, click to select one of the groups.



- Place them in your layout, as shown here.

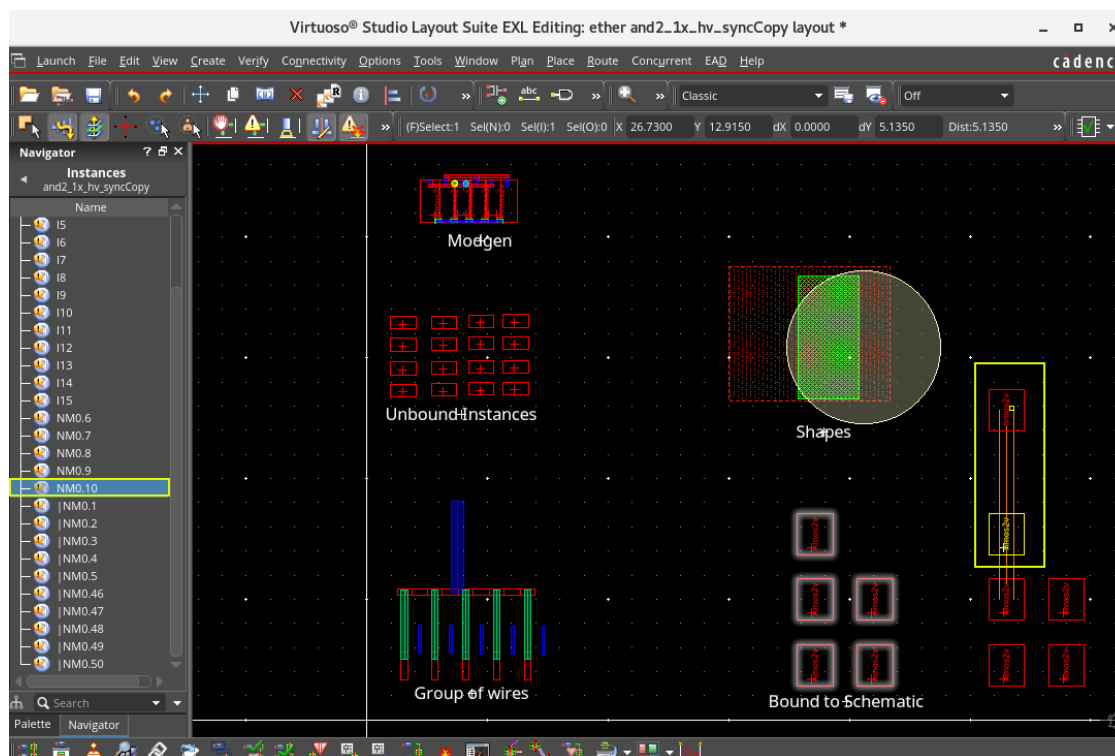


Note: When creating clones, as the command searches the schematic, the instances which are created in the layout are bound to the schematic instances. This is **NOT** the case with the *Synchronous copy* option.

The components in the clone are generated as “ungrouped components” in the layout view with **Free Objects** option. You can check the same by moving individual components as shown in the next step.

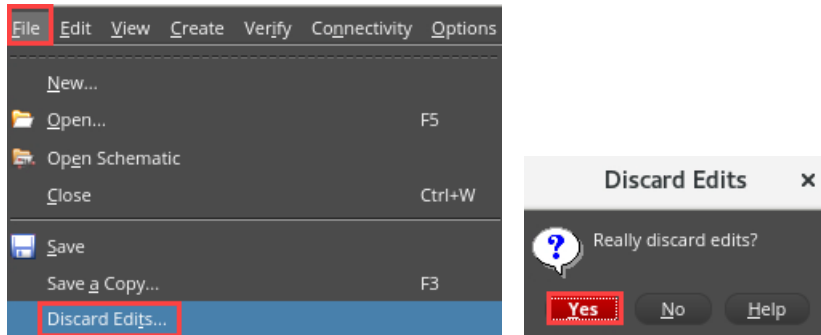
4. Select each instance of the newly placed clone (**NM0.10**, **NM0.9**, **NM0.8**, **NM0.7**, and **NM0.6** in this case) and try to move it.

Notice that the instances move individually, as shown here. This is because they are generated as “ungrouped components” with the **Free Objects** option in the *Create Clones* as group box in the *Create Options* section in the Generate Clones form.



5. Close the Generate Clones form.

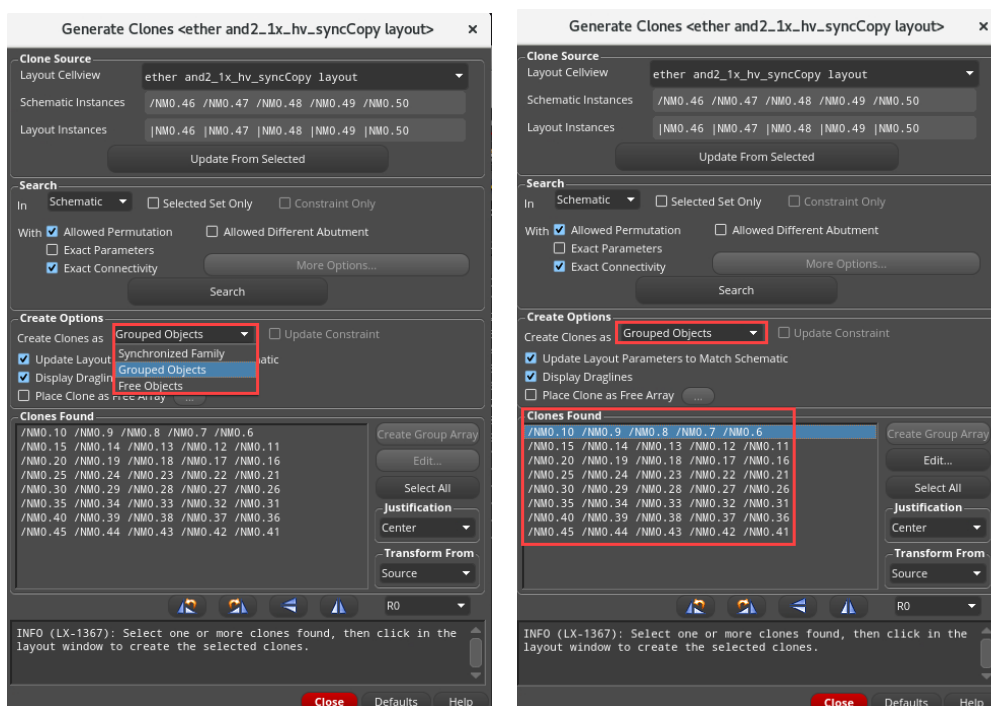
- In the layout window, choose the **File – Discard Edits** menu command and click the **Yes** button in the *Discard Edits* form to discard the edits you made.



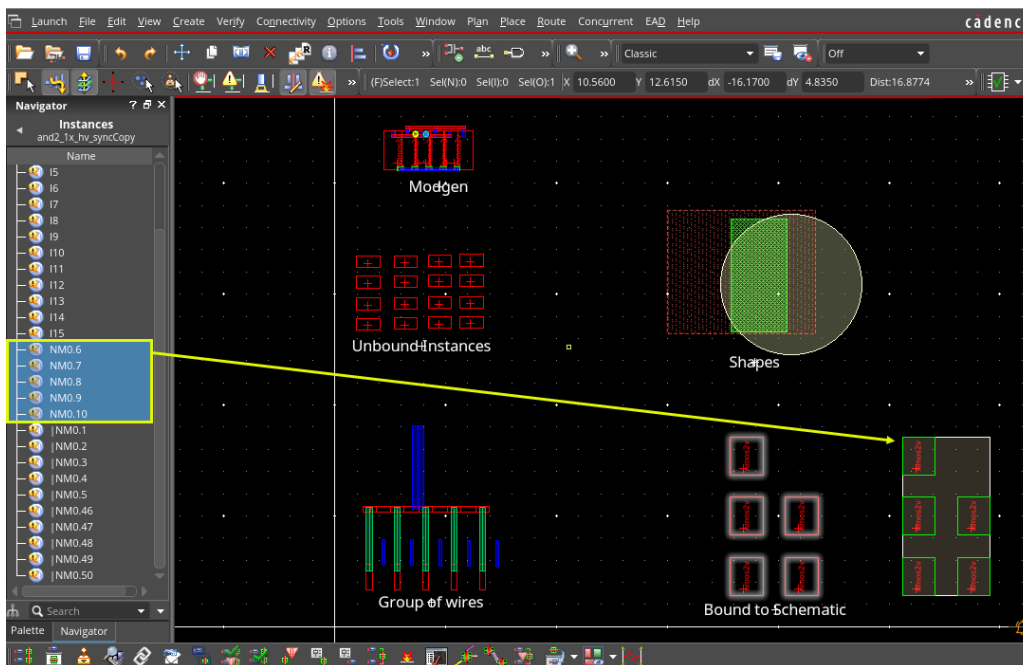
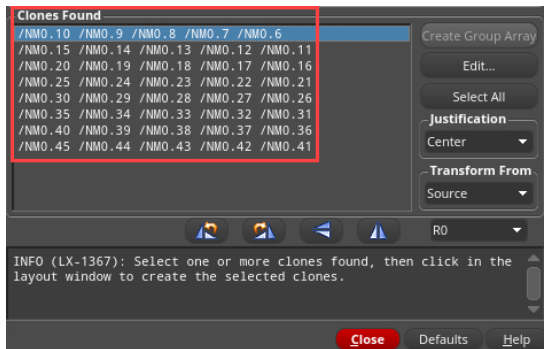
Generating Clones as Grouped Objects

Next, you create clones as **Grouped Objects**.

- In the layout window, select the 5 instances **NM0.46** through **NM0.50** again in the Navigator assistant.
- Invoke the **Generate Clones** form again and click the **Search** button to list the matching structures in the schematic again in the *Clones Found* pane.
- Select **Grouped Objects** in the *Create Clones as* group box in the *Create Options* section in the Generate Clones form to create clones as Grouped Objects.



4. In the Generate Clones form, in the *Clones Found* pane, click to select one of the groups and place them in your layout.

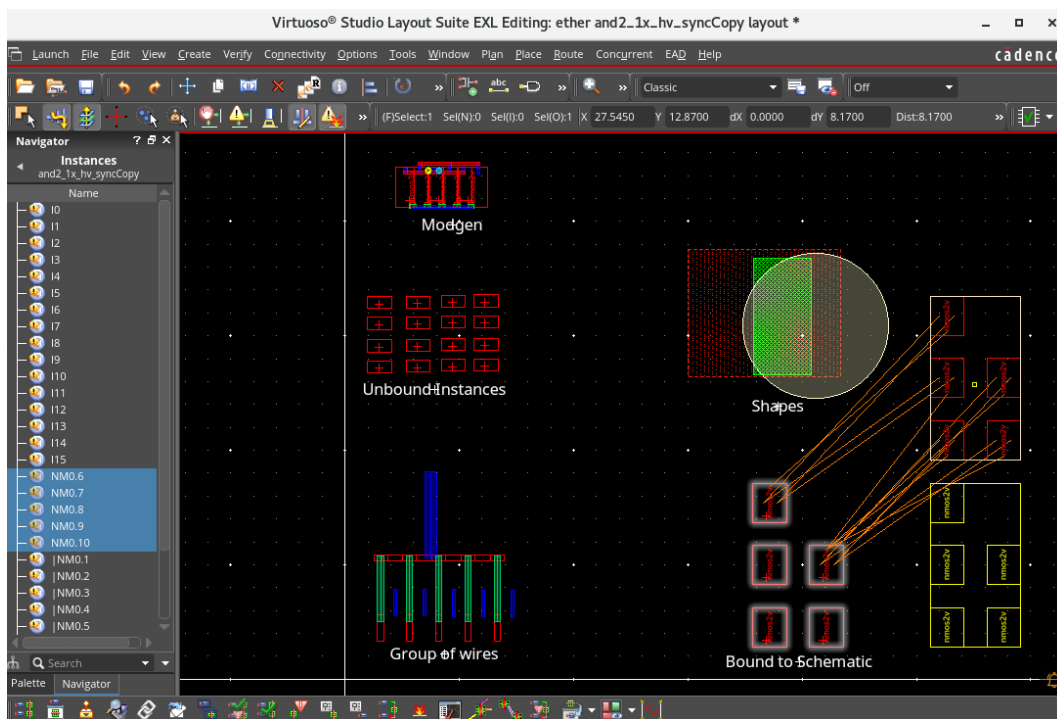
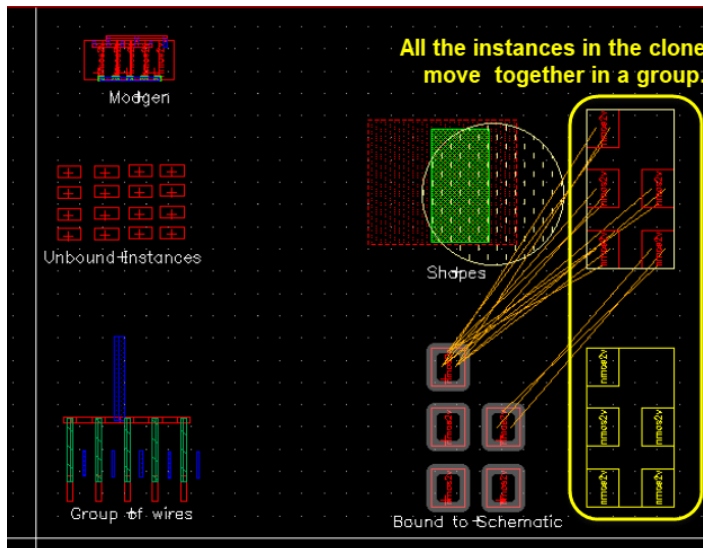


The clone is created as a “group”. Clones that are generated as groups are constrained in the ways they can be moved and rotated. All the devices in the clone move together in a group. You can check the same by moving the clone as shown in the next step.

Note: When creating clones, as the command searches the schematic, the instances which are created in the layout are bound to the schematic instances. This is **NOT** the case with *Synchronous copy* option.

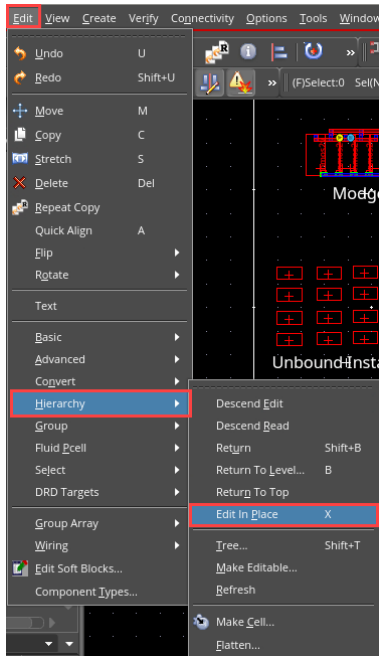
5. Select the newly placed clone and try to move it.

Notice that all the instances in the clone move together in a “group”. This is because they are generated as “grouped components” with the **Grouped Objects** option.

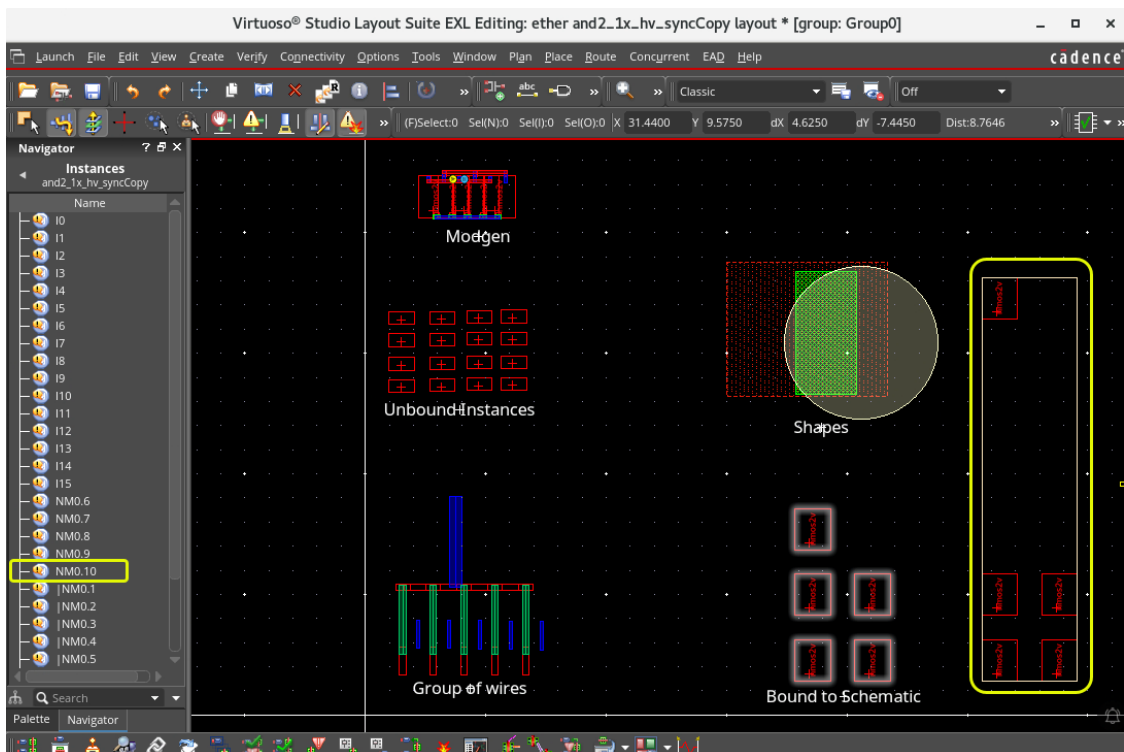


Now you examine the **Grouped Objects** behavior of the generated clone with **Edit In Place (EIP)** command.

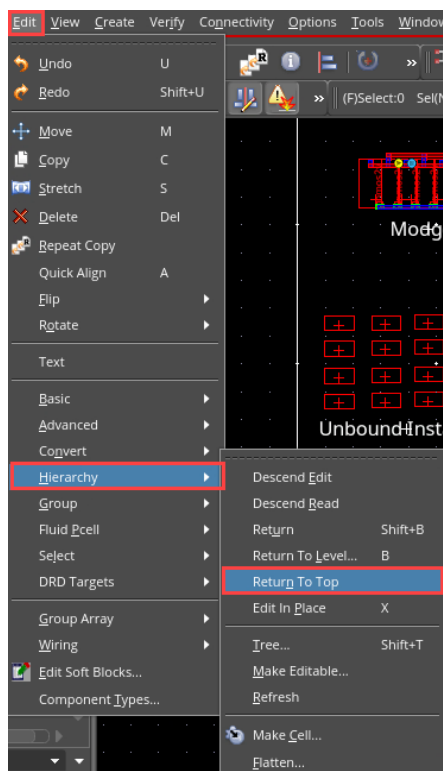
- In the layout canvas, select the generated clone and choose the **Edit – Hierarchy – Edit In Place** menu command or press the bindkey **X** to edit the clone in EIP mode.



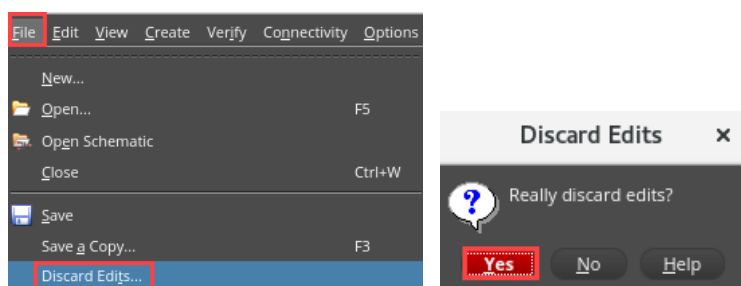
- In the layout window, select an instance in the clone, say for example **NM0.10**, and try to move it. You can see all 4 instances **NM0.9**, **NM0.8**, **NM0.7**, and **NM0.6** in the clone stay together as a group with **NM0.10**.



8. In the layout window, return to the top of the hierarchy by choosing the **Edit – Hierarchy – Return To Top** menu command.



9. Close the Generate Clones form.
10. In the layout window, choose the **File – Discard Edits** menu command and click the **Yes** button in the *Discard Edits* form to discard the edits you made.

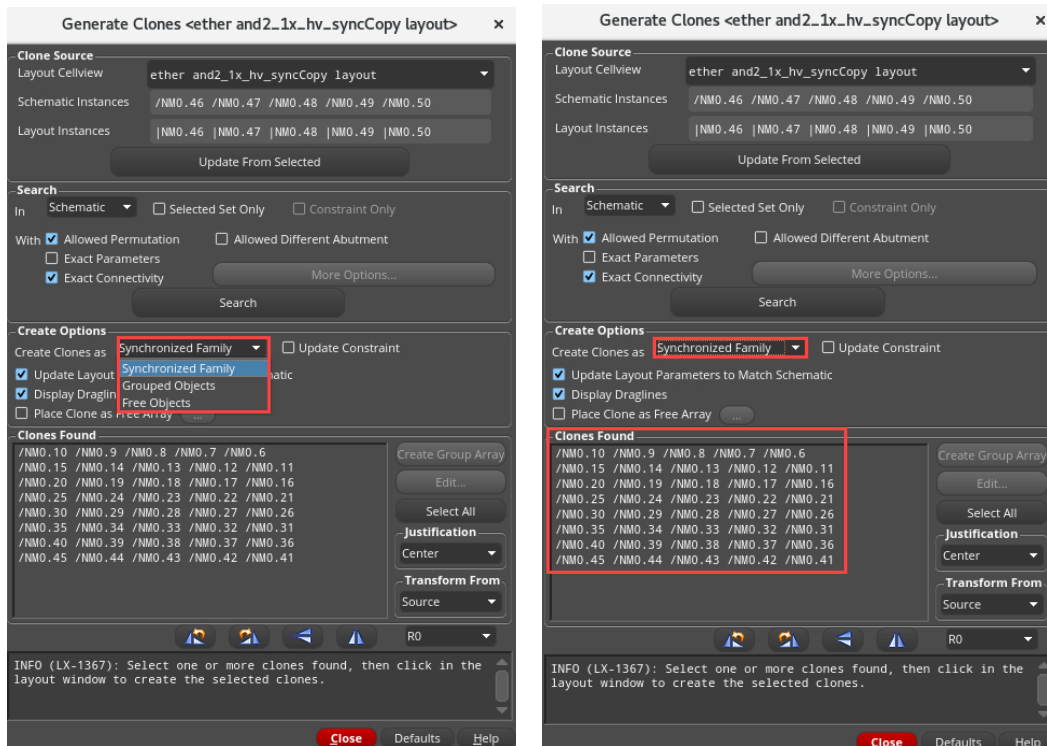


Generating Clones as Synchronized Family

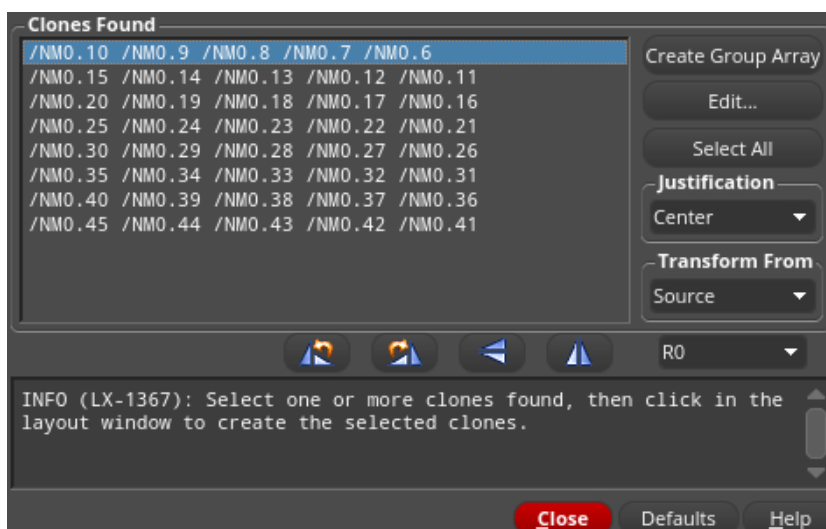
Next, you create clones as **Synchronized Family**.

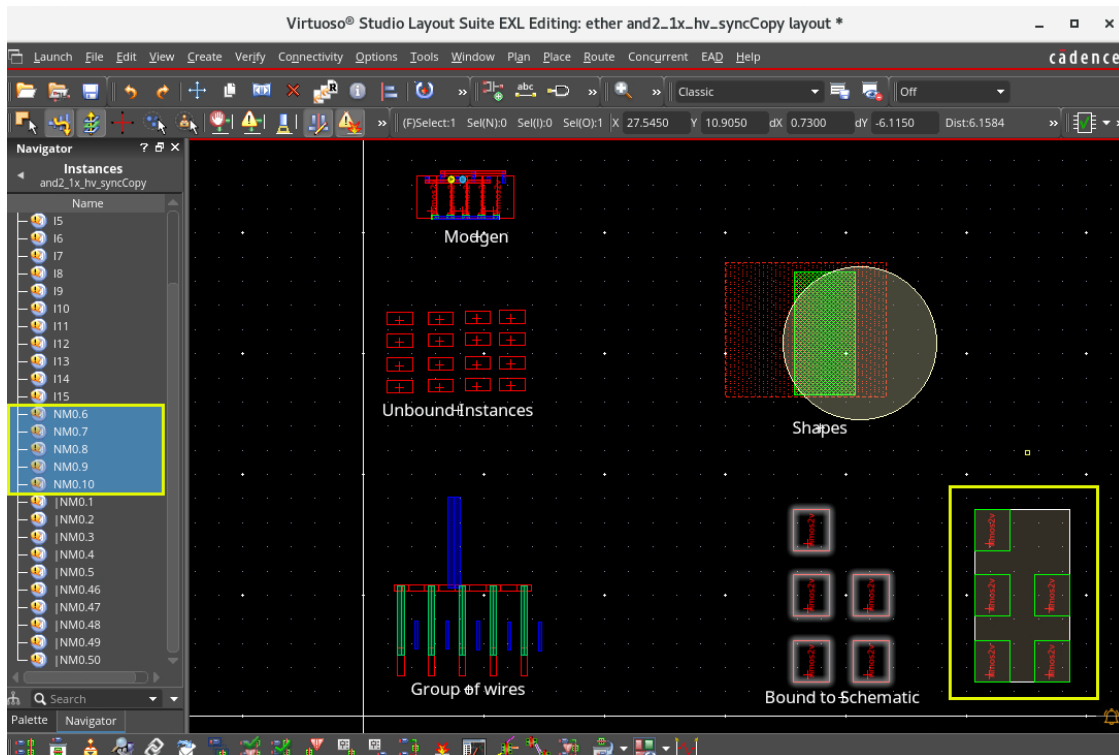
1. In the layout window, select the 5 instances **NM0.46** through **NM0.50** again in the Navigator assistant.

2. Invoke the **Generate Clones** form again and click the **Search** button to list the matching structures in the schematic again in the *Clones Found* pane.
3. Select **Synchronized Family** in the *Create Clones as* group box in the *Create Options* section in the Generate Clones form to create clones as Synchronized Family.



4. In the Generate Clones form, in the *Clones Found* pane, click to select one of the groups and place them in your layout.





If you make a change to one member of a **Synchronized Family**, all the other members are automatically changed in the same way.

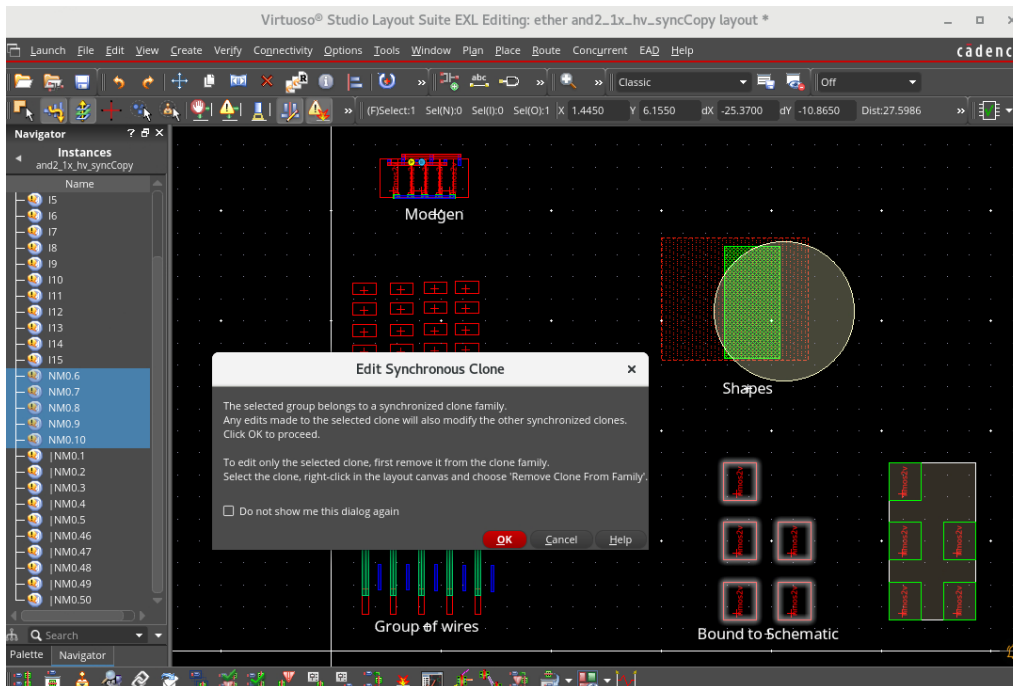
Note: When creating clones, as the command searches the schematic, the instances that are created in the layout are bound to the schematic instances. This is **NOT** the case with *Synchronous copy* option.

Next, you make an edit to show what is meant by Synchronous Clone.

Synchronous Clones cannot be edited from the top level. You need to run the **Edit In Place (EIP)** command.

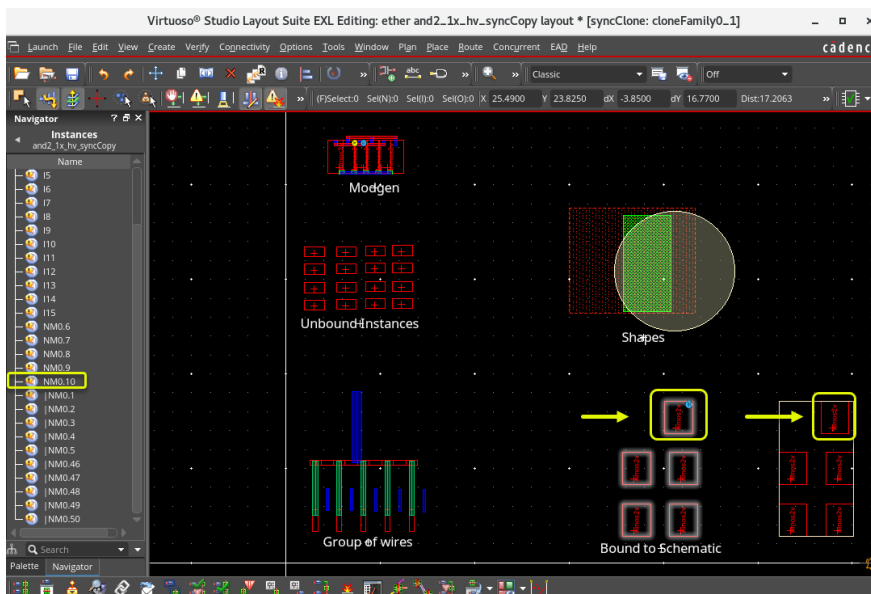
5. In the layout canvas, select the generated clone and choose the **Edit – Hierarchy – Edit In Place** menu command or press the bindkey **X** to edit the clone in EIP mode.

The Edit Synchronous Clone form appears.



6. Click **OK** in the Edit Synchronous Clone form.
7. In the layout window, inside the group selected for **EIP**, select and move the top instance to the right.

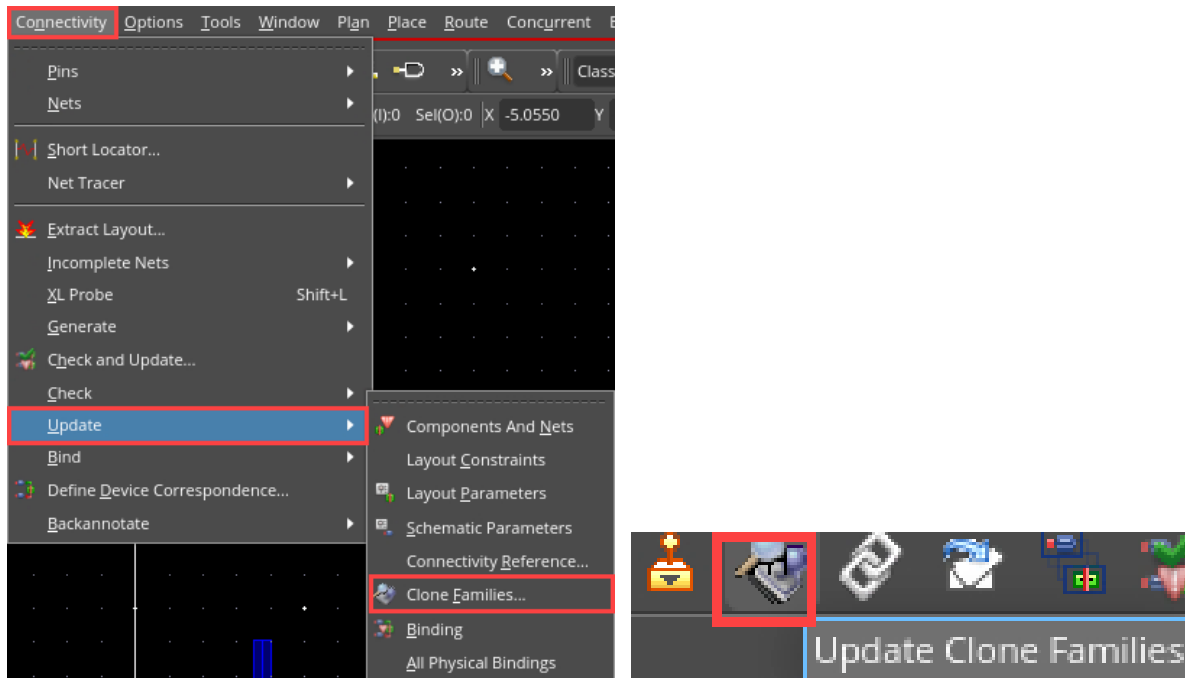
Notice that the changes made in one clone (on the right) are automatically made in the other clone (on the left) of the family, as shown here.



8. In the layout window, return to the top of the hierarchy by choosing the **Edit – Hierarchy – Return To Top** menu command.

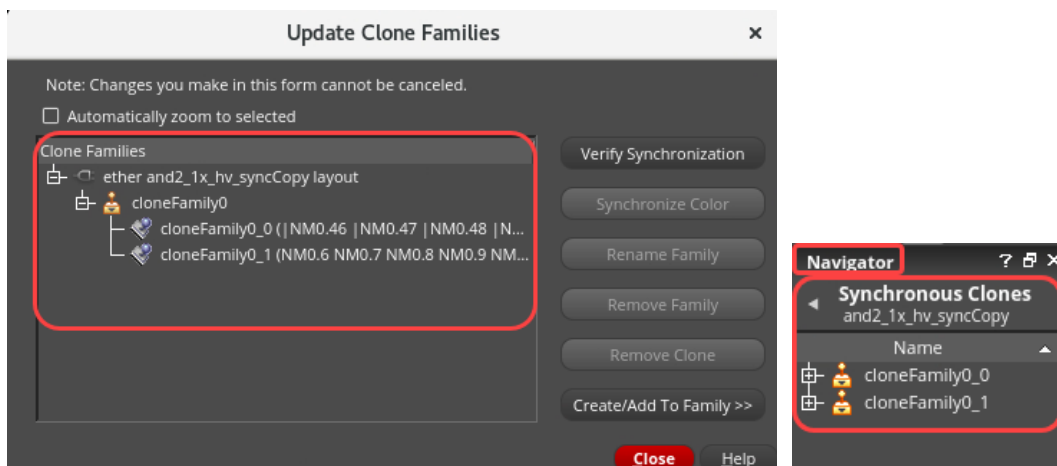
After synchronous clones are created, they can be managed with the *Update Clone Families* command.

9. In the layout canvas, choose the **Connectivity – Update – Clone Families** menu command or click the **Update Clone Families** icon in the Layout EXL toolbar to invoke the Update Clone Families form, as shown here.



In the **Update Clone Families** form, notice that two synchronous clones namely *cloneFamily0_0* and *cloneFamily0_1* have been created in the same family *cloneFamily0*, as shown in the graphic.

10. In the layout canvas, in the Navigator assistant, select **Synchronous Clones** in the *GROUPS* category to view the clones.



11. Close the **Generate Clones** and **Update Clone Families** forms and keep the cellviews open.

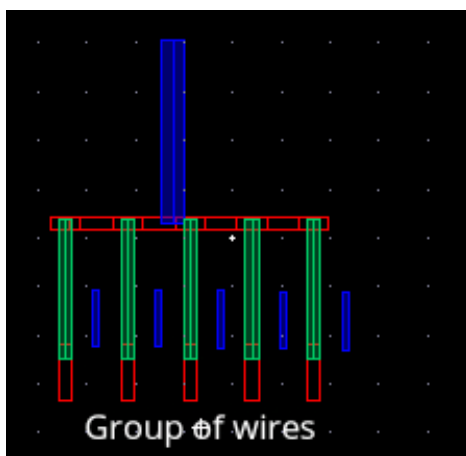
Generating Synchronous Copy Clones of Wires

The synchronous cloning functionality requires that at least one instance in the source be in the schematic. Users have requested the capability to create synchronous groups that **ONLY** contain physical objects, meaning objects that do **NOT** exist in the schematic.

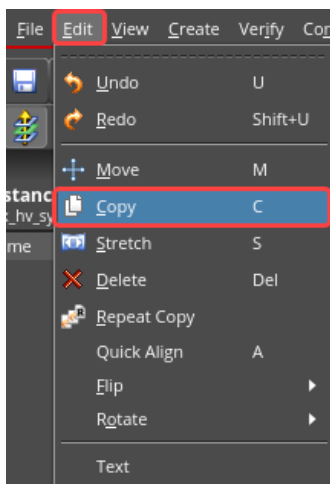
Starting with the release of IC 6.1.5, this has been possible by using the *Synchronous copy* functionality, which is available only in the **XL** and **EXL** tiers of the Virtuoso software.

You are going to select a group of wires (which exist only in the layout) and copy them using the *Synchronous copy* option in the Copy form. (It is available with the checkbox option.)

1. In the same cellview which is already kept open, select the area around the “Group of wires” in the layout, as shown here.

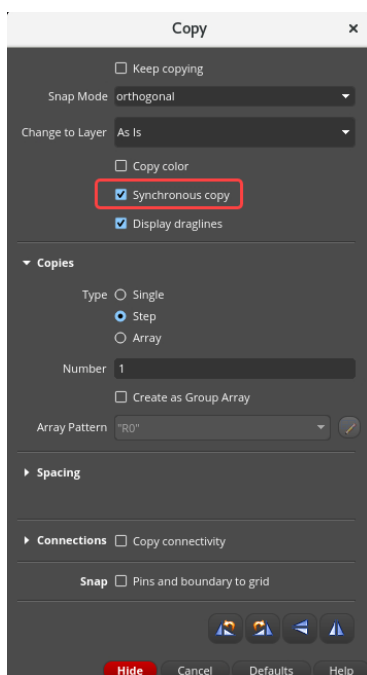


2. In the layout canvas, choose the **Edit – Copy** menu command or click the **Copy** icon in the Edit toolbar or press the bindkey **C** to invoke the Copy form.

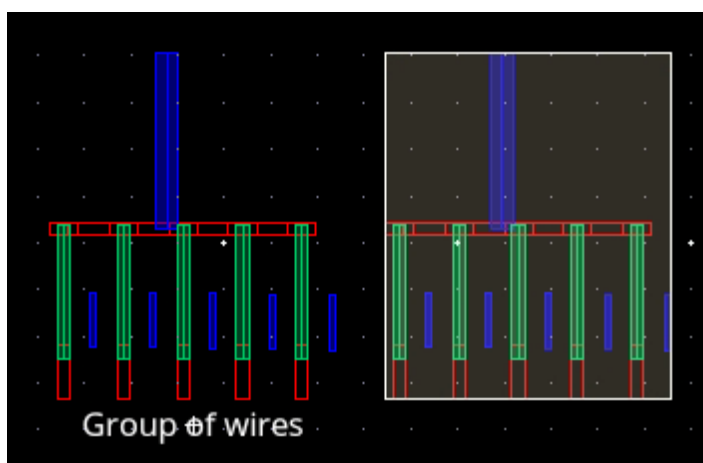


The Copy form appears. If it does not appear, press **F3** on the keyboard to open the form.

3. In the Copy form, click the **Synchronous copy** checkbox, as shown here.



4. Place a copy of the selected wires next to the original wires, as shown here.



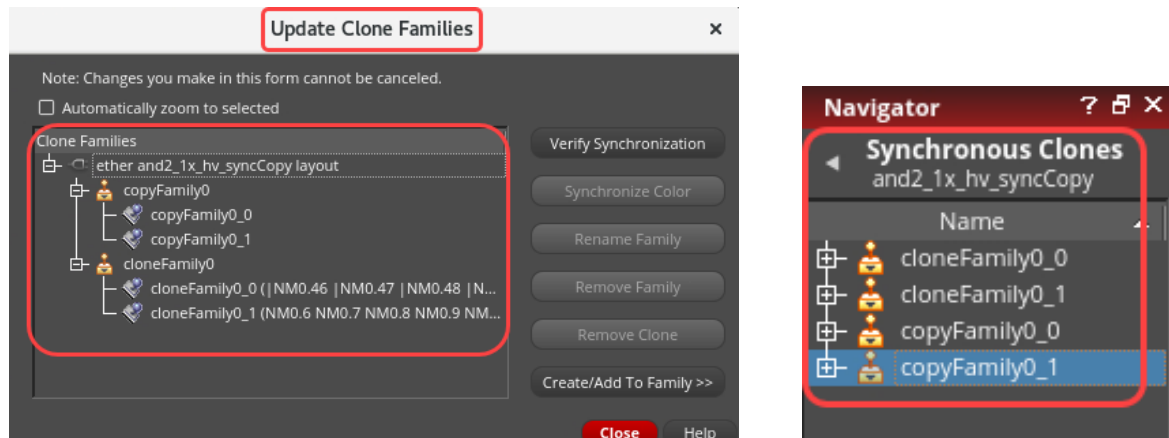
Note: Each time the Copy command finishes, the *Synchronous copy* option reverts to being unchecked (unselected). This state avoids accidental or inadvertent synchronous copying.

Important: Both the source (original wires on the left) and the target (copied wires on the right) are created as synchronous clones.

5. Open the Update Clone Families form by choosing the **Connectivity – Update – Clone Families** menu command or by clicking the **Update Clone Families** icon in the Layout EXL toolbar in the layout canvas.

In the **Update Clone Families** form, notice that two synchronous copy clones of wires namely *copyFamily0_0* and *copyFamily0_1* have been created in the same family *copyFamily0*, as shown in the graphic.

6. In the layout canvas, in the Navigator assistant, select **Synchronous Clones** in the *GROUPS* category to view the clones.



You have observed that the synchronous clones can be created by the *Synchronous copy* functionality and can be managed in the same way as those created by cloning.

Note: It is important to understand that an object created by *Synchronous copy* will **NOT** be bound to schematic instances. There are cases where using the *Synchronous copy* functionality are advantageous, such as in the last case. However, if there are layout instances in the source that are bound to schematic instances, you will want to choose cloning over copying because created instances will be bound to schematic instances.

7. Close the **Update Clone Families** form and keep the cellviews open.

Examining Synchronous Behavior in the Synchronous Copy Clones of Wires Created by the Copy Command

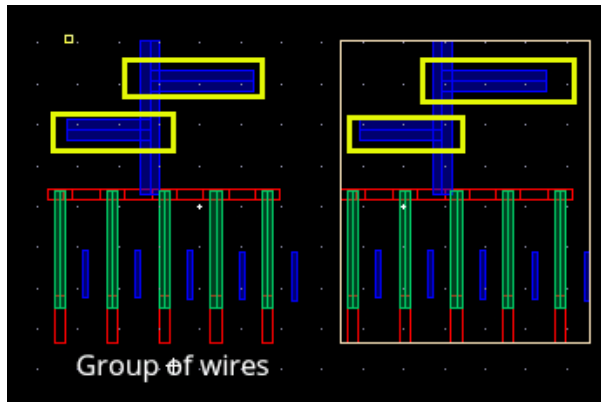
If you make a change to one member of a Synchronous Copy Clones, all the other members are automatically changed in the same way.

You make an edit on the newly created Synchronous Copy Clone to show what is meant by Synchronous Copy Clone.

Synchronous Copy Clones cannot be edited from the top level. You need to run the **Edit In Place (EIP)** command.

1. In the layout canvas, select the generated synchronous copy clone and choose the **Edit – Hierarchy – Edit In Place** menu command or press the bindkey **X** to edit the synchronous copy clone in EIP mode.

2. Click on the newly created synchronous copy clone family (if not already selected).
3. Click **OK** in the Edit Synchronous Clone form.
4. In the layout canvas, choose the **Create – Wiring – Wire** menu command or use the **Create Wire** icon in the Create toolbar or use the bindkey **P** to make a few edits in the newly created copy clone.
 - a. Add **metal1** spines on the newly created synchronous copy clone, as shown here.



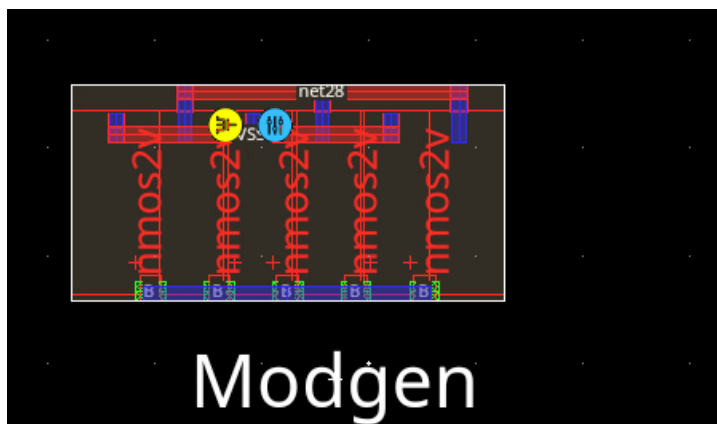
Notice that those same edits are replicated in the other clone of the clone family.

5. As always, when you use the **Edit In Place** command, you must return to the top level of the layout hierarchy, so choose the **Edit – Hierarchy – Return To Top** menu command in the layout window.

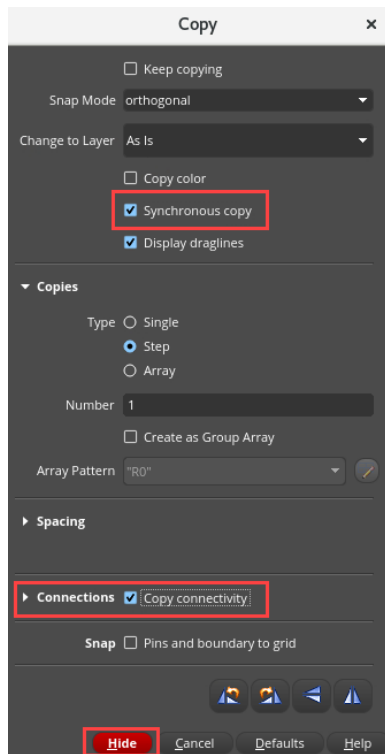
Generating Synchronous Copy Clones of Modgen

Next, you explore the *Synchronous copy* functionality on the modgen in the layout canvas.

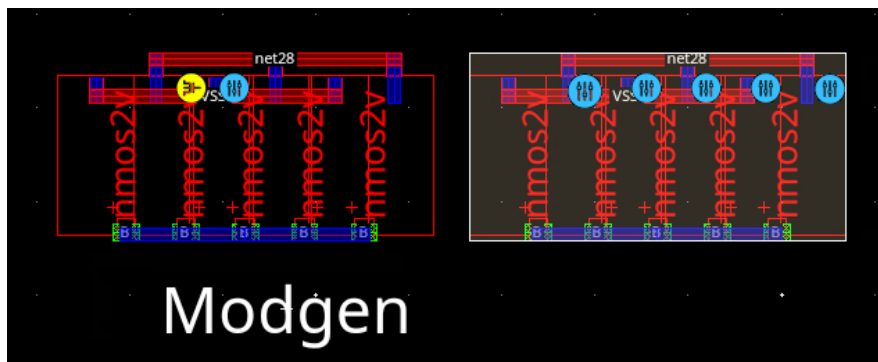
1. Try *Synchronous copy* option on the modgen in the layout.
 - a. In the same cellview which is already kept open, select the modgen.



- b. In the layout canvas, choose the **Edit – Copy** menu command or click the **Copy** icon in the Edit toolbar or press the bindkey **C** to invoke the Copy form.
The Copy form opens. If it does not appear, press **F3** to open the form.
- c. In the Copy form, select the **Synchronous copy** option.



- d. Place a copy of the selected modgen next to the original modgen, as shown here.



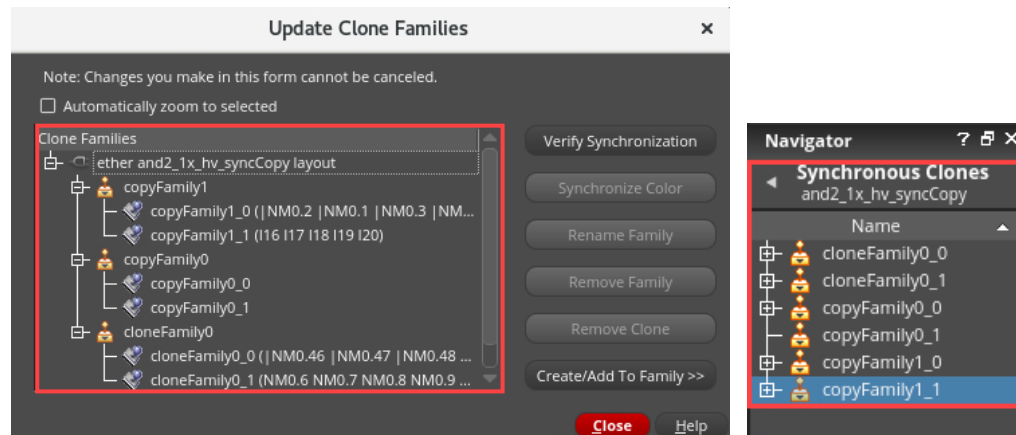
Note: You can observe that each time after the copy operation finishes, the *Synchronous copy* option reverts to being unchecked (unselected). This state avoids accidental or inadvertent synchronous copying.

Important: Both the source (original modgen on the left) and the target (copied modgen on the right) are created as synchronous clones.

- e. Open the Update Clone Families form by choosing the **Connectivity – Update – Clone Families** menu command or by clicking the **Update Clone Families** icon in the Layout EXL toolbar in the layout canvas.

In the **Update Clone Families** form, notice that two synchronous copy clones of modgen namely *copyFamily1_0* and *copyFamily1_1* have been created in the same family *copyFamily1*, as shown in the graphic.

- f. In the layout canvas, in the Navigator assistant, select **Synchronous Clones** in the **GROUPS** category to view the clones.



You have observed that the synchronous clones can be created by the *Synchronous copy* functionality and can be managed in the same way as those created by cloning.

- g. Close the **Update Clone Families** form and keep the cellviews open.

Examining Synchronous Behavior in the Synchronous Copy Clones of Modgen Created by the Copy Command

If you make a change to one member of a Synchronous Modgen Clones, all the other members are automatically changed in the same way.

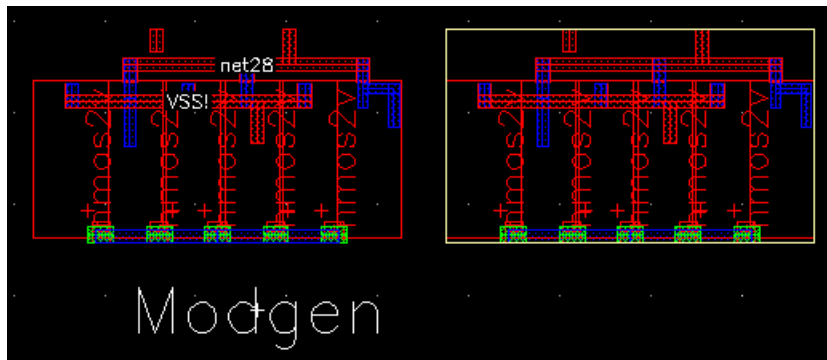
You make an edit on the newly created Synchronous Modgen Clone to show what is meant by Synchronous Copy Clone of Modgen.

Synchronous Modgen Clones cannot be edited from the top level. You need to run the **Edit In Place (EIP)** command.

1. In the layout canvas, select the generated synchronous modgen clone and choose the **Edit – Hierarchy – Edit In Place** menu command or press the bindkey **X** to edit the synchronous modgen clone in EIP mode.
2. Click on the newly created modgen clone family (if not already selected).
3. Click **OK** in the Edit Synchronous Clone form.

4. In the layout canvas, choose the **Create – Wiring – Wire** menu command or use the **Create Wire** icon in the Create toolbar or use the bindkey **P** to make a few edits in the newly created modgen clone.
 - a. Add some **metal** spines on the newly created synchronous modgen clone as shown here.

Notice that those same edits are replicated in the other modgen clone of the clone family.



5. In the layout window, choose the **Edit – Hierarchy – Return To Top** menu command to return to the top level of the layout.
6. You can close the schematic and layout views now if you do not want to do the optional extended lab exercises. Do **NOT** save.

Optional Extended Exercises on Unbound Instances and Shapes

1. The layout view also contains a group of “Unbound Instances”. Try exploring *Synchronous copy* functionality on these using the same procedure as before.
The instances in both of the two clones will be unbound i.e., no counterpart in the schematic.
2. The layout view also contains a group of “Shapes”. *Synchronous copy* can also be applied to shapes. Try exploring *Synchronous copy* functionality on these shapes.
3. Close the schematic and layout views. Do **NOT** save.
4. Keep the Virtuoso session running for the next lab.

Summary

In this lab, you

- ◆ Generated clones as Free Objects
- ◆ Generated clones as Grouped Objects
- ◆ Generated clones as Synchronized Family
- ◆ Generated synchronous copy clones of wires
- ◆ Examined synchronous behavior in the synchronous copy clones of wires created by the copy command
- ◆ Generated synchronous copy clones of modgen
- ◆ Examined synchronous behavior in the synchronous copy clones of modgen created by the copy command



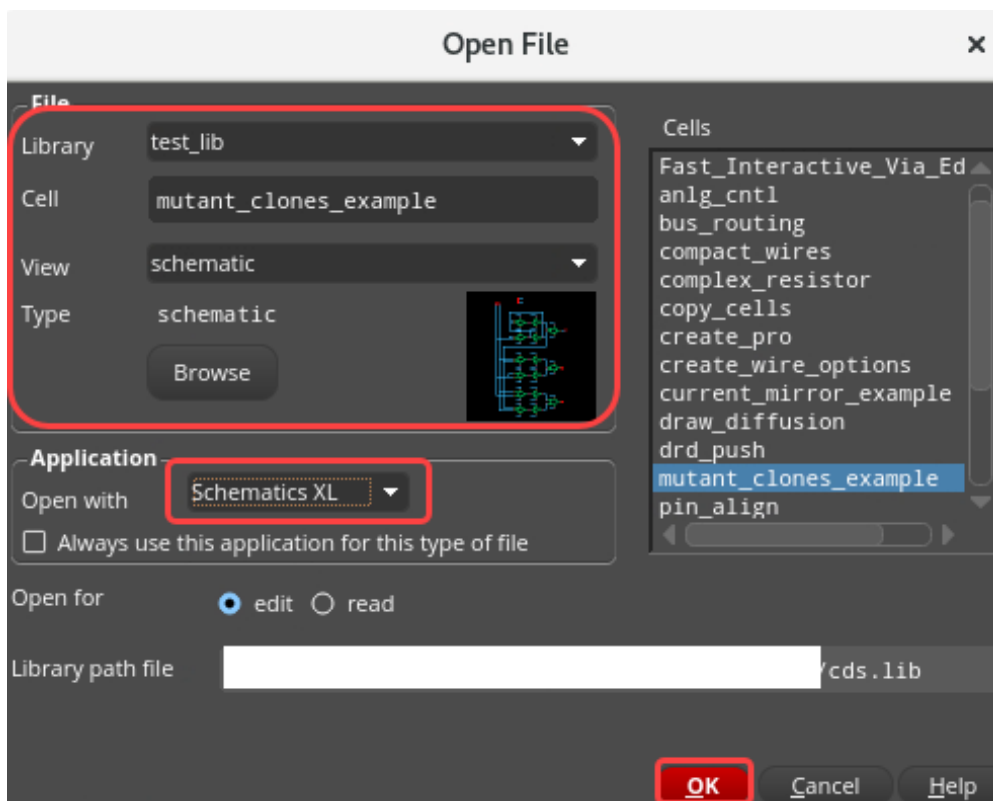
Lab 2-2 Generating Mutant Clones

Objective: To generate the clone structures that have a different connectivity.

In this lab, you create clones when the connectivity is not identical, which is called *mutant clones*.

1. In the CIW, choose **File – Open** menu command and open the following schematic view in XL mode.

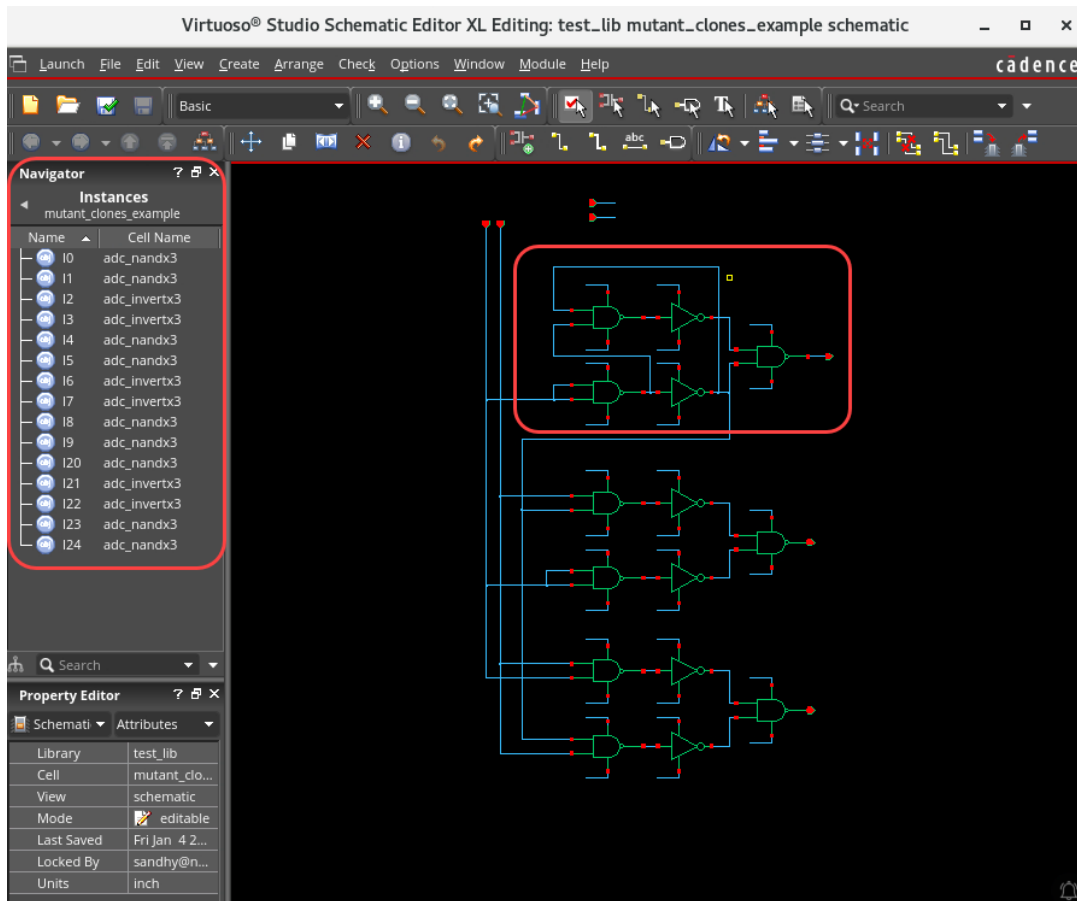
Library	Cell	View
test_lib	mutant_clones_example	Schematic XL



The *test_lib/mutant_clones_example/schematic* cellview opens for editing.

In this schematic, there are three similar logic blocks. All three use the same gates, but the connections are slightly different.

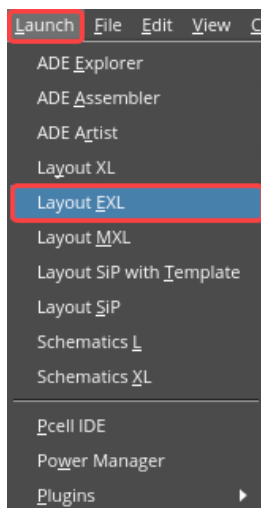
In the Navigator, choose **Types**, where you will find the device details, as shown below.



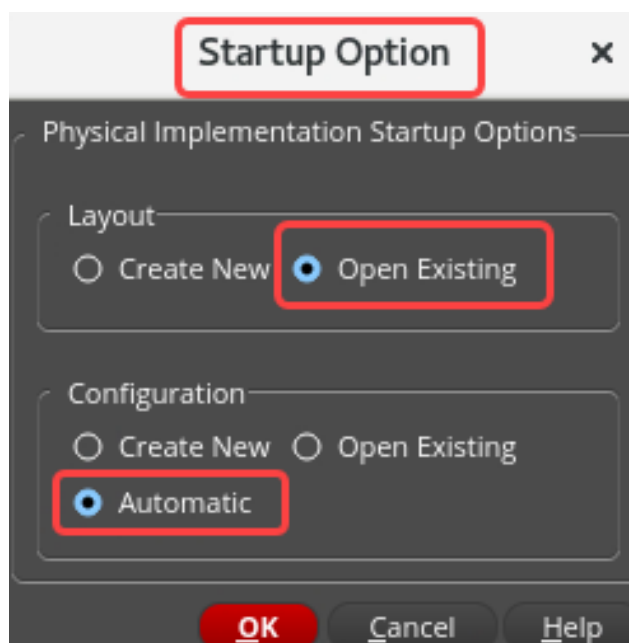
The first block highlighted in red in the schematic above is already laid out, and you will use it to generate clones for the other two blocks.

In this case, you have *Mutant Clones*. They will **NOT** be identical to the reference.

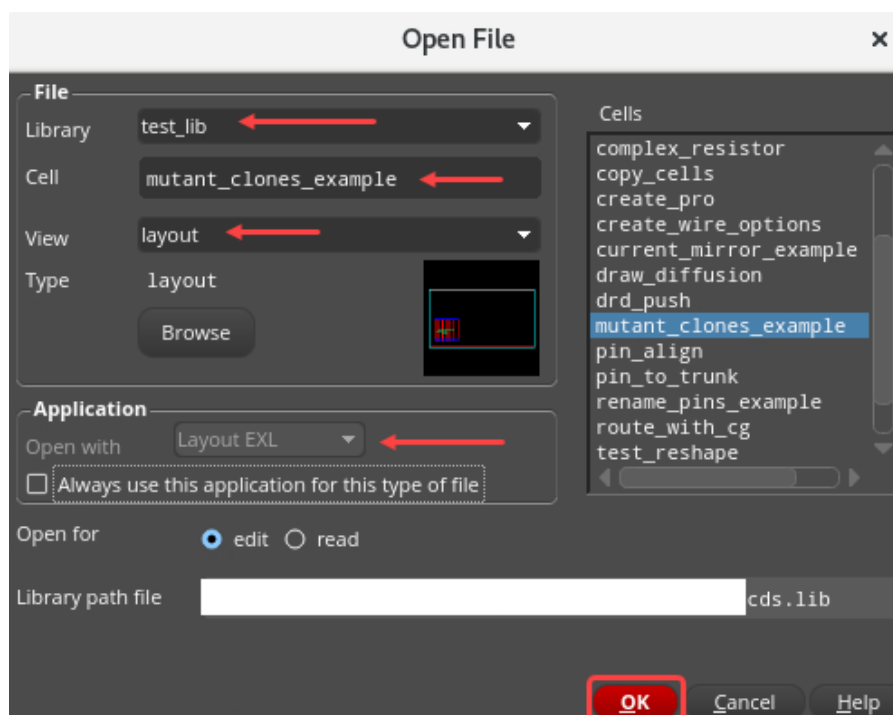
2. In the schematic, choose **Launch – Layout EXL** to bring up the *Startup Option* menu.



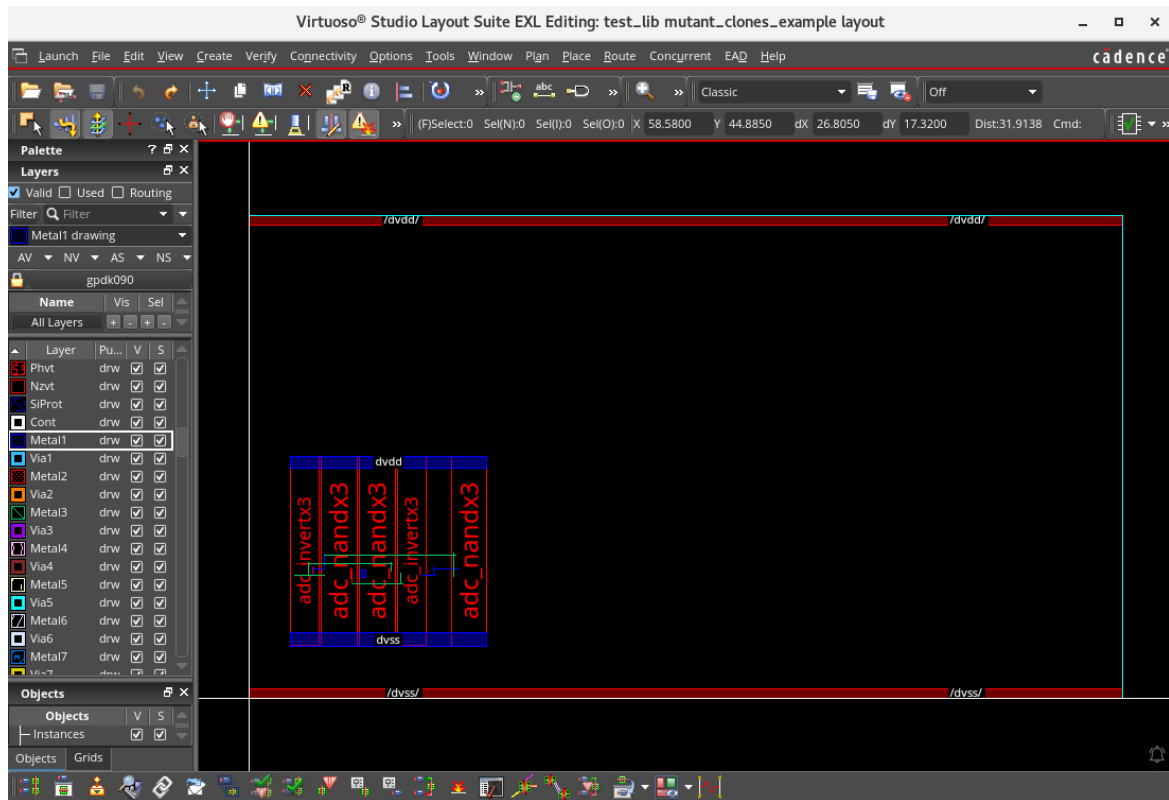
3. Select the options, as shown here, to open an existing layout view and click **OK**.



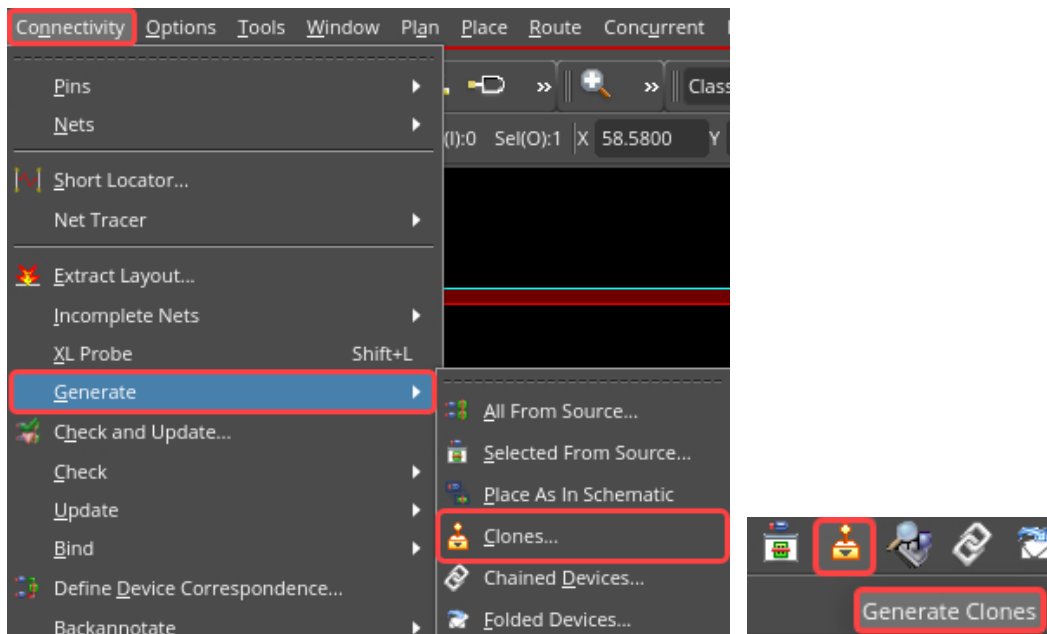
4. Fill out the *Open File* form, as shown here, and click **OK**.



The Virtuoso Layout Suite EXL palette opens with *test_lib/mutant_clones_example/layout* view.



5. Select the five stdcells in the schematic which are grouped together. Use the **Connectivity – Generate – Clones** menu command or click the **Generate Clones** icon in the Layout EXL toolbar to start the Clone command.

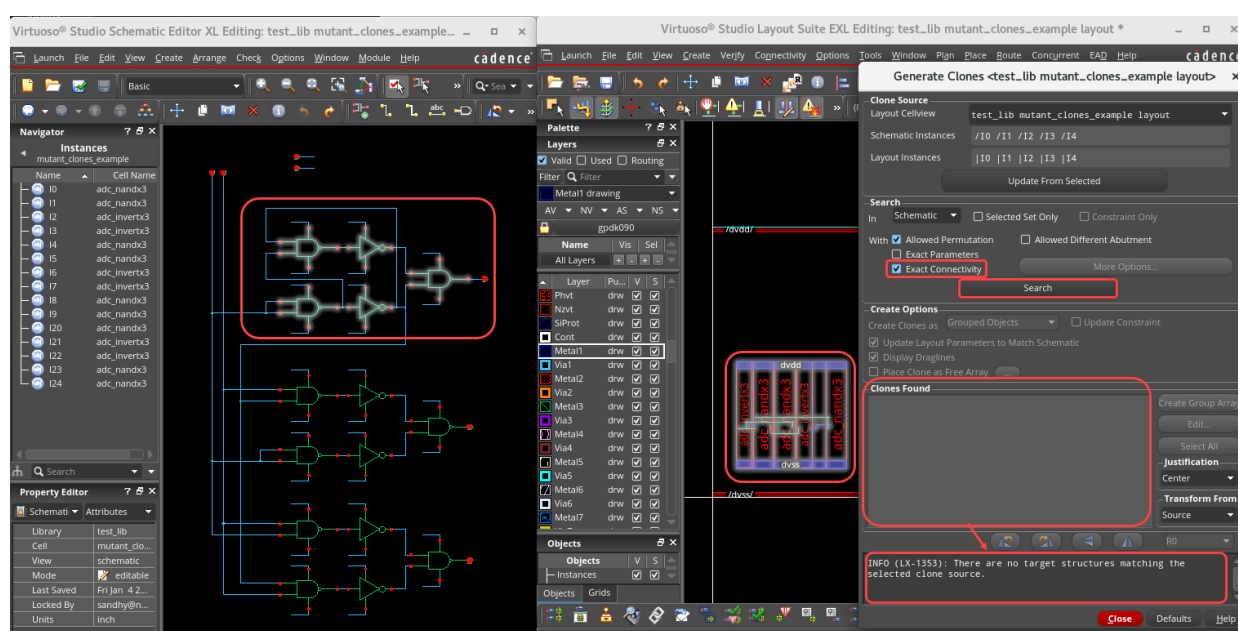


Generating Mutant Clones with *Exact Connectivity* Turned ON

You explore generating the mutant clones with the *Exact Connectivity* option turned **ON** in the Generate Clones form.

1. In the Generate Clones form, do the following:
 - a. Keep the *Exact Connectivity* option turned **ON**.
 - b. Keep the other options as default settings.
 - c. Click the **Search** button.

In this case, no matching target is found in the *Clones Found* pane, as shown next.



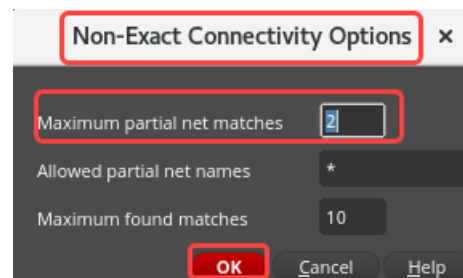
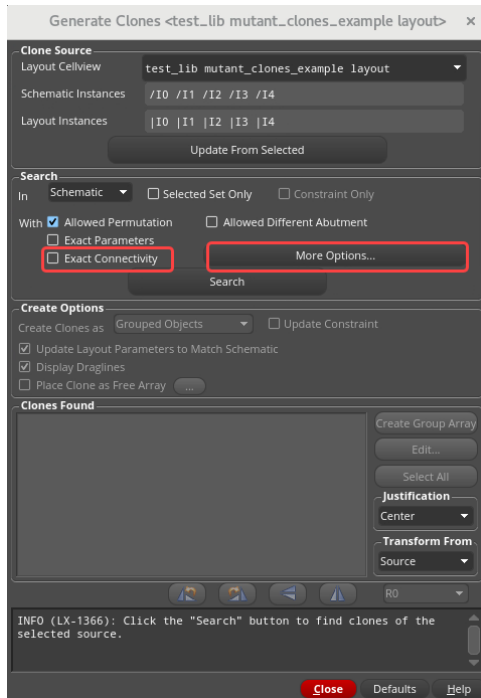
Generating Mutant Clones with *Exact Connectivity* Turned OFF with Relax Match as 2

It is possible to “relax” the matching condition by turning the *Exact Connectivity* option **OFF**. In this case, it is also important to specify how many “differences” you allow. If there are too many differences, then the clones probably do not correspond. You explore this option now.

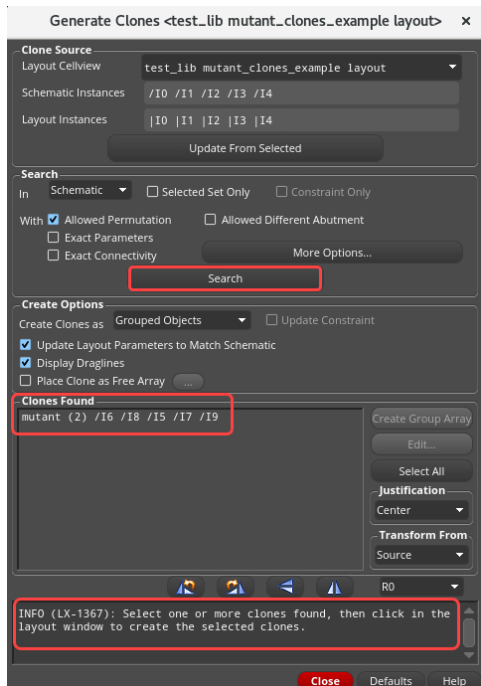
1. In the Generate Clones form, do the following:
 - a. Deactivate the *Exact Connectivity* checkbox.
 - b. Click the **More Options** button to display the *Non-Exact Connectivity Options* form, as shown here.

- c. In the Non-Exact Connectivity Options form, specify **2** as *Maximum partial net matches*.
- d. Click **OK**.

This means that only blocks that have **2** or **less differences** will be taken as matching clones.



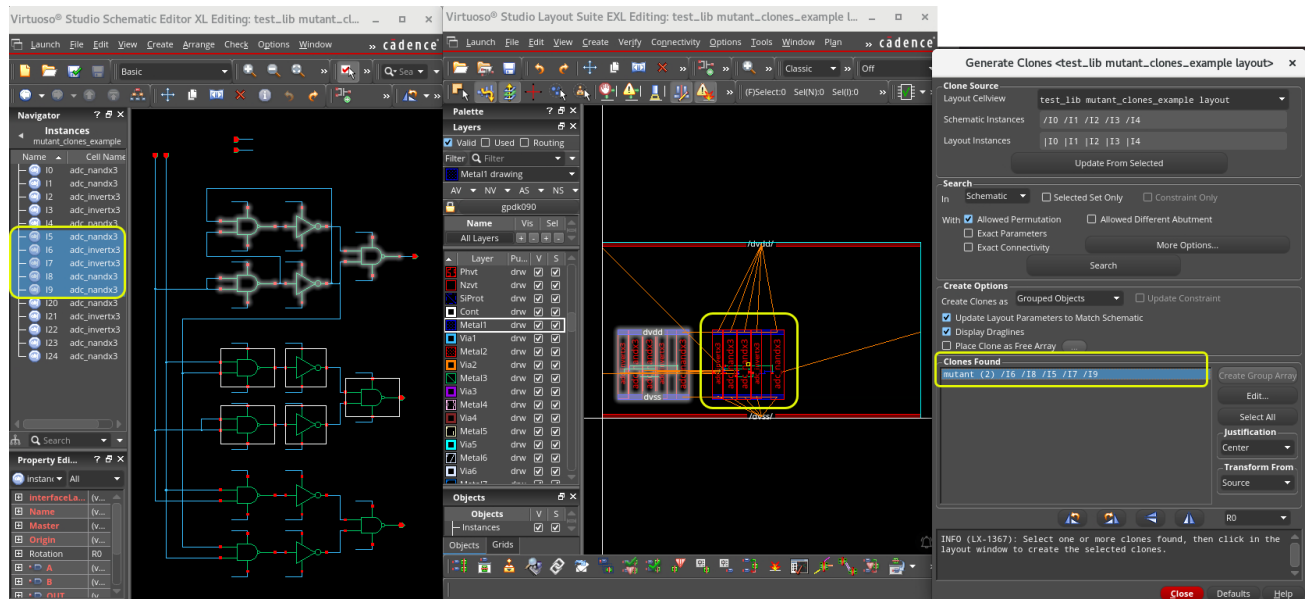
2. In the Generate Clones form, click the **Search** button again to find the matches.



You can see that **one** match has been found in the *Clones Found* pane.

Important: The number in parentheses following the word **mutant** reports the number of connectivity differences found in this clone. In this case, there are **2** connectivity differences found in this clone.

3. Click to select this group in the *Clones Found* pane and place them in your layout, as shown here.



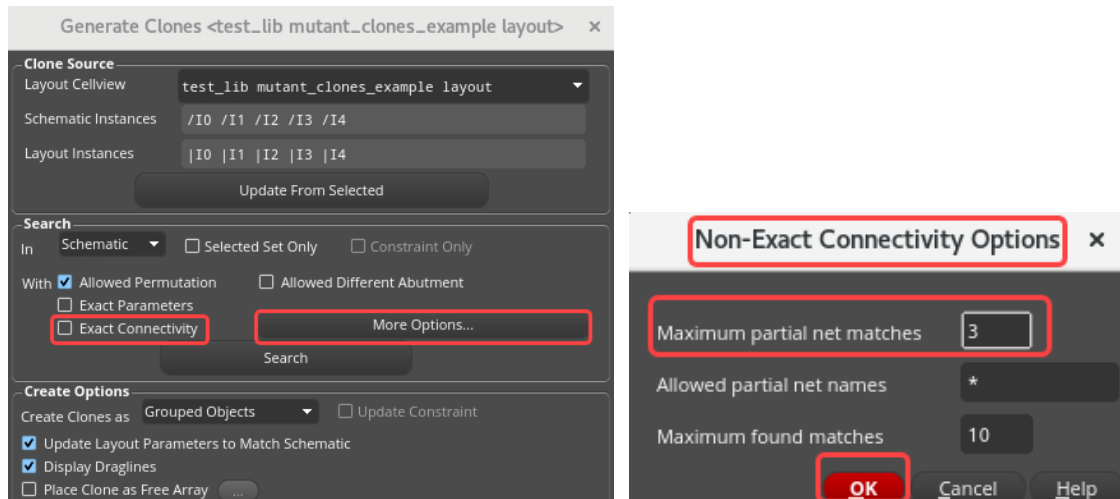
4. In the layout canvas, choose **Edit – Undo** to undo the edits.

Generating Mutant Clones with *Exact Connectivity* Turned OFF with Relax Match as 3

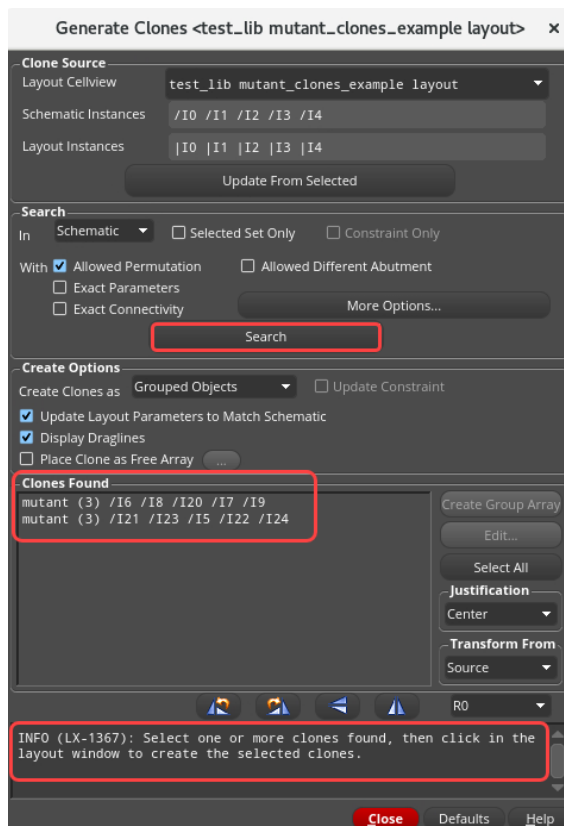
Now you explore generating the mutant clones with the *Exact Connectivity* option turned **OFF** with the relax match option as **3** in the Generate Clones form.

1. In the Generate Clones form, do the following:
 - a. Make sure that the *Exact Connectivity* checkbox is deselected.
 - b. Click the **More Options** button again to display the *Non-Exact Connectivity Options* form, as shown here.
 - c. In the Non-Exact Connectivity Options form, increase the *Maximum partial net matches* value to **3**.
 - d. Confirm by clicking **OK**.

This means that only blocks that have **3 or less differences** will be taken as matching clones.



2. In the Generate Clones form, click the **Search** button again to find the matches.

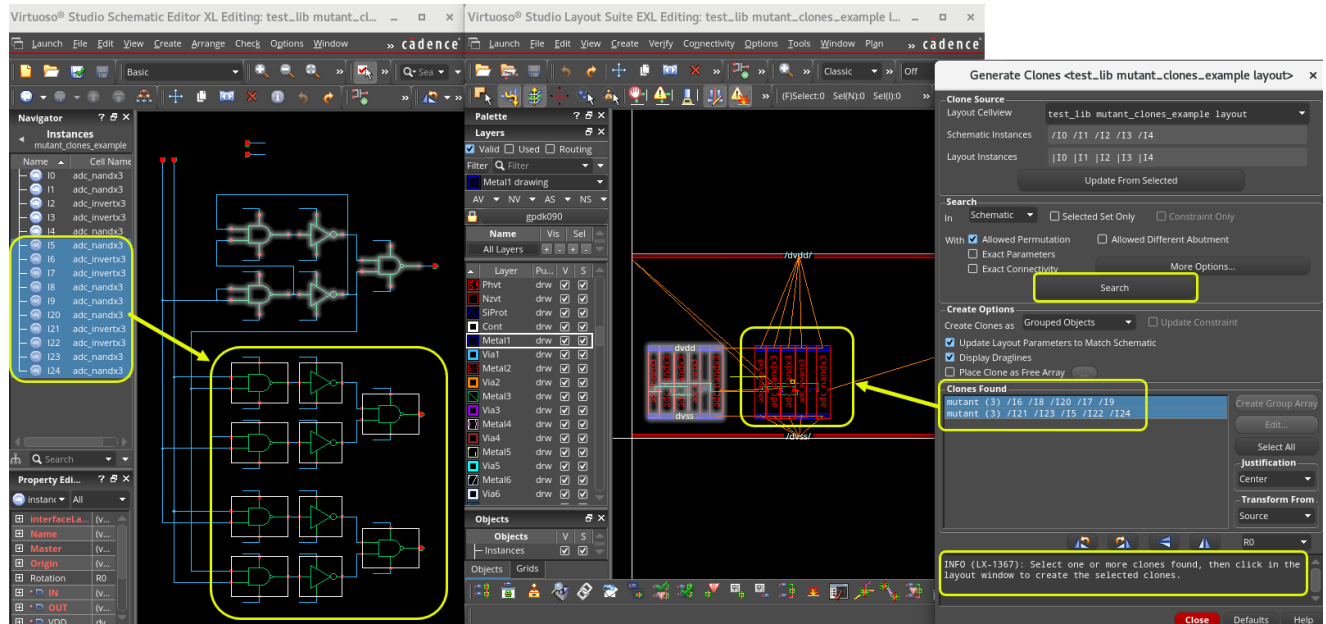


This will lead to the recognition of the missing group of gates.

You can see that **two** missing group of gates are found in the *Clones Found* pane.

Important: The number in parentheses following the word **mutant** reports the number of connectivity differences found in these clones. Here there are **3** connectivity differences found in these clones.

- Click to select these groups in the *Clones Found* pane and place them in your layout, as shown here.



Creating Mutant Clones

In some cases, it is helpful to use the **Mutant Clones**. The small differences in the connectivity can be fixed later by hand, using the flight lines and the connectivity information automatically created by Virtuoso Layout Suite (EXL). The only pitfall is that relaxing the *Exact Connectivity* criteria might lead to false positives when searching for matching targets. Therefore, it is very important to proceed with the lowest possible numbers of differences.

- Close the Generate Clones form, schematic and layout views. Do **NOT** save.
- Keep the Virtuoso session running for the next lab.

Summary

In this lab, you

- Generated clones when the connectivity is not identical
- Set the tolerance when creating *mutant clones*



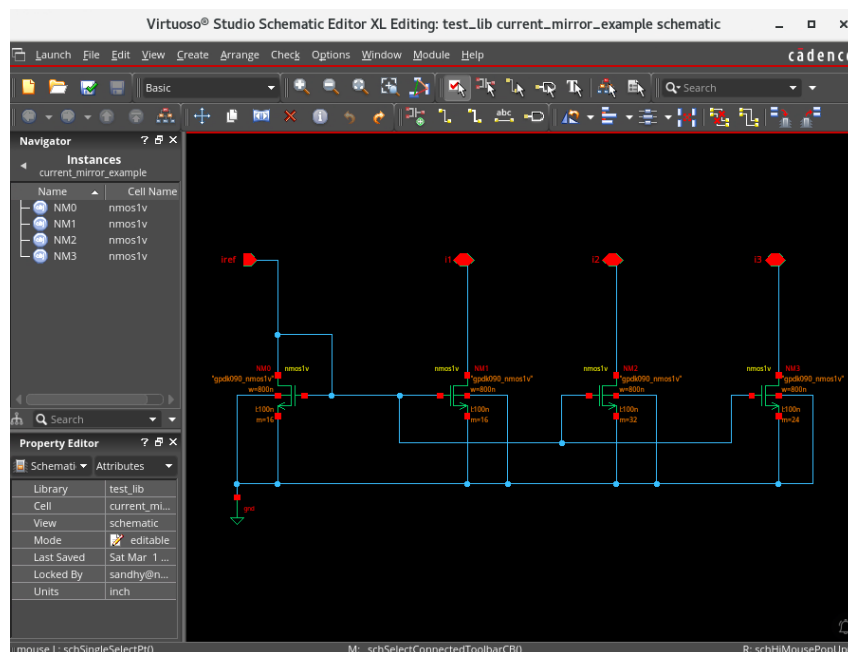
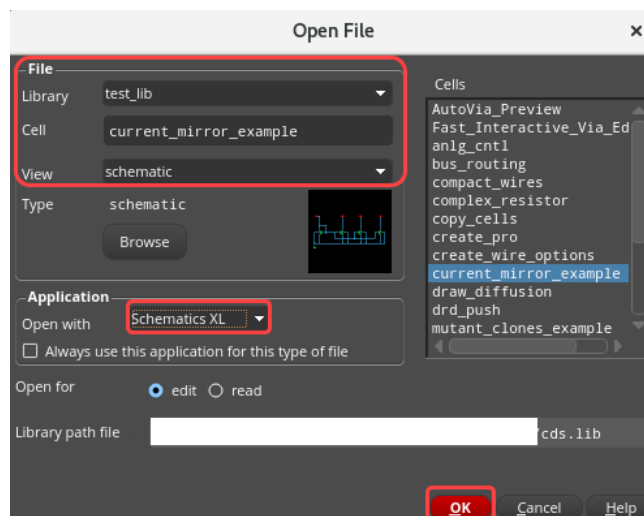
Lab 2-3 Cloning and Copying in VLS-EXL

Objective: To create a “repetitive” layout cell using the Clone and the Copy commands.

In this lab, you clone a structure with the “exact” connectivity and copy and map to reuse blocks that do not have matching connectivity.

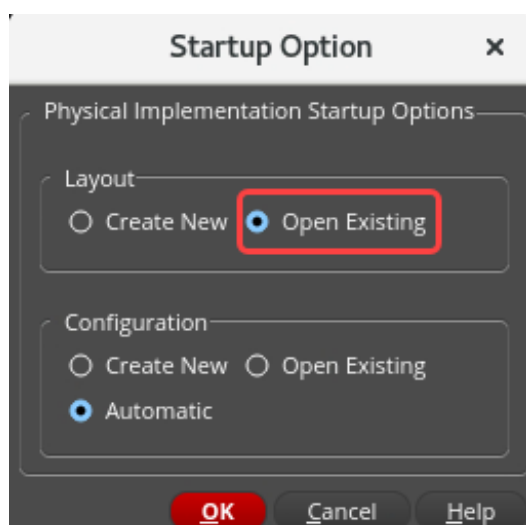
1. In the CIW, choose the **File – Open** menu command and open the following schematic view in XL mode.

Library	Cell	View
test_lib	current_mirror_example	schematic

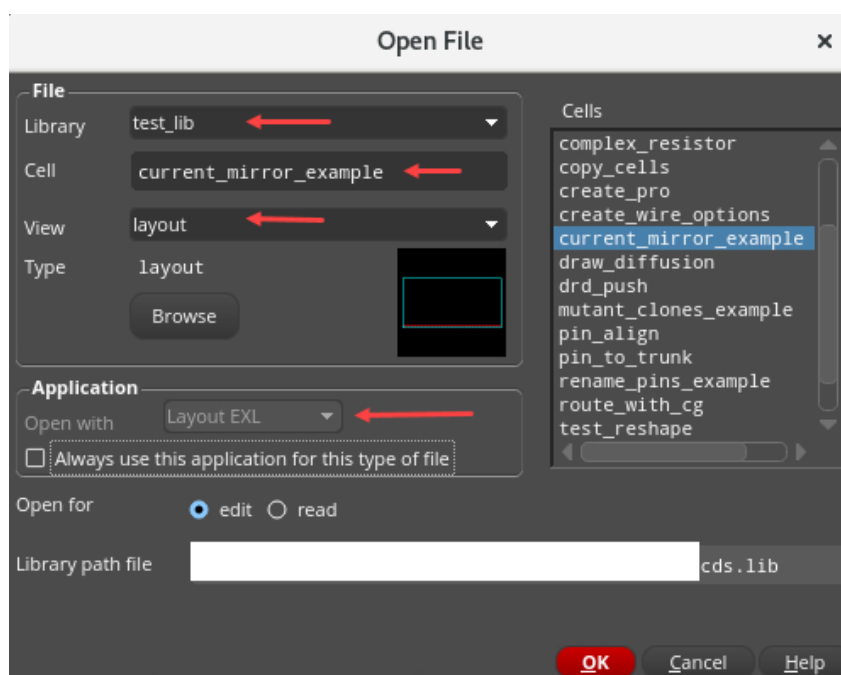


The `test_lib/current_mirror_example/schematic` cellview opens for editing. The circuit is a very simple current mirror.

2. In the schematic, choose **Launch – Layout EXL** to bring up the *Startup Option* menu.
3. Select the options, as shown here, to open an existing layout view and click **OK**.

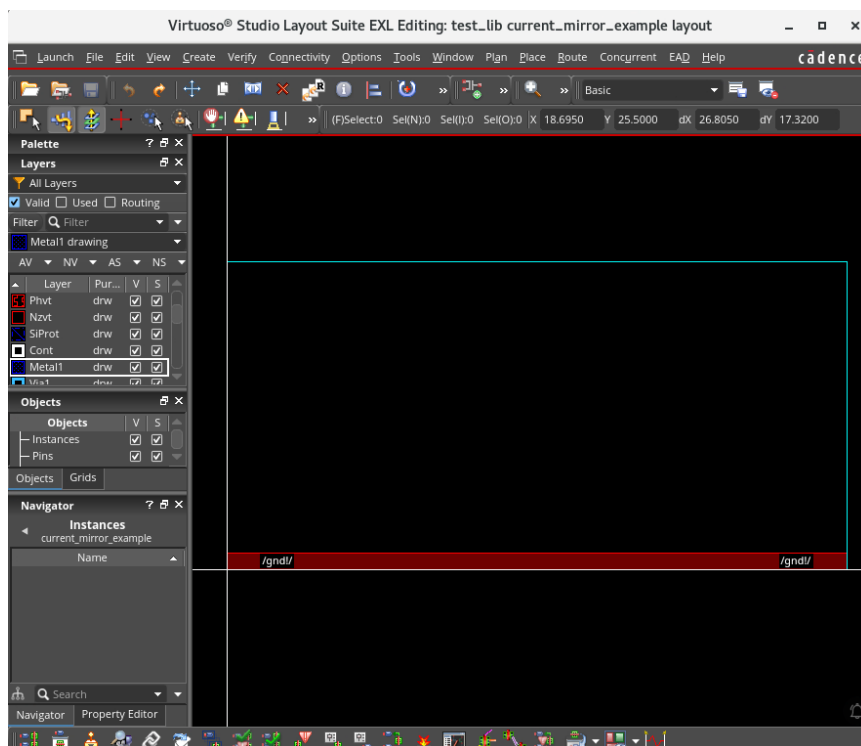


4. Fill out the *Open File* form, as shown here, and click **OK**.

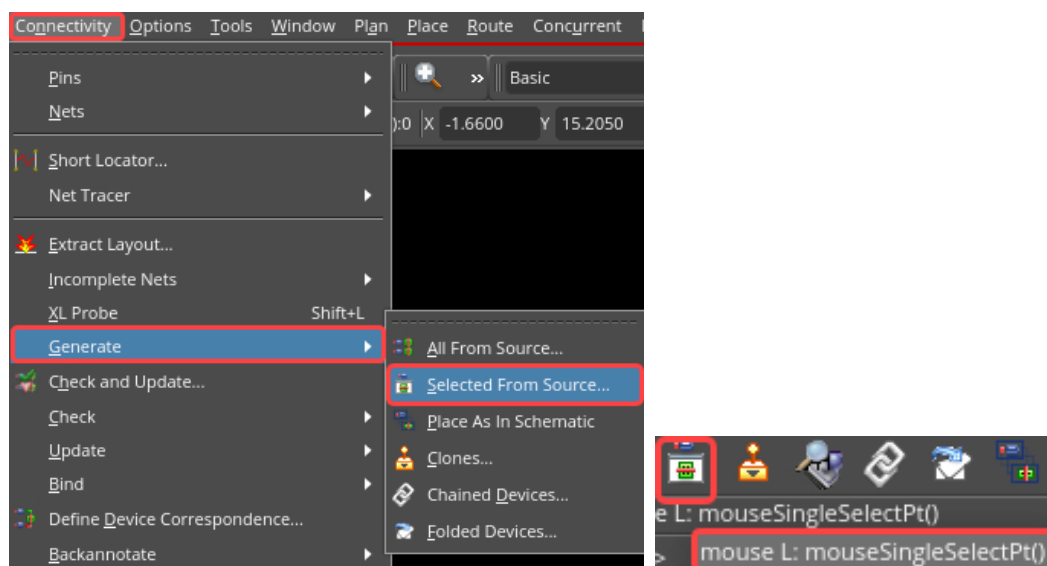


The Virtuoso Layout Suite EXL palette opens with `test_lib/current_mirror_example/layout` view.

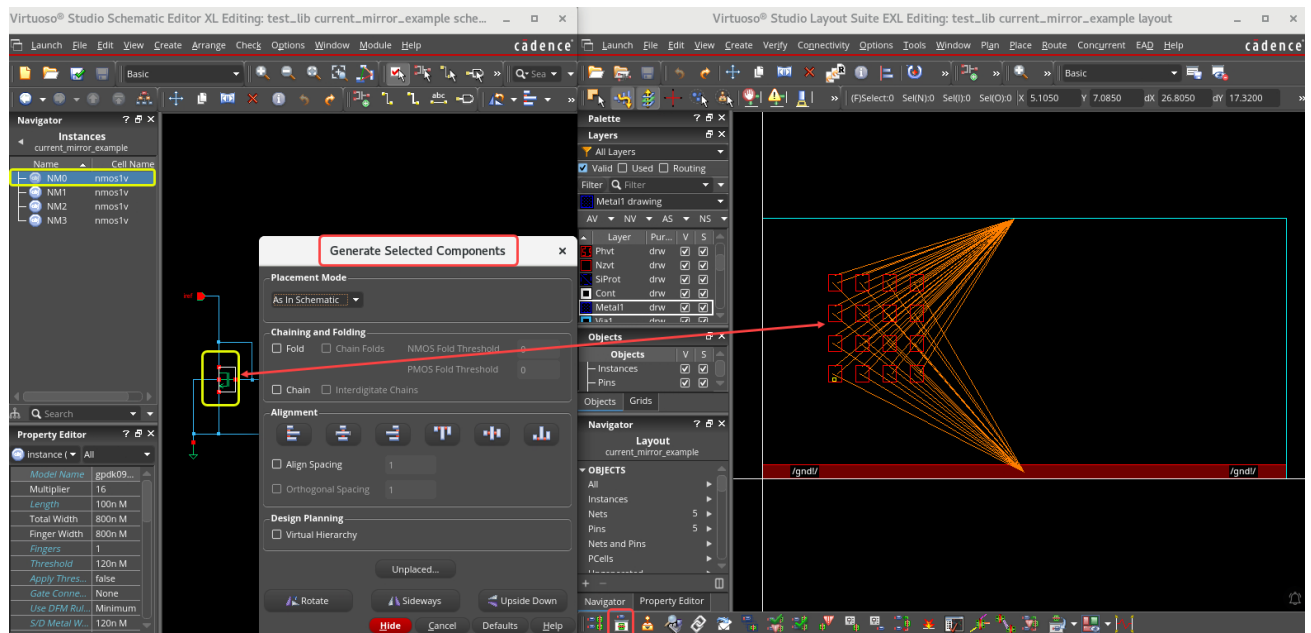
5. In the layout window, choose **Basic** from the *Workspace Chooser* to open the Navigator and the Property Editor assistants in the design canvas.



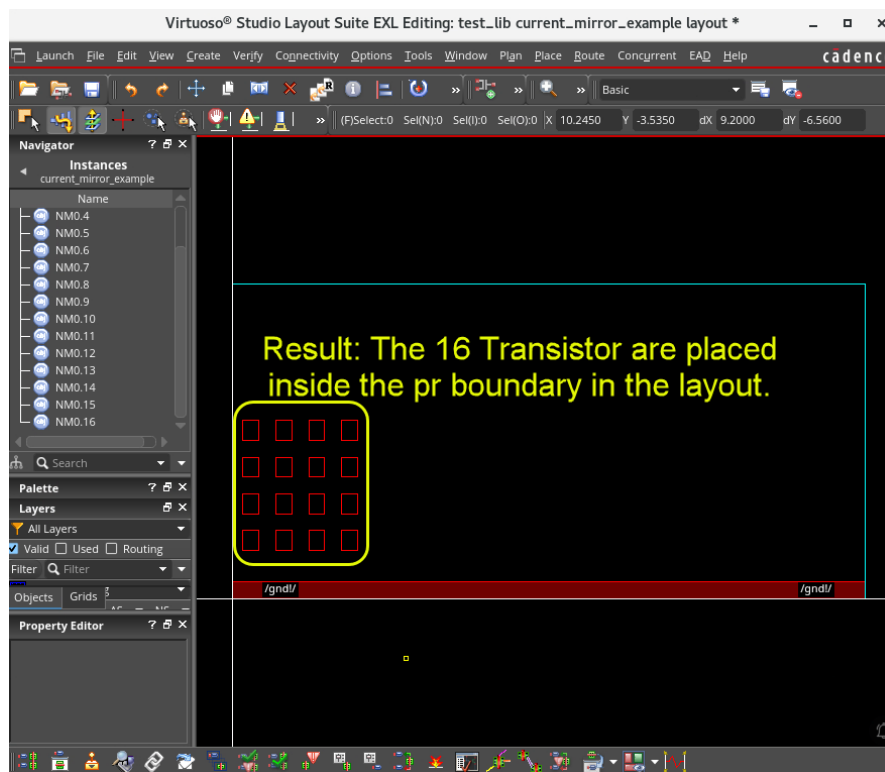
6. Create the first set of transistors: generate the reference transistor.
 - a. Select the instance **NM0** in the schematic window.
 - b. Generate selected from the source by choosing either the **Connectivity – Generate – Selected From Source** menu command, or by clicking the **Generate Selected From Source (GSFS)** icon in the Layout EXL toolbar to open the Generate Selected Components form.

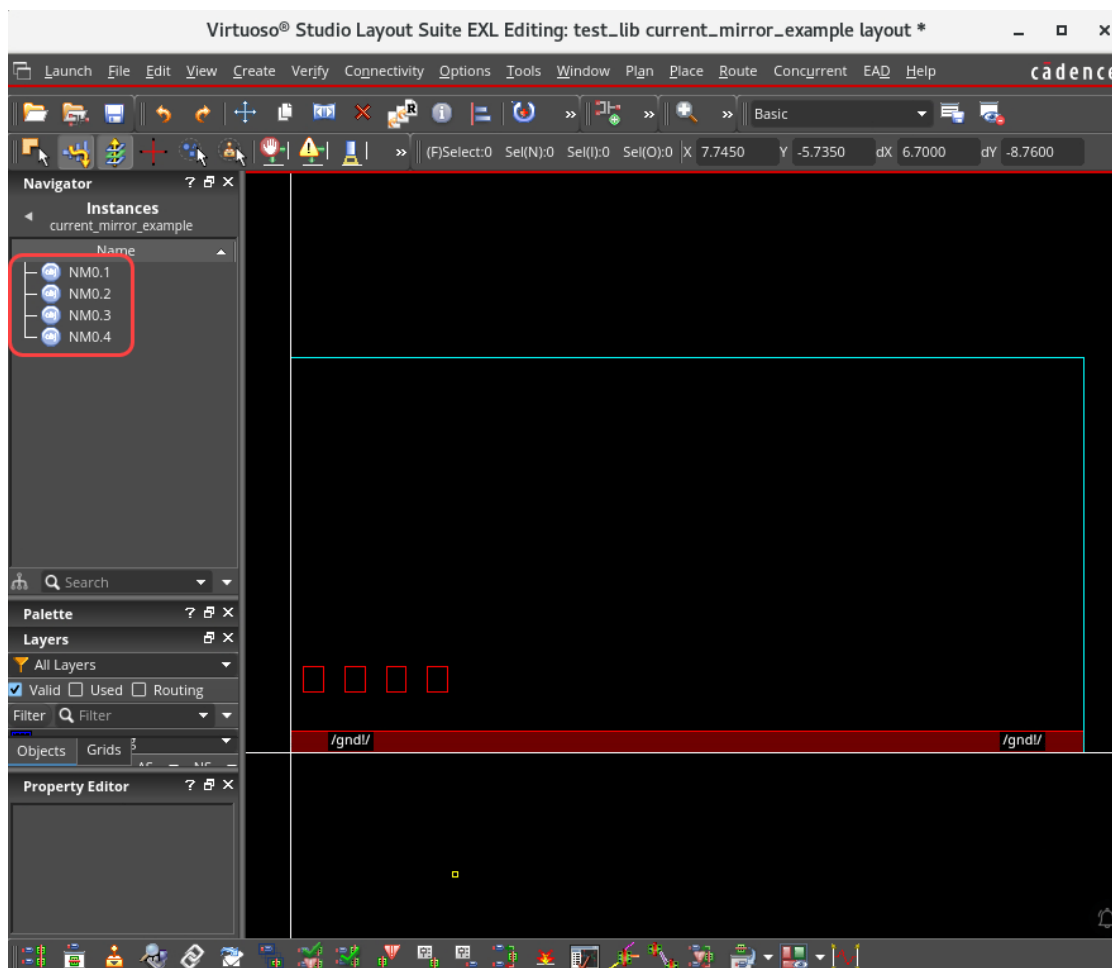


- c. Keep the default values in the Generate Selected Components form.
- d. Place the **16** transistors inside the *prBoundary*.



7. Because you intend to use a block of “4-fingers” as your building tile, you can delete the remaining 12 instances and only keep the 4 you need, as shown here.



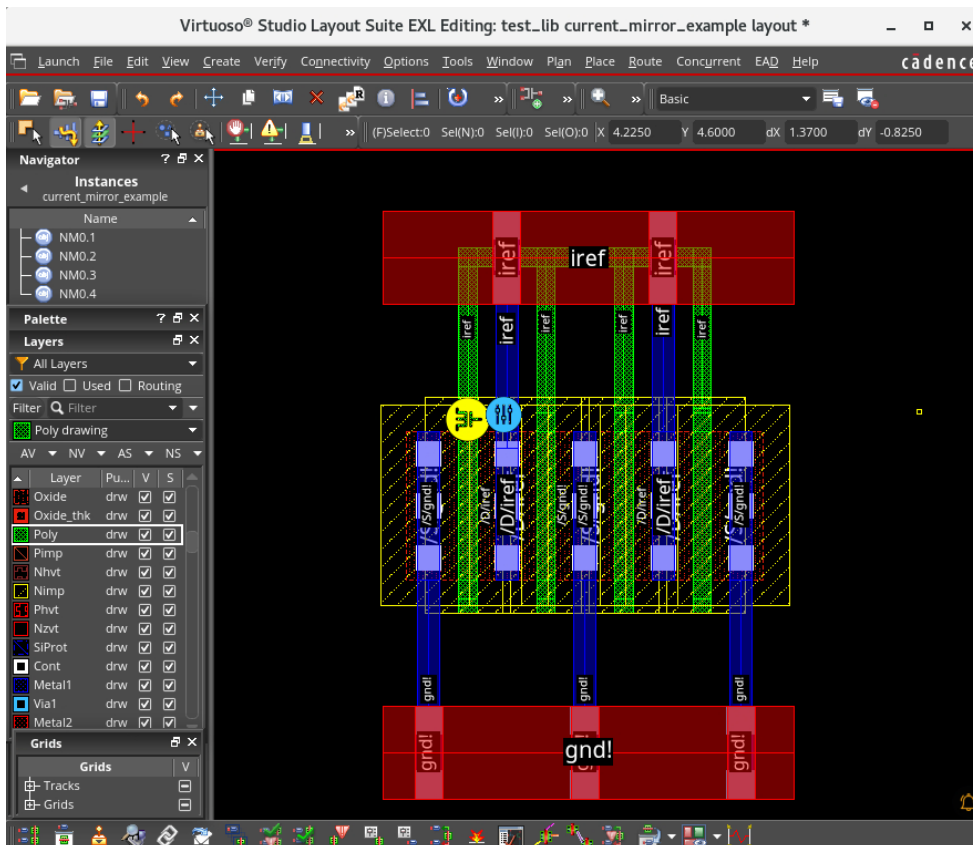
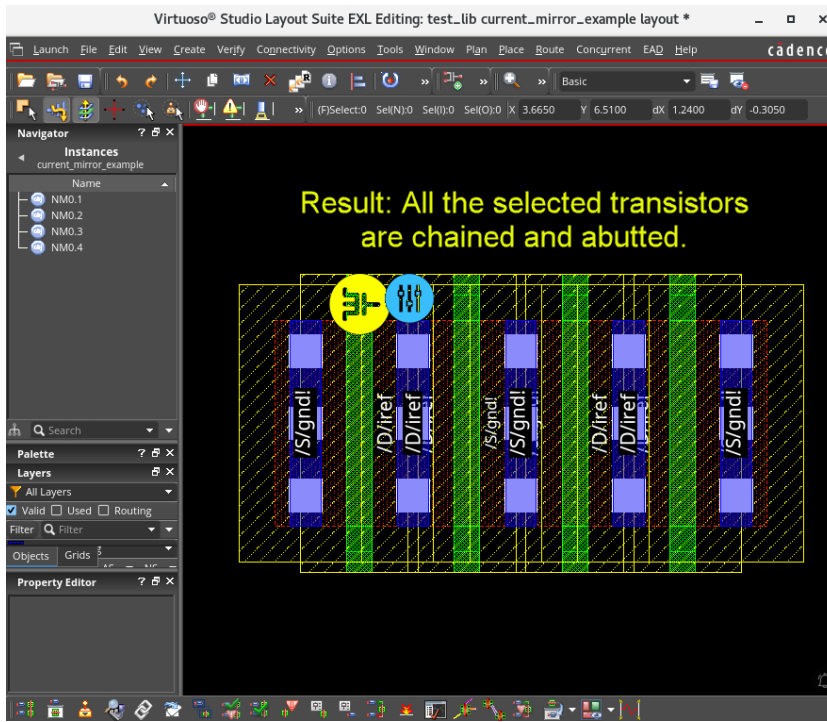


Note: The image above represents after deleting 12 instances.

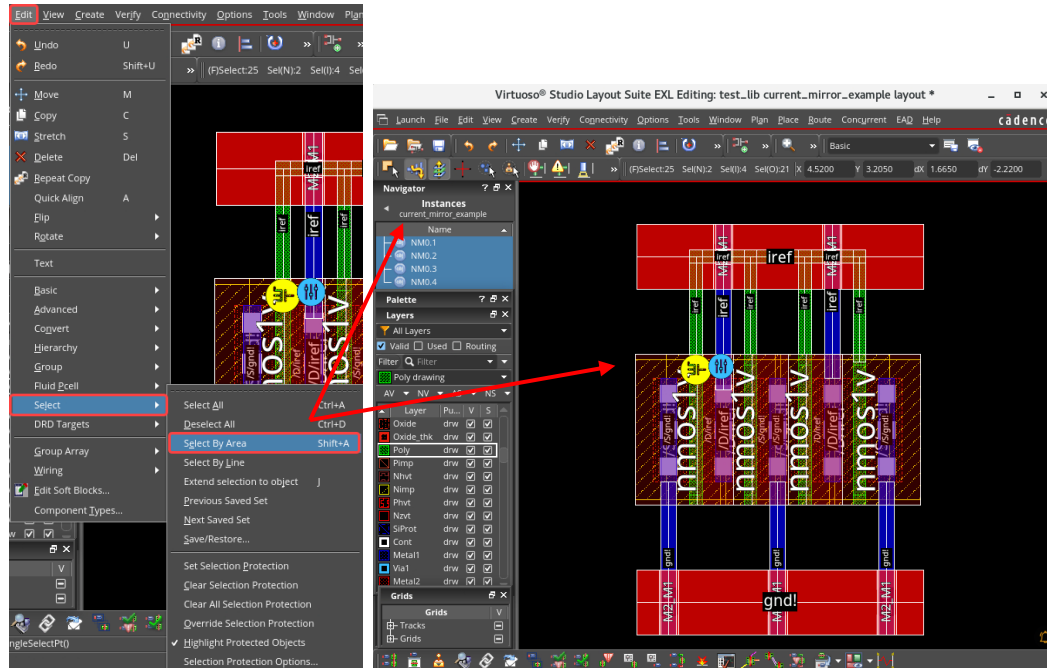
8. About the four transistors and connect them as shown in the graphic.

Note: Try to use *Auto Abutment*, *Quick Align*, and *Create Wire* as much as possible.

Tip: (**Shift+f** and see the physical view of the instances. Then Abut the devices with quick align (bindkey: 'a' option).

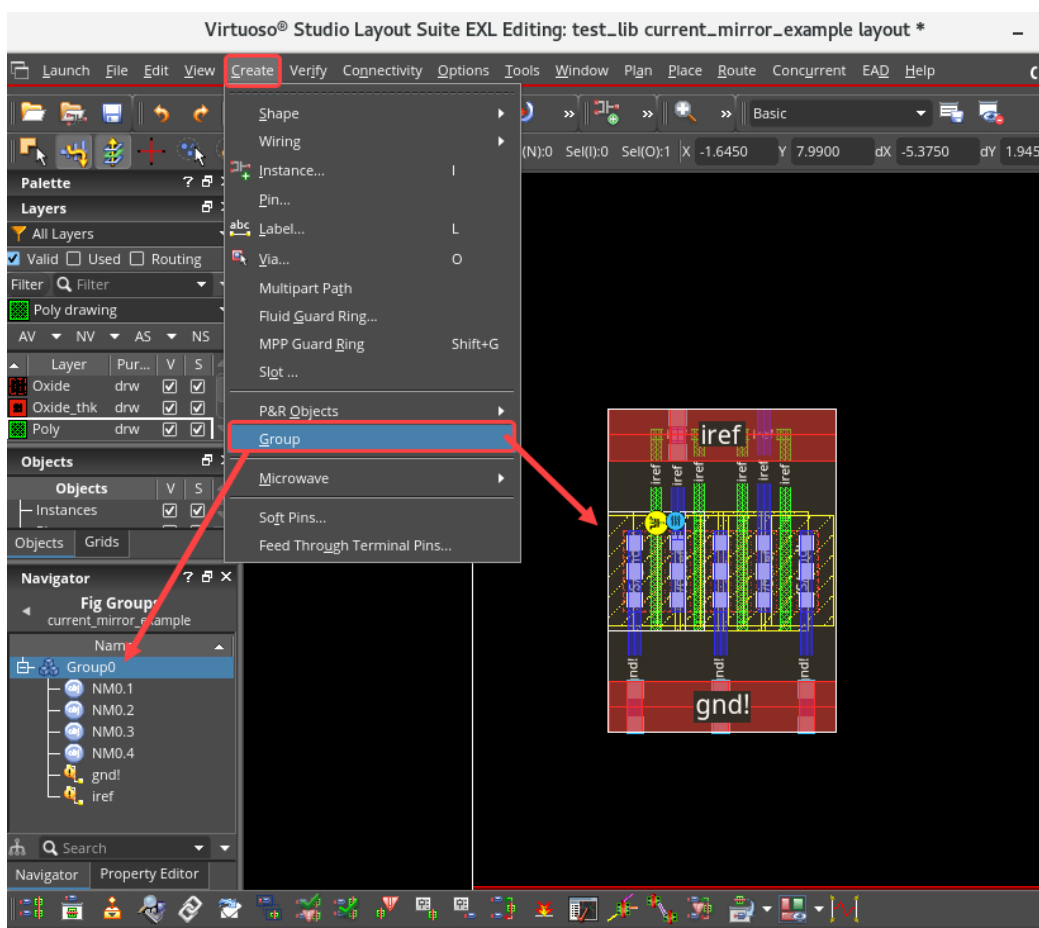


9. After the four transistors are properly connected, select the instances and the routing shapes in the design canvas and create a group.
 - a. Choose the **Edit – Select – Select By Area** menu command or press the bindkey **Shift+A** to select the 4 instances and the routing shapes in the design.



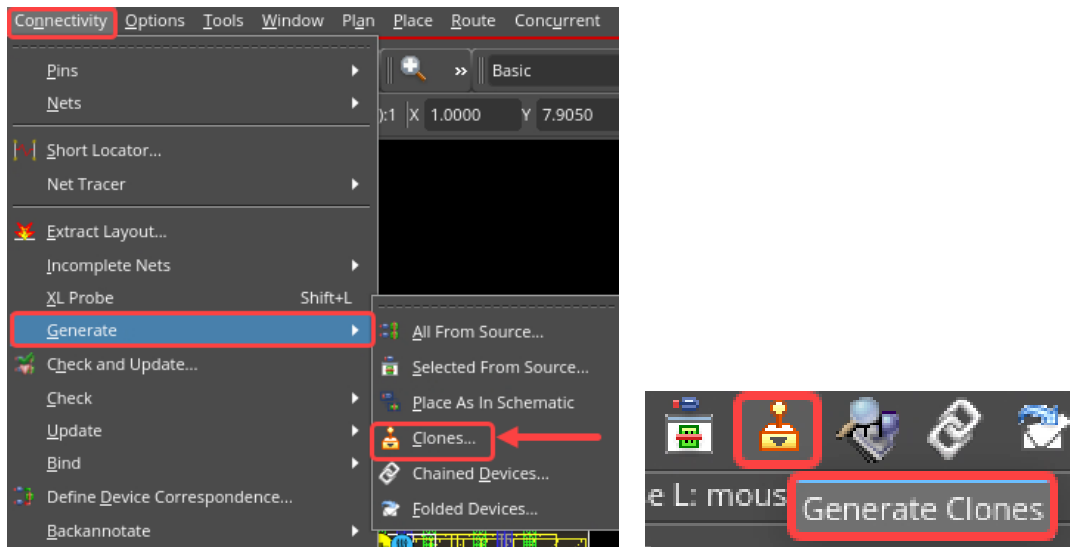
- b. Choose the **Create – Group** menu command in the layout canvas.
 - c. In the Navigator assistant, select **Fig Groups** in the *GROUPS* category to view the groups.

The new group “*Group0*” is displayed in the Navigator assistant.



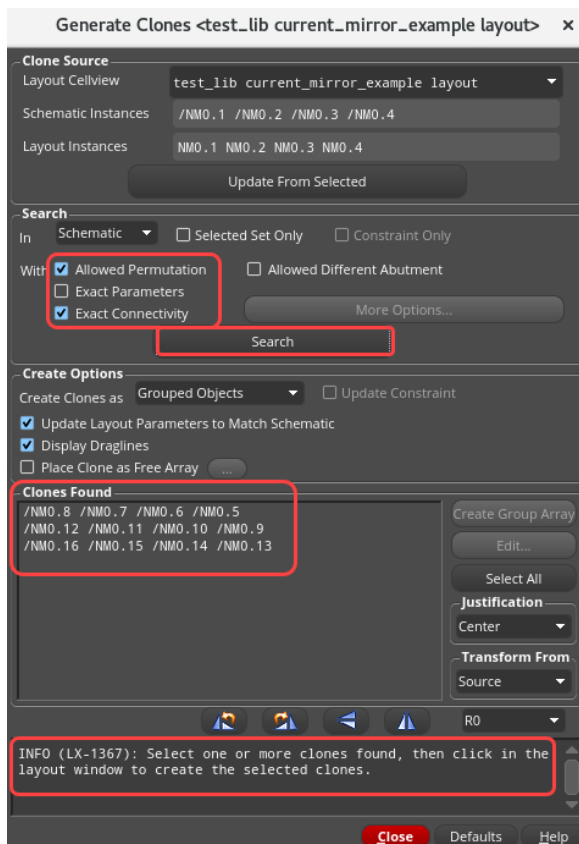
This group of *4fingers* will be the building block you use to create all the transistors in your layout cell.

10. Select the newly created *Group0* and choose the **Connectivity – Generate – Clones** menu command or choose the **Generate Clones** icon in the Layout EXL toolbar to start the Clone command, as shown here.



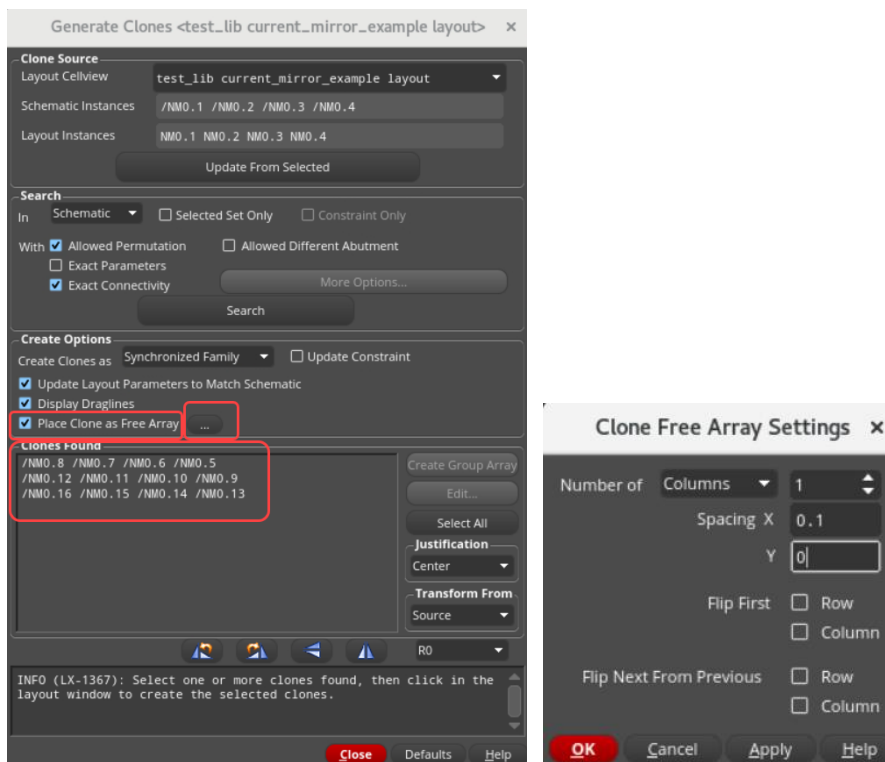
The Generate Clones form appears, as shown here.

11. Leave the default settings and click the **Search** button in the Generate Clones form.



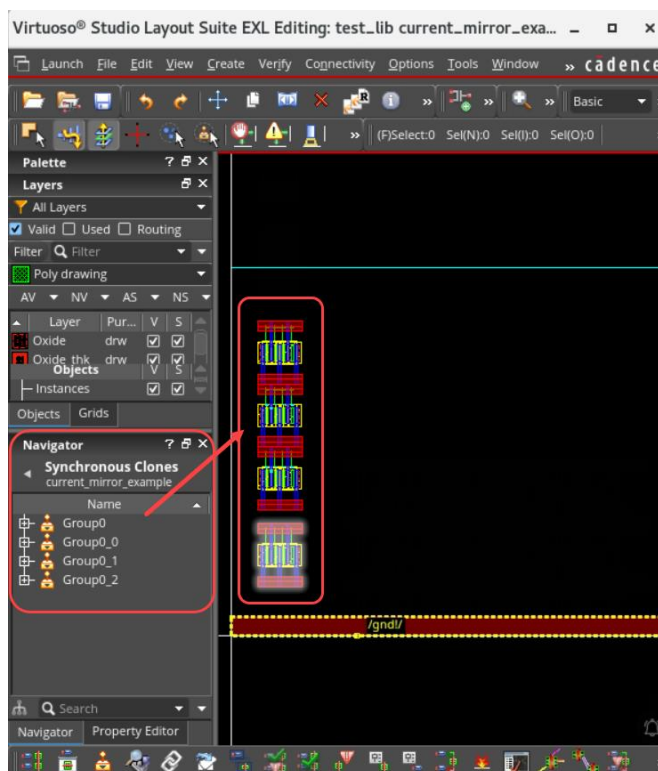
This action populates the *Clones Found* list with **12** instances. These are the same **12** instances you deleted previously.

12. Now you will place these matches found in the *Clones Found* pane from the Generate Clones form in the layout canvas.
 - a. Select all the cells listed using **Shift** and-or **Ctrl** together with the left-mouse-button,
 - b. Activate the **Place Clone as Free Array** and **Display Dragline** button.
 - c. Click the **Array Settings** button  to open the *Clone Free Array Settings* form.
 - d. Set the *Clone Array Settings* form, as shown here.
 - e. Click **OK** in the *Clone Array Settings* form.



13. Keep selecting all these matches found, place these three groups of four transistors one on top of the other which forms a column automatically, as shown here.

Note: In the Navigator assistant, select **Synchronous Clones** in the *GROUPS* category to view the clones created.

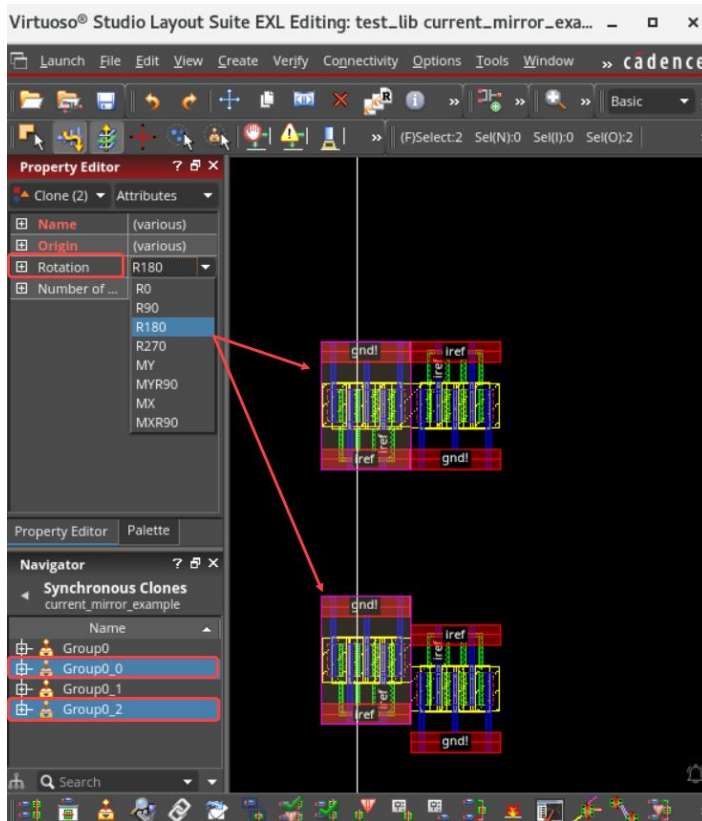


14. When all transistors are placed, click **Close** in the Generate Clones form.

The four clone groups *Group0*, *Group0_0*, *Group0_1*, and *Group0_2* are now “roughly” aligned in a column, but their placement is not ideal: You would like to “mirror” some of the transistors to have a better connection between the **VSS** rails.

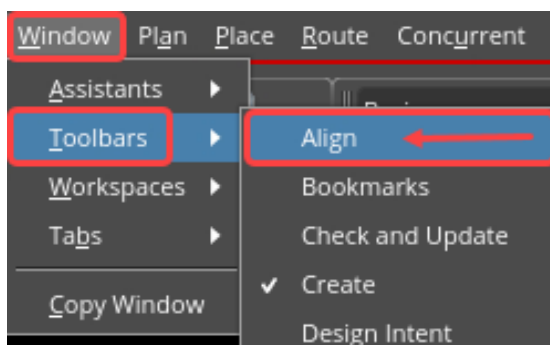
This operation can be easily done since you are using *groups*. All the instances and routing shapes are grouped together, and you can specify an orientation for them.

15. Select the 2nd and the 4th clone groups *Group0_0* and *Group0_2* and change their orientation to **R180**, as shown in the graphic.



The 2nd and the 4th transistors are mirrored now.

16. You need to use the *Alignment* toolbar. In the layout window, choose the **Window – Toolbars – Align** menu command to display it in case it is hidden.

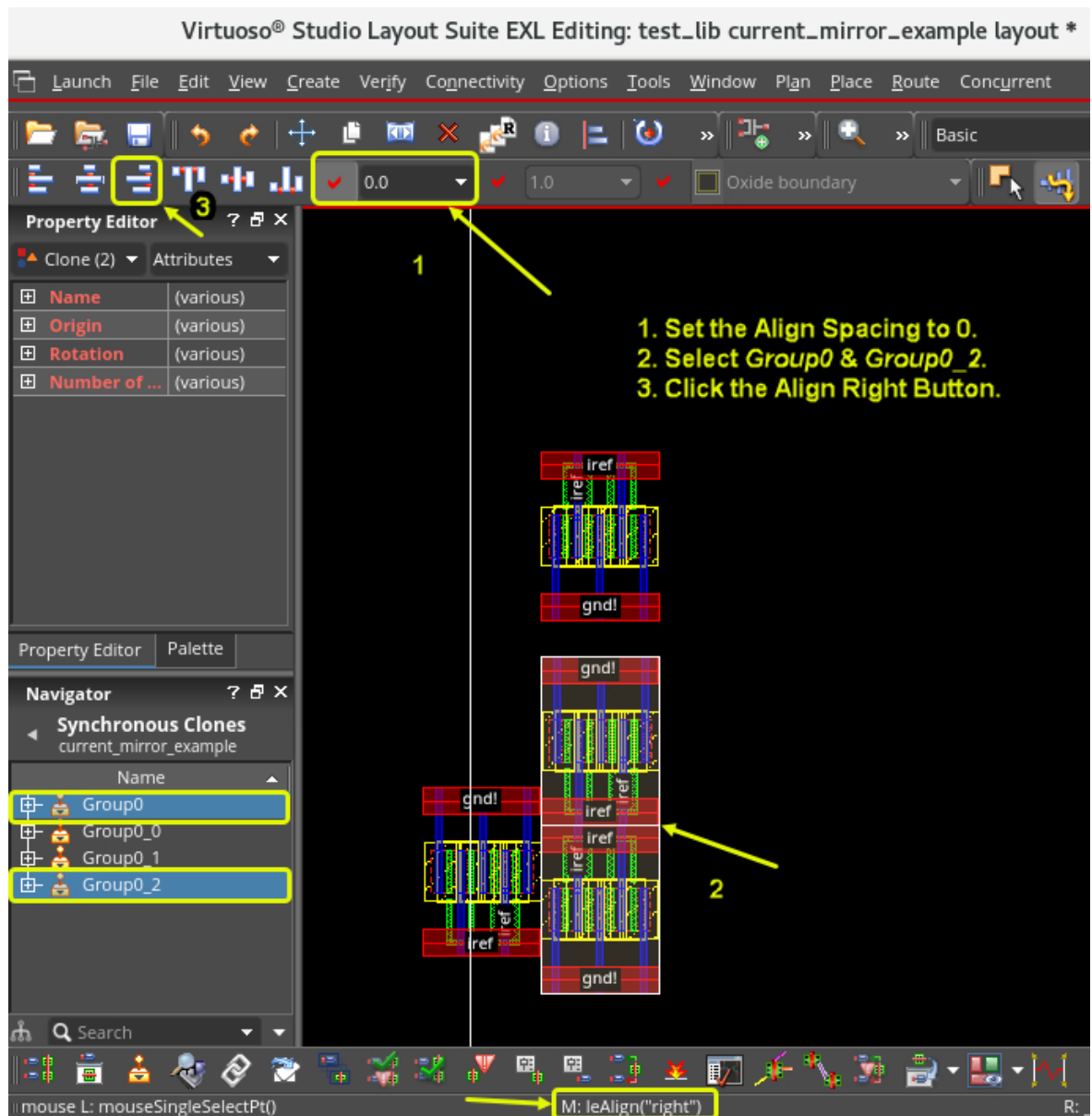


17. In the layout window, set the following in the *Alignment* toolbar.

- a. Activate and set the *Align Spacing* to **0**, so that you have abutting groups.

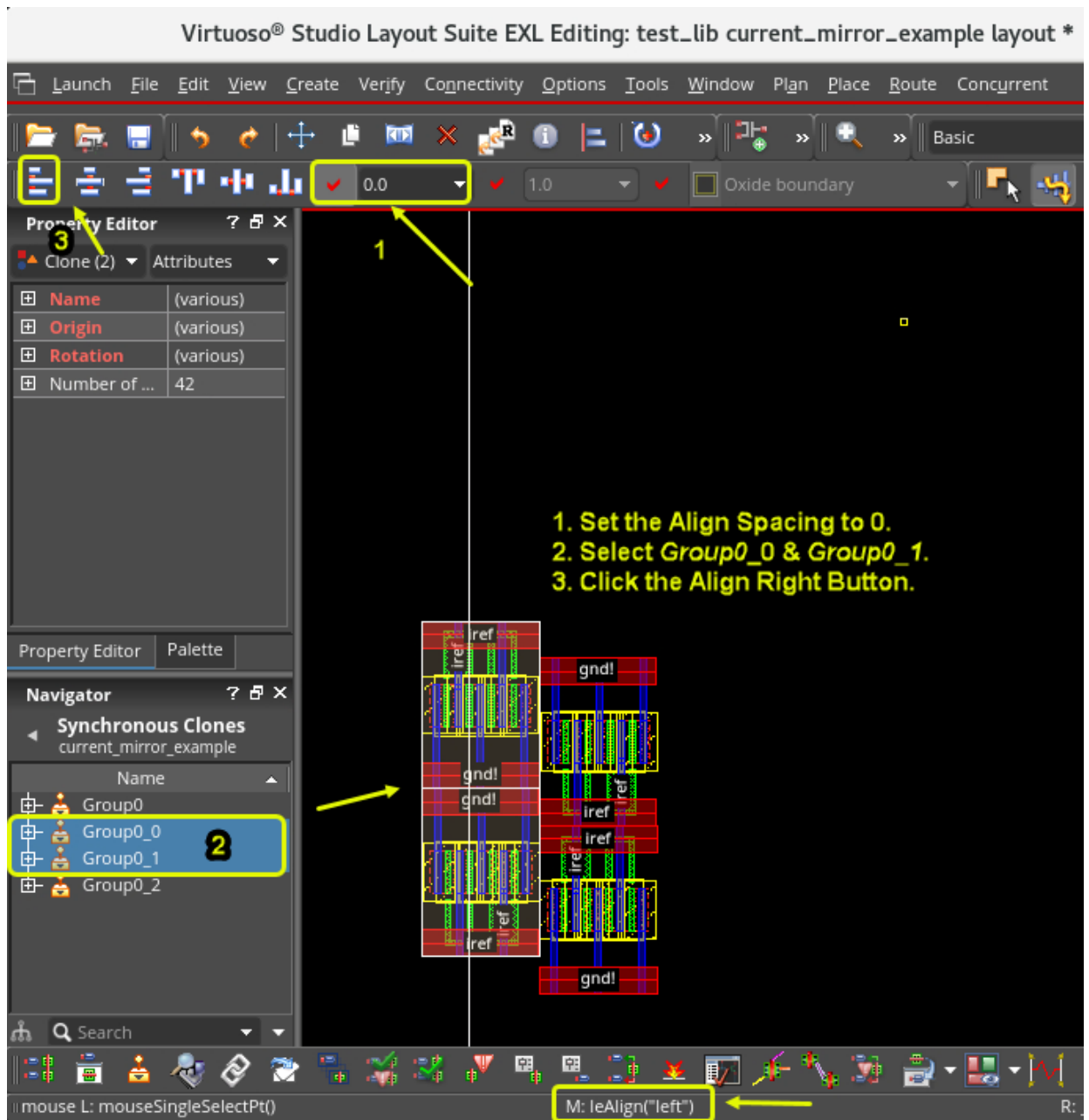


- b. Select *Group0* & *Group0_2* and click the **Align Right** button.



Group0 & *Group0_2* are aligned to the right now.

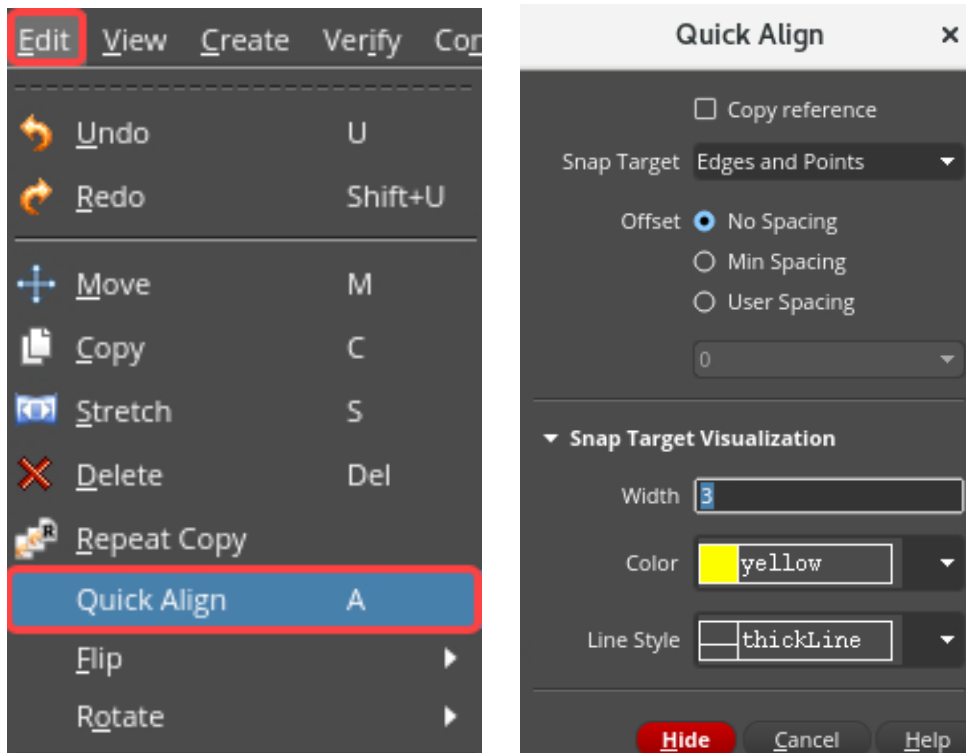
- c. Select *Group0_0* & *Group0_1* and click the **Align Left** button.



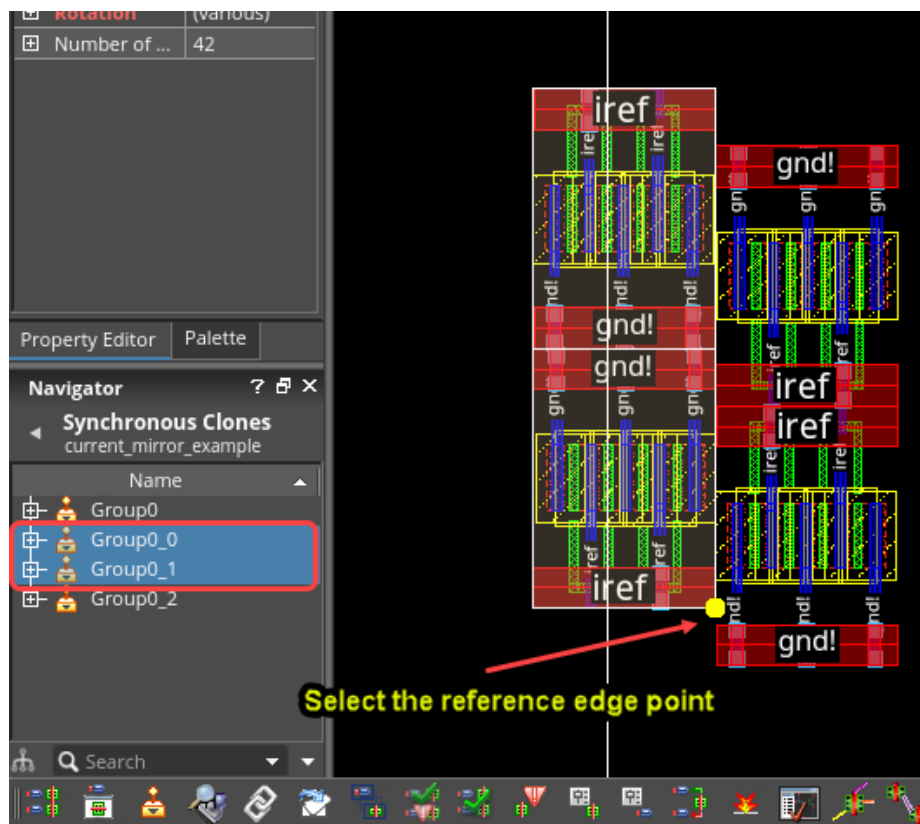
Group0_0 & *Group0_1* are aligned to the left now.

18. Now, with the **2 upper clone families** selected, choose the **Edit – Quick Align** menu command or press the bindkey **A** to invoke the Quick Align form.

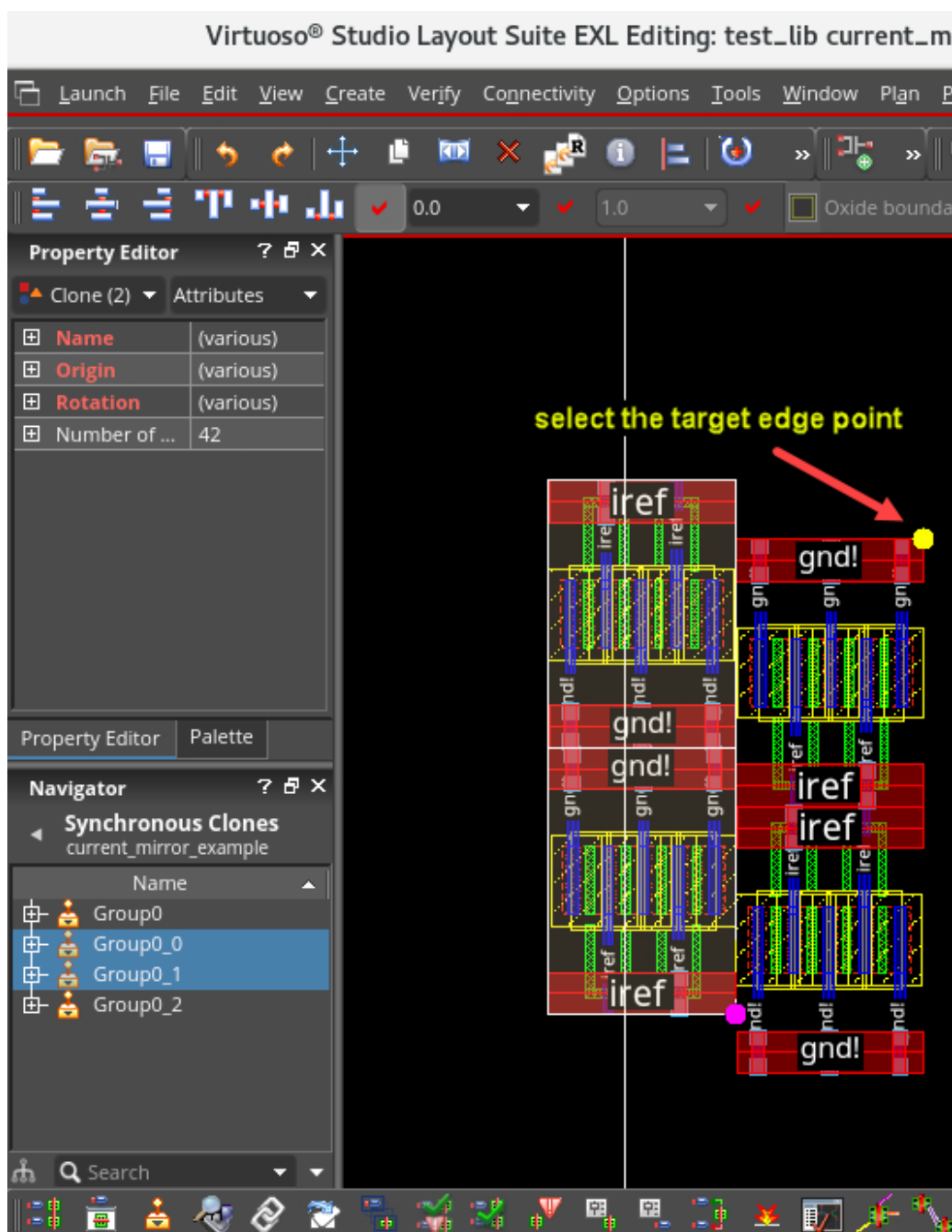
If the form does not open, press the **F3** key to open the form.



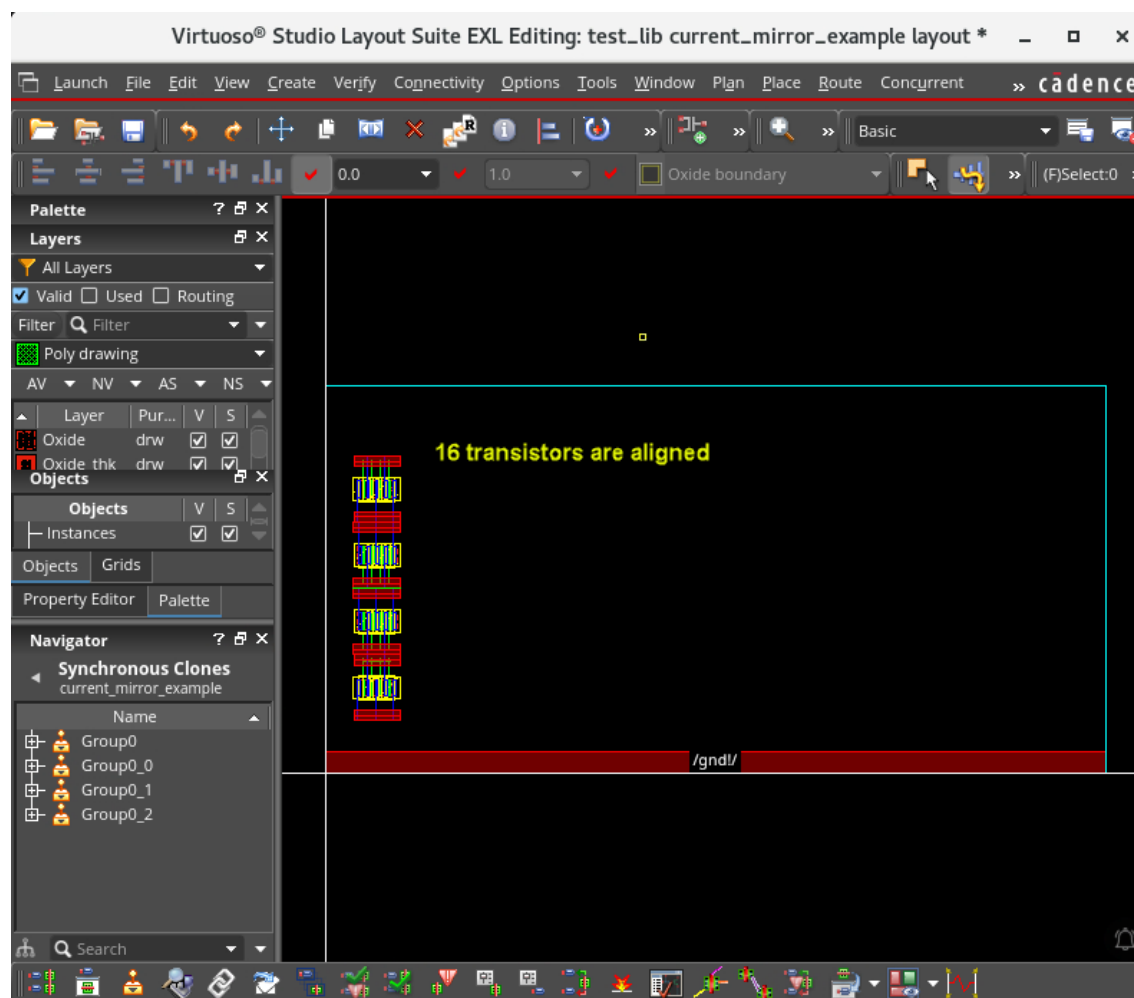
- a. Select the reference edge point WITHIN the selection set.



- b. Select the target edge point.

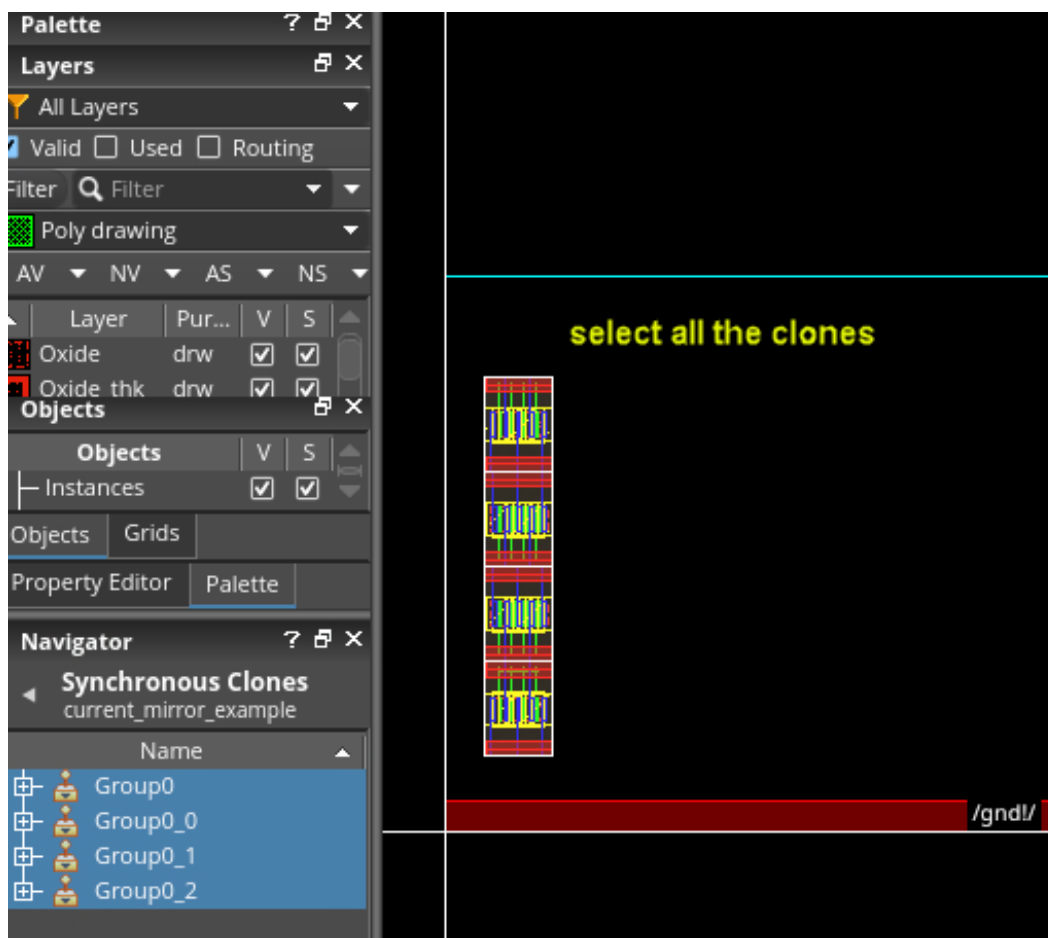


This creates exactly the aligned structure you wanted to achieve, as shown here.

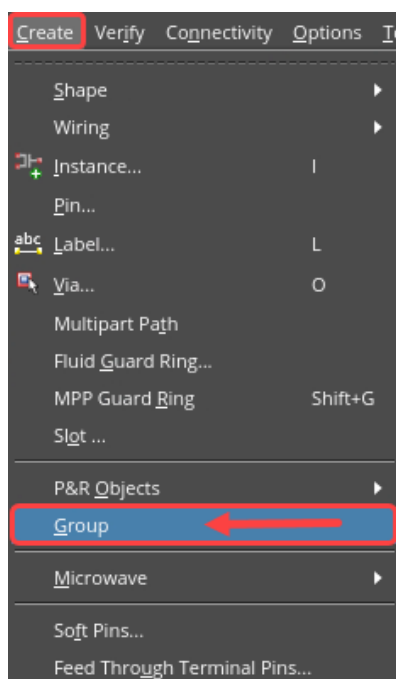


19. You can now group together the **16 transistors** you just aligned in the layout canvas.

- a. Select all the clones in the layout canvas.

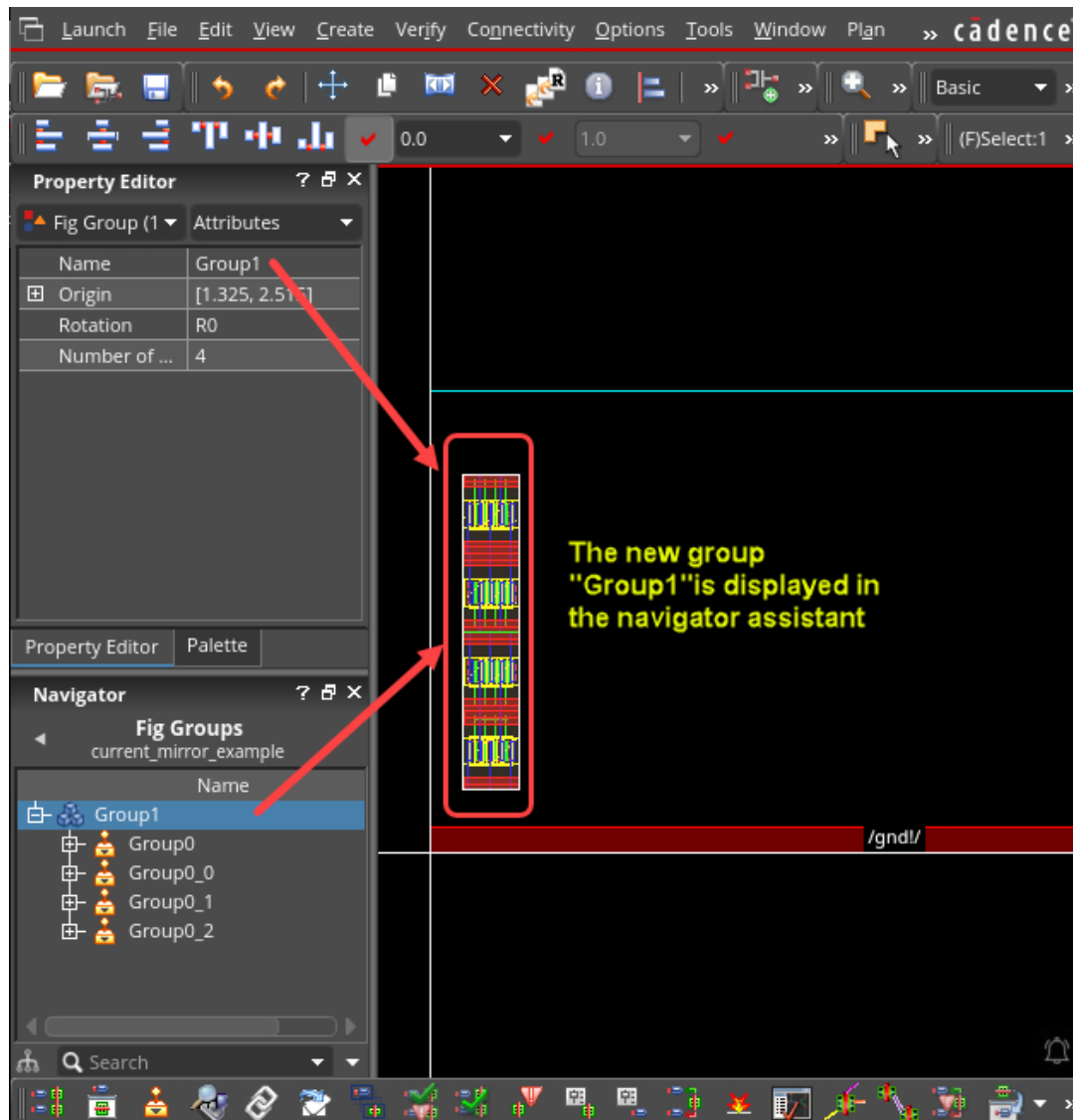


- b. Choose the **Create – Group** menu command in the layout canvas.



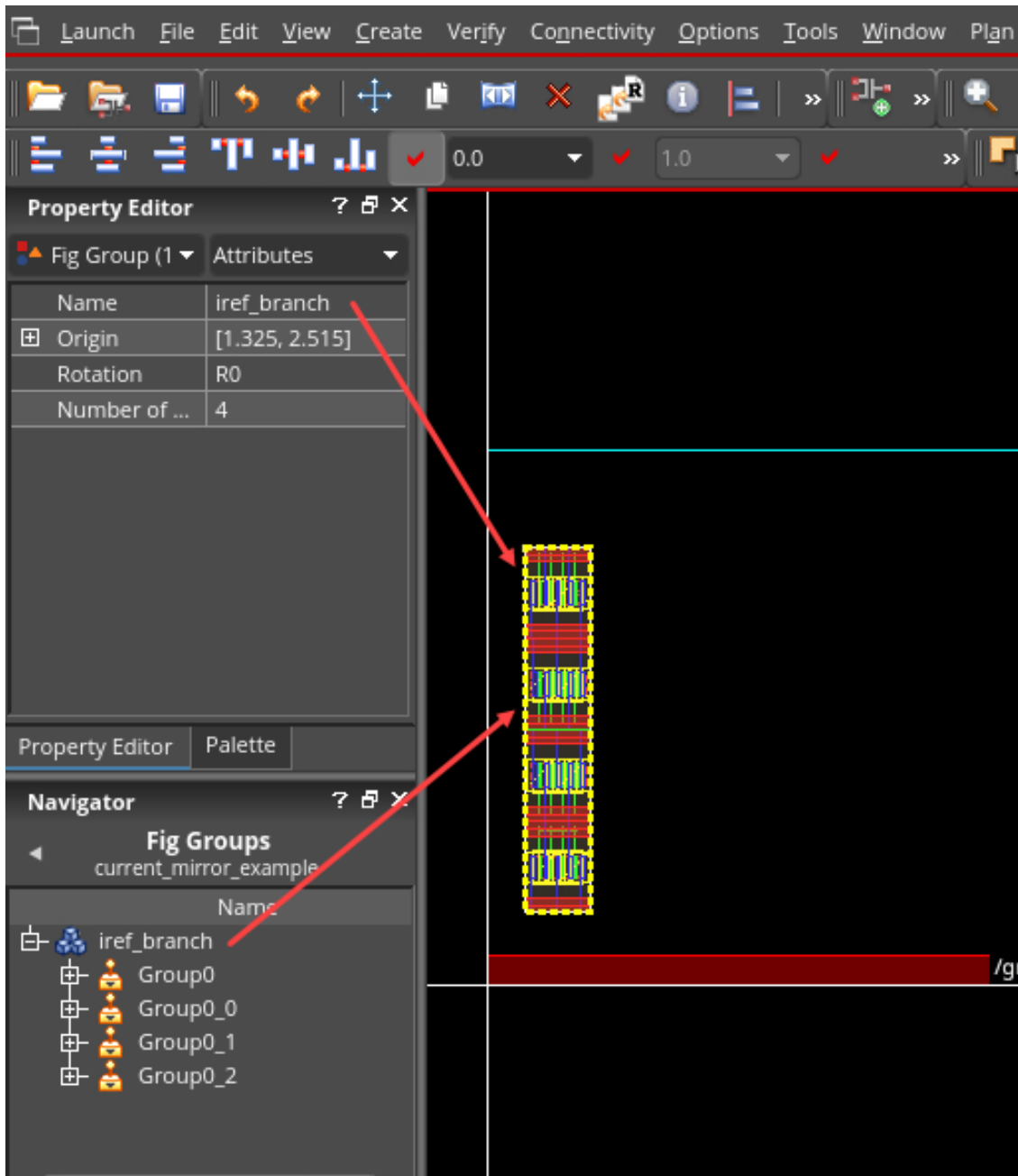
- c. In the Navigator assistant, select **Fig Groups** in the *GROUPS* category to view the groups.

The new group “Group1” is displayed in the Navigator assistant.

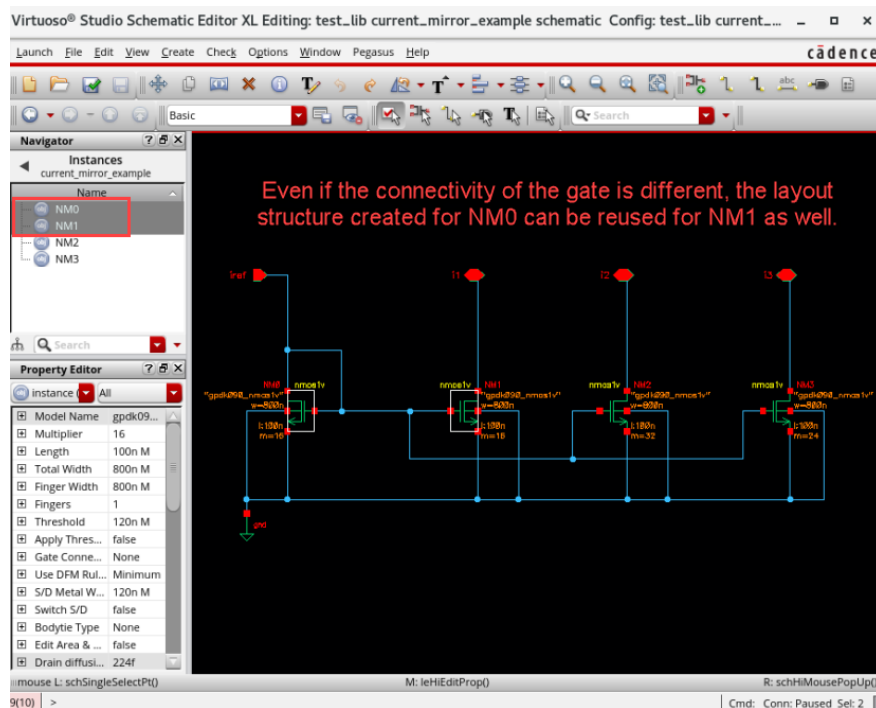


20. Rename the group *Group1* as *iref_branch*.

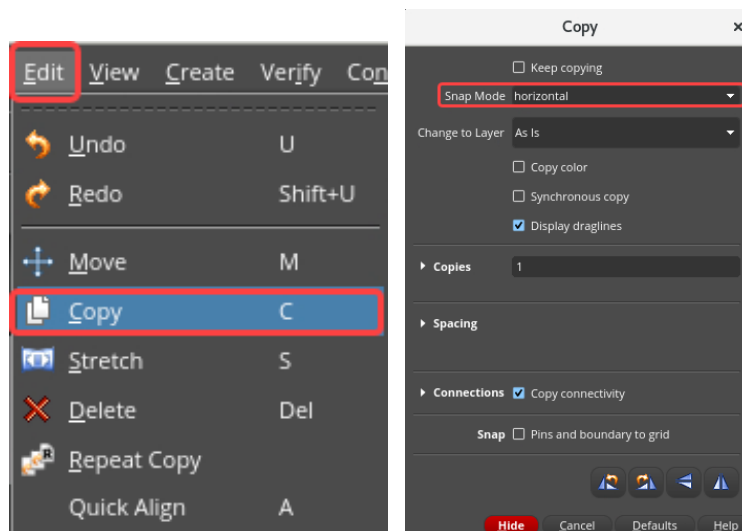
Try to always use meaningful names for your groups. You can easily rename them using the **Property Editor** assistant.



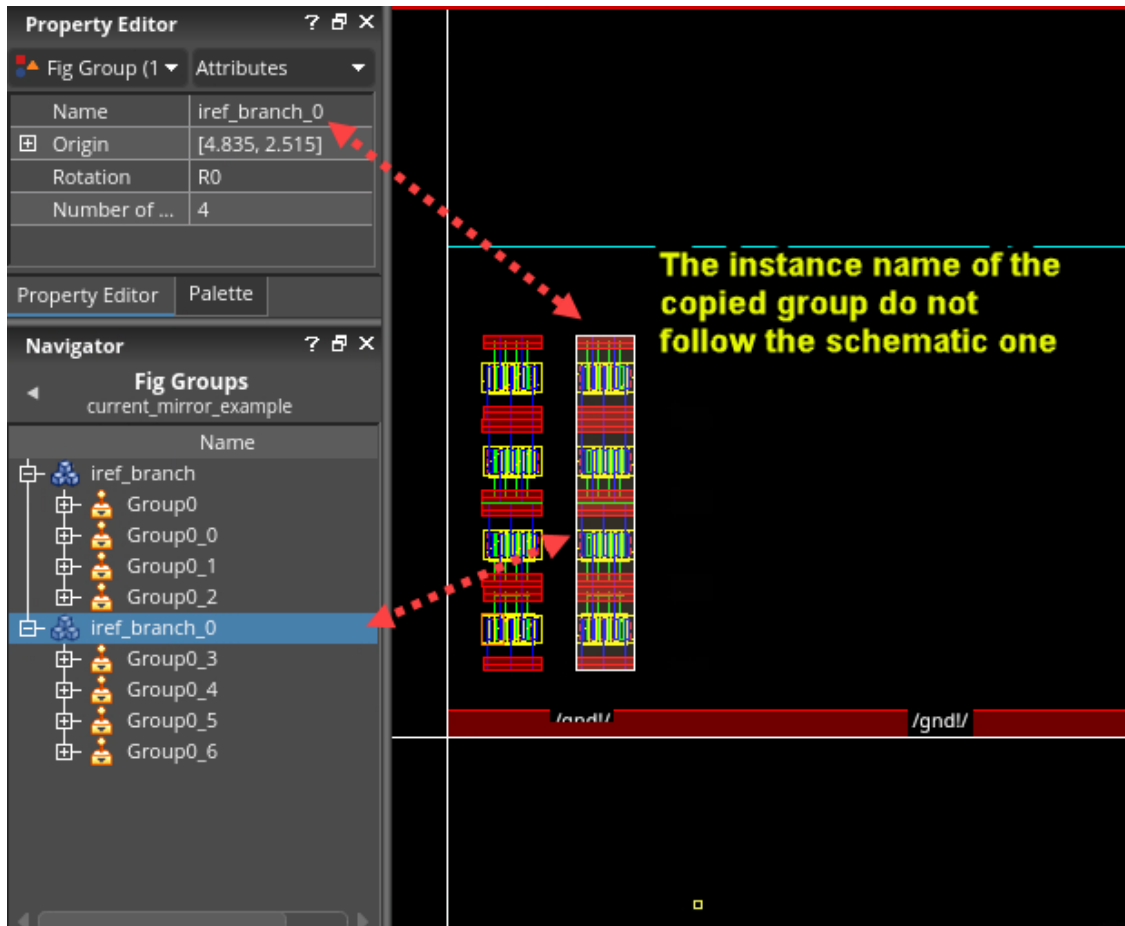
You need to create the second transistor, the one used for the *i1_branch*. In this case, you cannot clone it, because the connectivity is not identical. The gate connection is different.



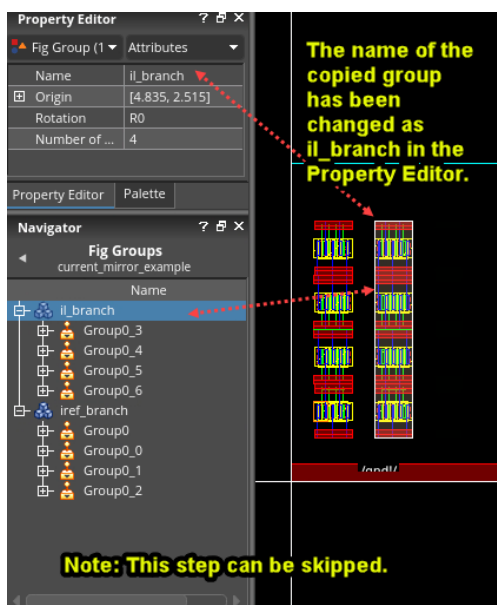
21. In the layout canvas, proceed with a plain **Copy** and then you can manually map the instances.
 - a. Select the *iref_branch* group.
 - b. Choose the **Edit – Copy** menu command or click the **Copy** icon in the Edit toolbar or press the bindkey **C** to invoke the Copy form.
 - c. Select the **Snap Mode** as horizontal, as shown here, and make a copy in the layout window.



The instance names of the copied group do not follow the schematic ones. In this case, it is named as *iref_branch_0*.



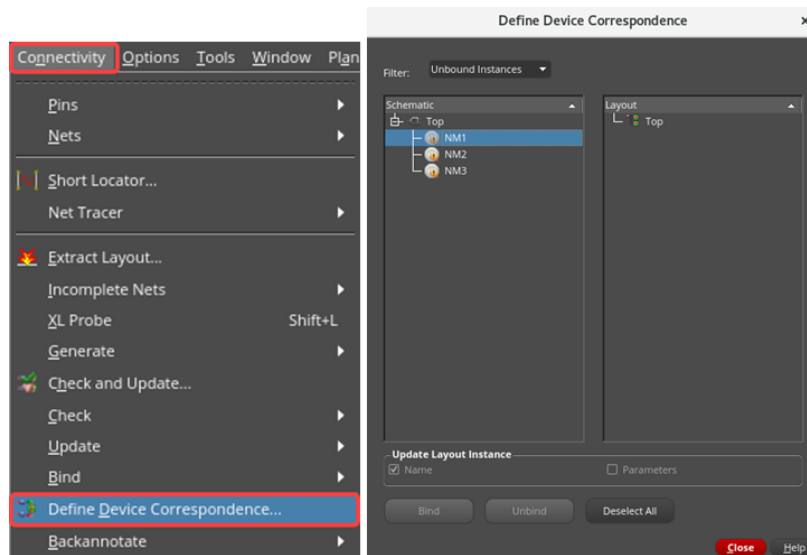
- d. Change the name of the copied group as *il_branch* using the Property Editor assistant.



The resulting copy is not “understood” by Layout-EXL. Notice the instance names in the schematic and the layout windows.

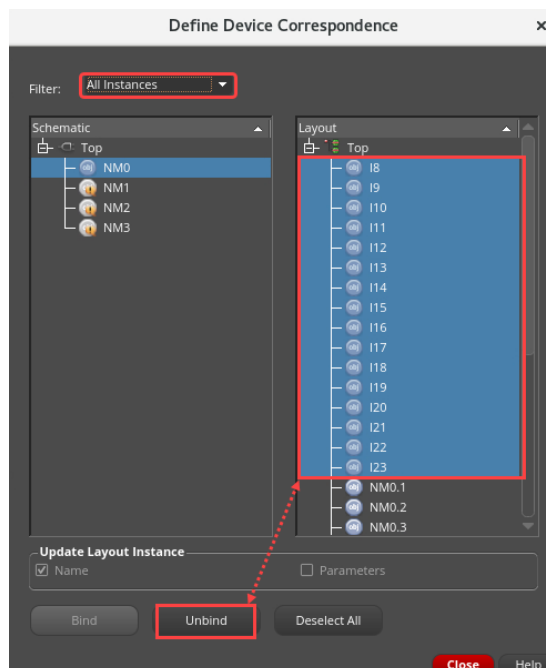
22. You will manually map the instances now.

- a. In the layout canvas, choose the **Connectivity – Define Device Correspondence** menu command, or click the **Define Device Correspondence (DDC)** icon in the Layout EXL toolbar to open the Define Device Correspondence form.

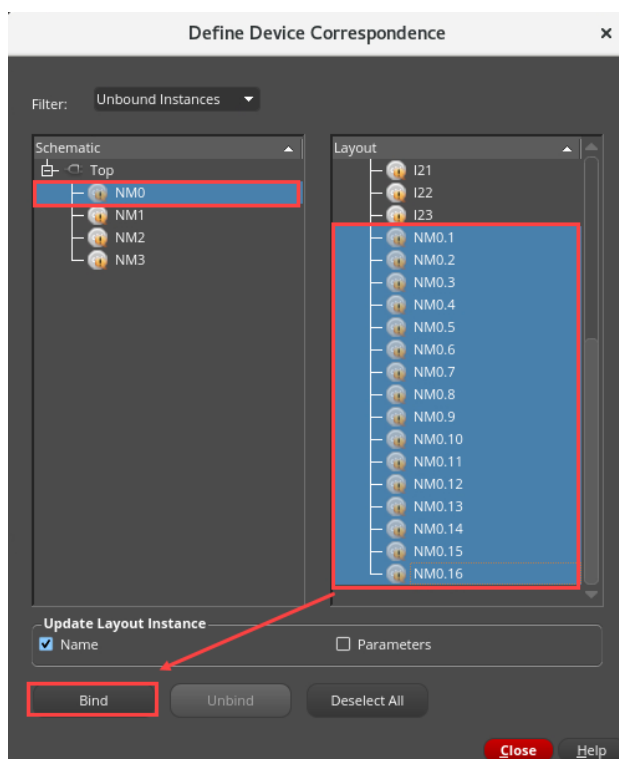


Note: If you don't find any unbind instances in the layout side, the following step will help you to bind and unbind the instances.

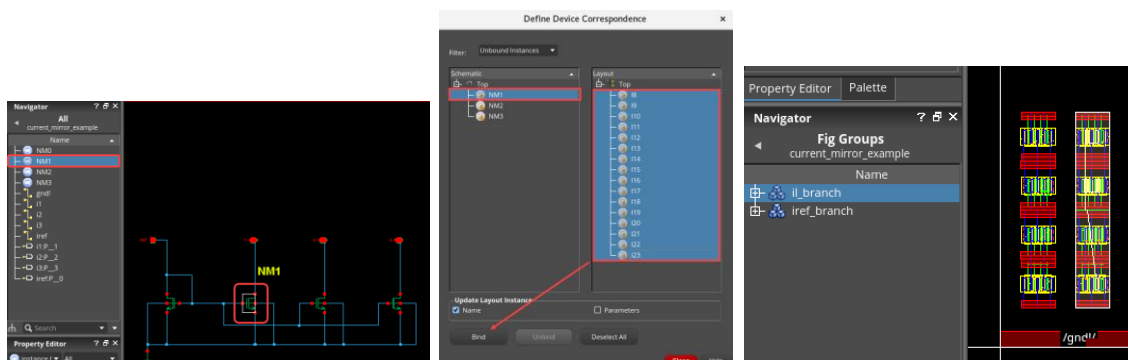
- Go to **All instances**, and **unbind** the **I8 to I23** instances from **NM0**



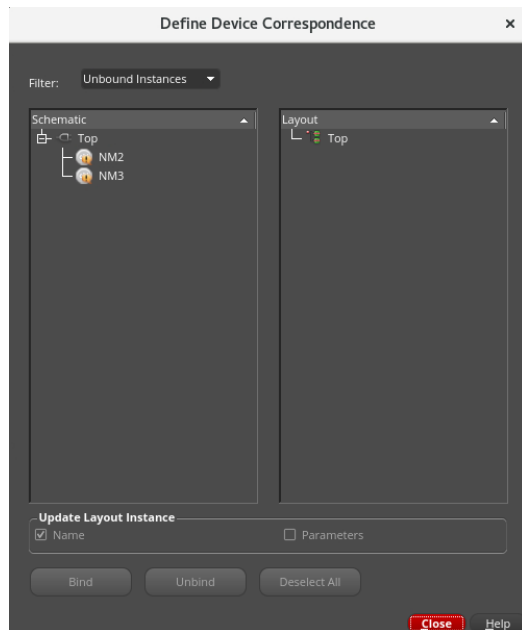
- Select **NM0** in schematic side, choose **NM0.1 to NM0.16** instances and **Bind**.



- Now, select **NM1** on the schematic side (this is the transistor you are currently building) and all of the “unknown” instances on the layout side.
- Then click the **Bind** button.



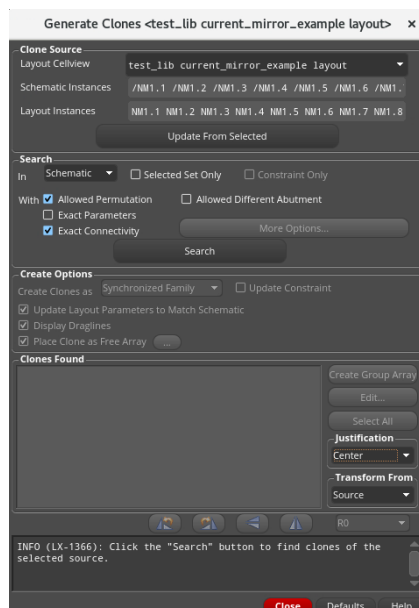
- d. In the DDC form, you can check that the mapping is done between **NM1** in the schematic window and the newly copied instances in the layout window, as shown here.



- e. Close the DDC form now.

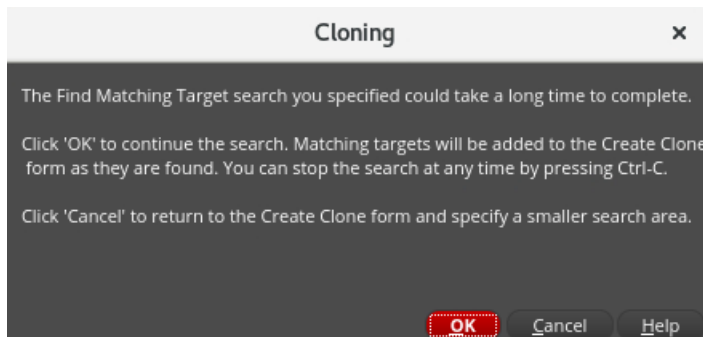
23. You can now clone the just created structure. The next transistor **NM2** is simply “two-times” **NM1**.

- a. In the layout window, select the **NM1** structure (the group called *i1_branch*) and choose the **Connectivity – Generate – Clones** menu command, or click the **Generate Clones** icon in the Layout EXL toolbar to invoke the Generate Clones form.

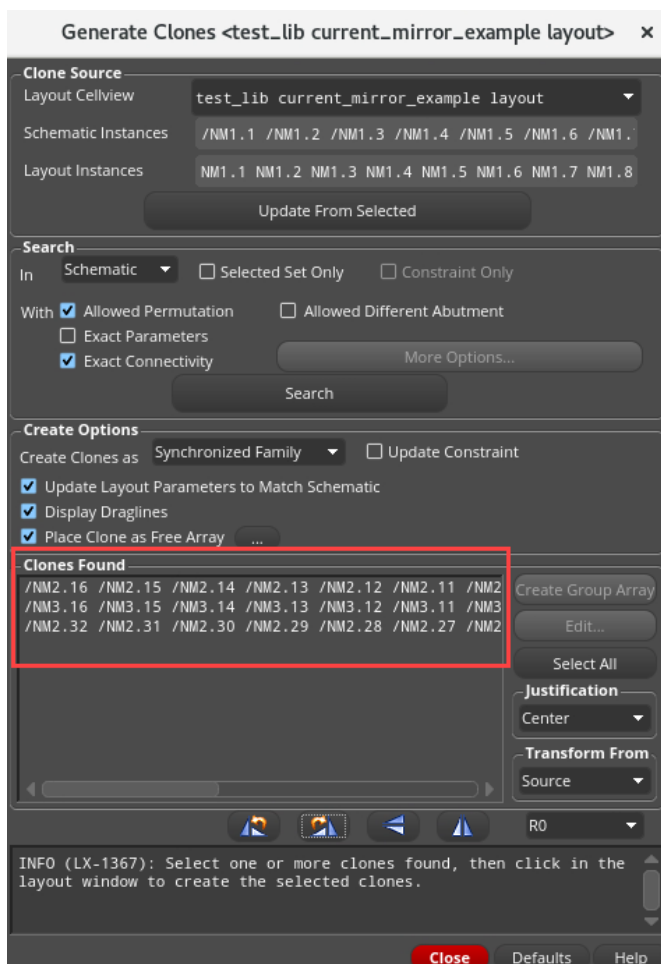


24. Leave the default settings and click the **Search** button.

- a. If a message form is displayed, just confirm by clicking **OK** to start the search.



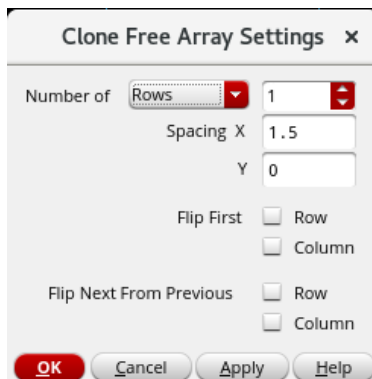
Three groups of transistors are found in the *Clones Found* pane, as shown here.



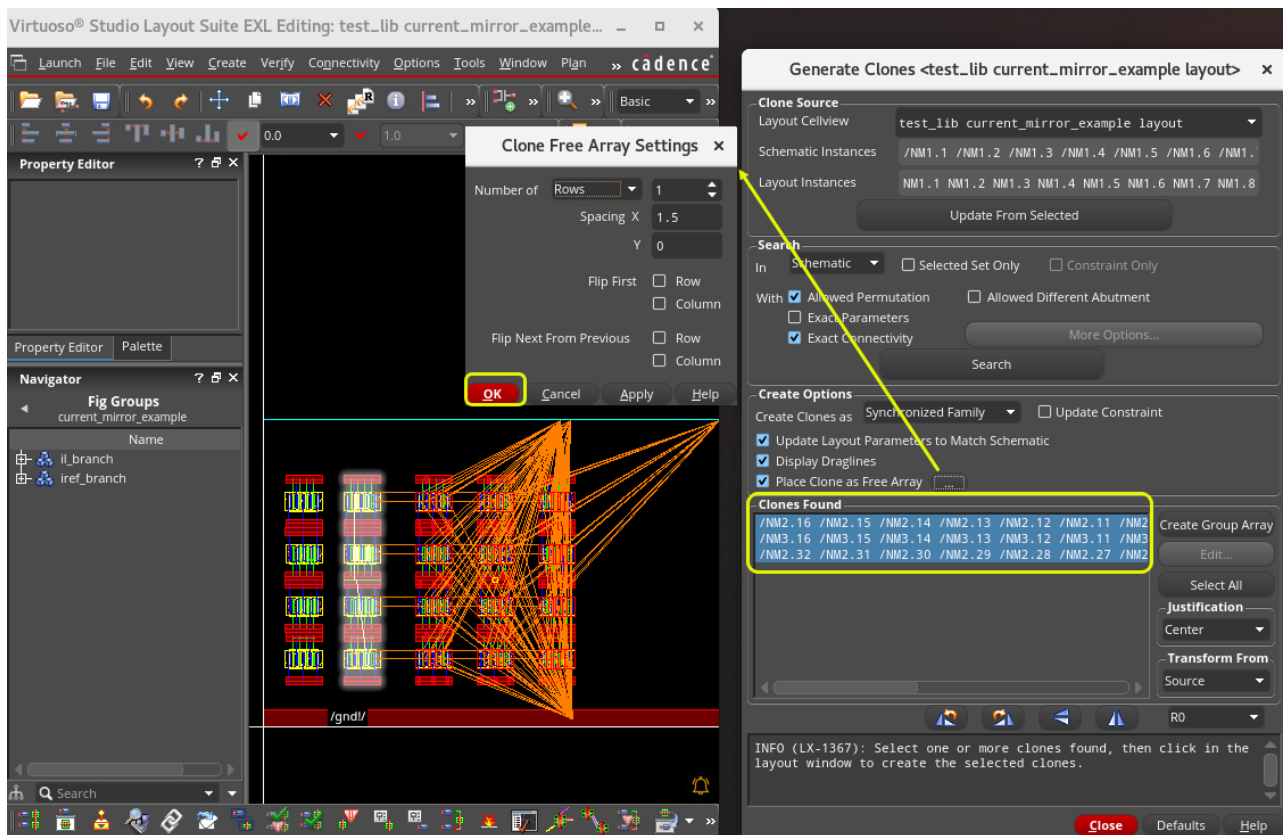
- The first 16 instances of **NM2**
- The second 16 instances of **NM2** (this is a 32x transistor)
- 16 of the 24 instances which compose **NM3** (this is a 24x transistor)

25. Set the following options in the Generate Clones form:

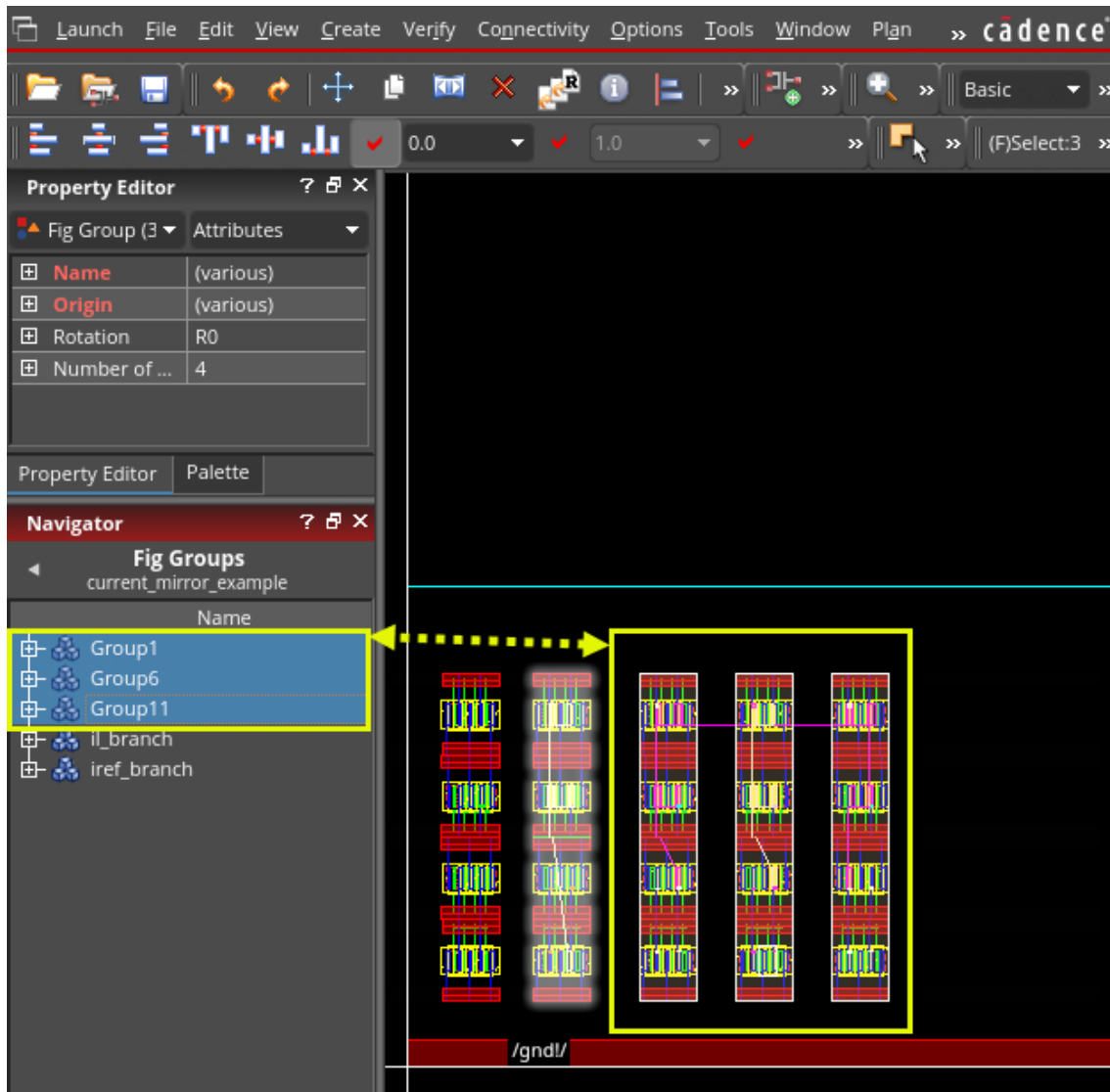
- Select all the cells listed using **Shift** and-or **Ctrl** together with the left-mouse-button.
- Make sure the **Place Clone as Array** field is enabled.
- Click the **Array Settings**  button to open the *Clone Array Settings* form.
- Set the *Clone Array Settings* form, and click **OK** in the form.



26. Select **Grouped Objects** in the *Create Clones as* group box in the *Create Options* section to create clones as Grouped Objects. (You don't need synchronous clones.)



27. Place the three groups of transistors one close to the other and click **Close** to close the Generate Clones form and remove the highlight from the reference group.

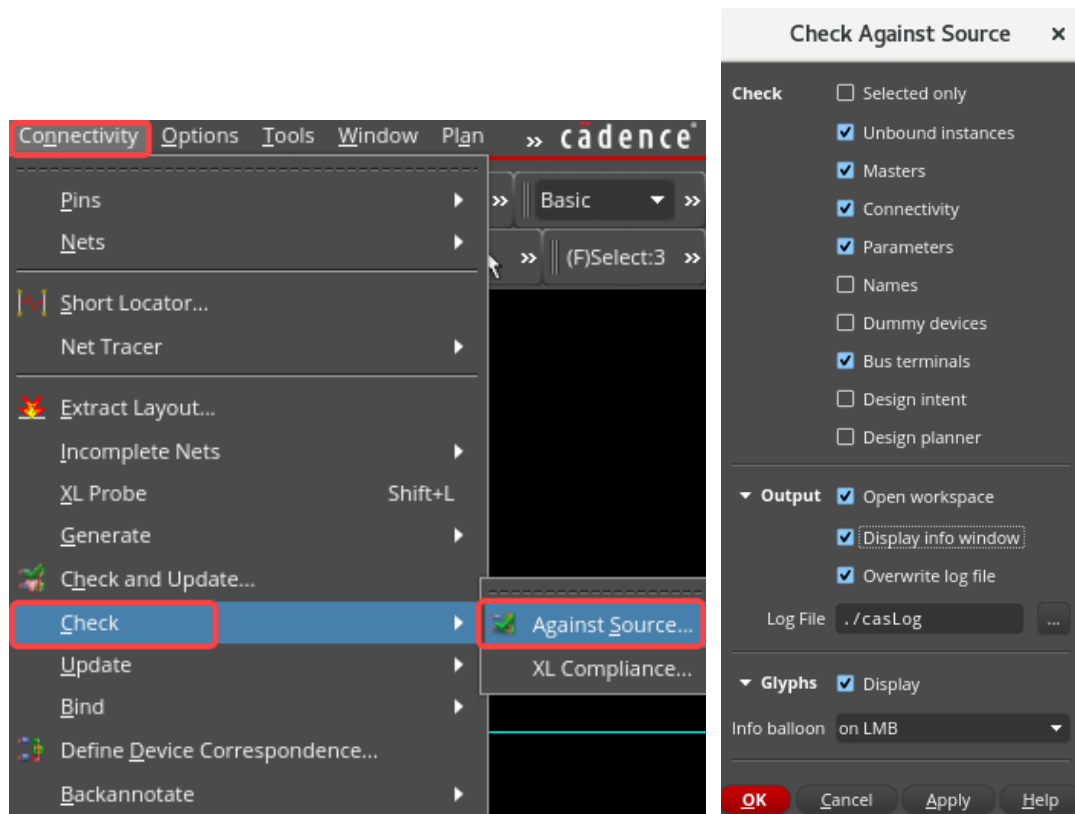


28. Now you will check the layout cellview against the schematic source cellview.

You will use the *Check Against Source* command to show any differences between the schematic and the layout.

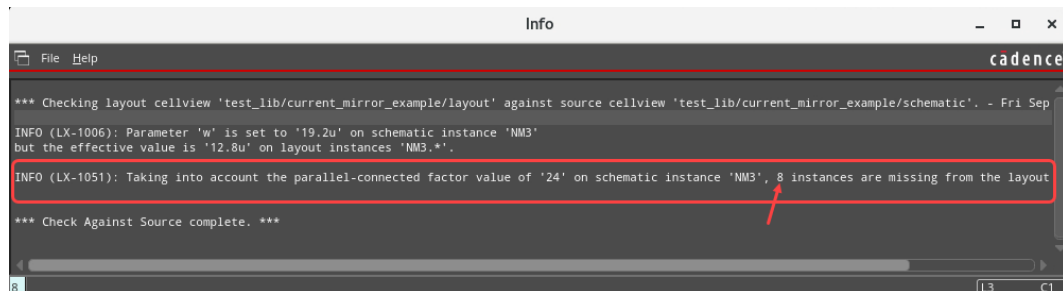
- In the layout canvas, choose the **Connectivity – Check – Against Source** menu command, or click the **Check Against Source (CAS)** icon in the Layout EXL toolbar to open the Check Against Source form.

- b. Keep the options, as shown here.



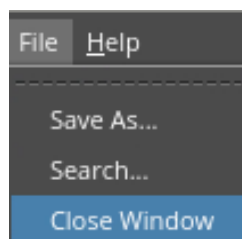
- c. Click **OK** in the CAS form.

The *Info* window opens with information about the differences between the layout and the new schematic. You can check the *casLog* file as well from the current working directory.



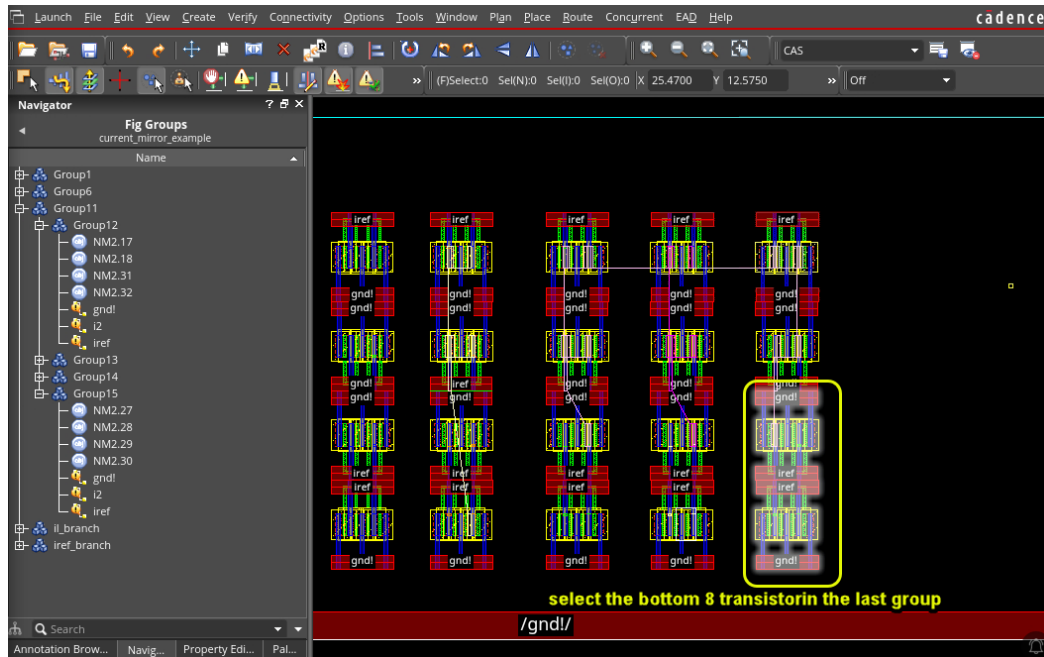
You can observe that **8** instances are missing from the layout.

- d. Choose **File – Close Window** to close the *Info* window.

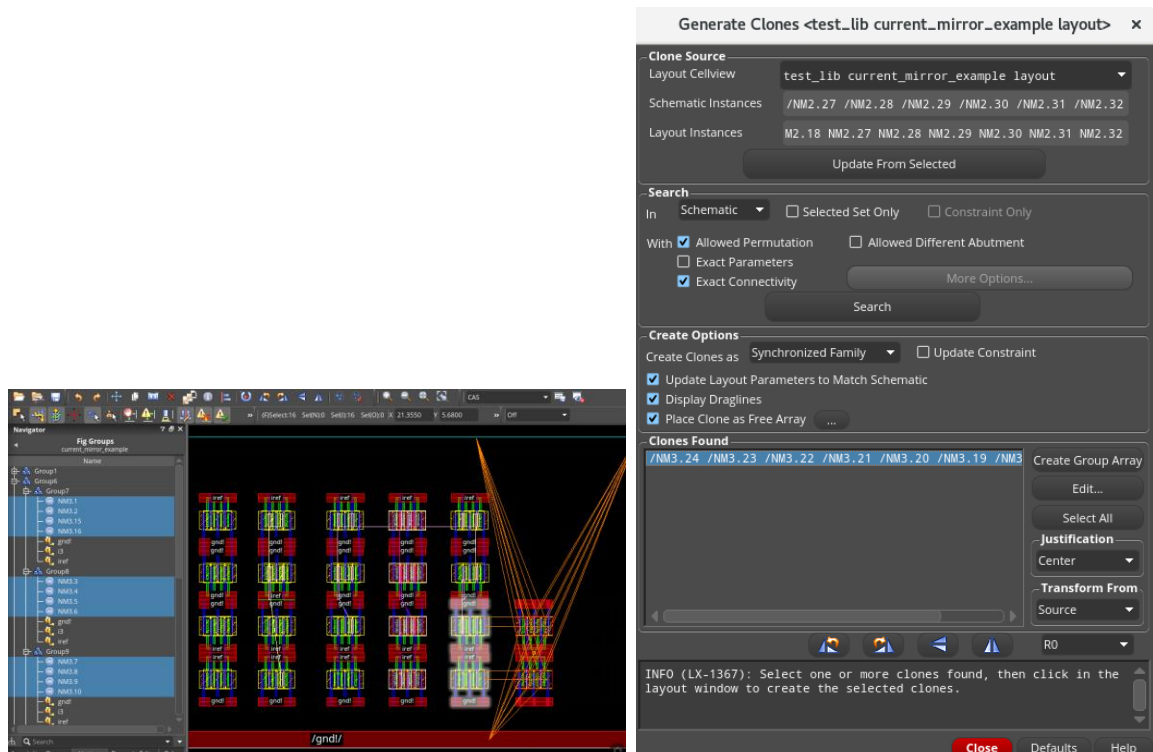


29. Because you are still missing **8 transistors**, follow these steps in the layout window.

- Click the **Transparent Group** toolbar icon.
- Select the **bottom 8 transistors** in the last group, as shown here.

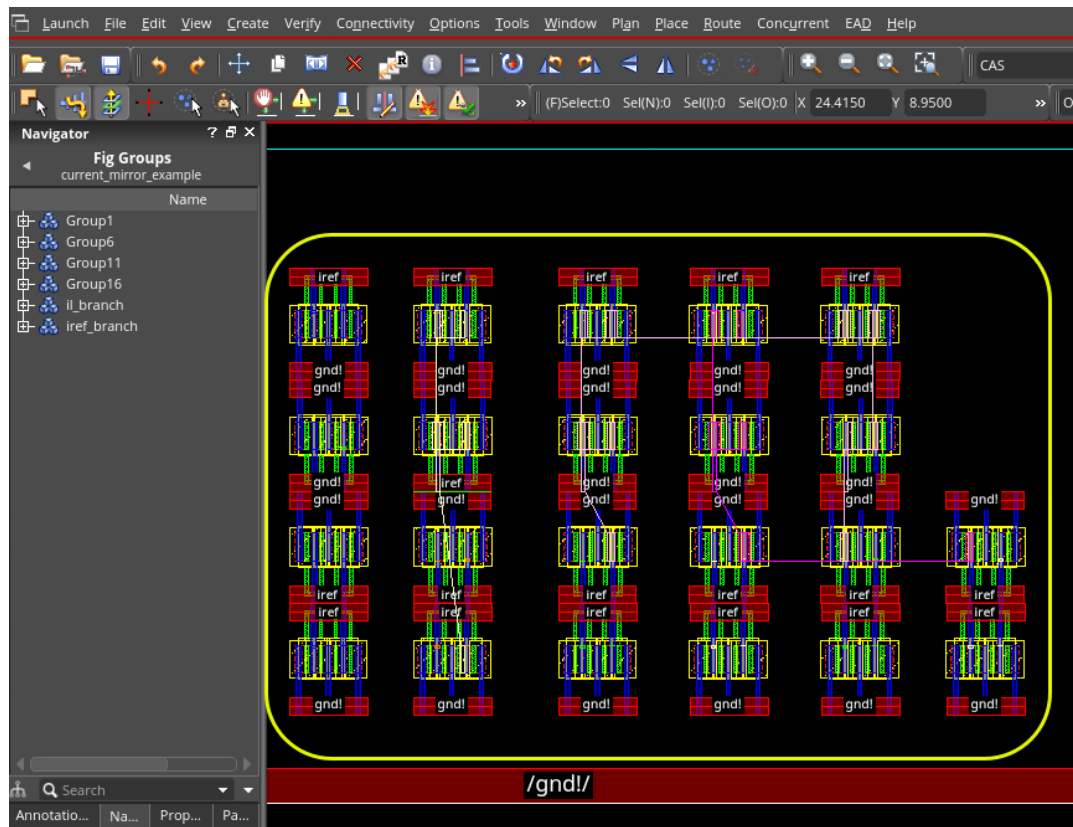


- Clone them by choosing the **Connectivity – Generate – Clones** or by clicking the **Generate Clones** icon. Click the **Search** button and place the matches found.

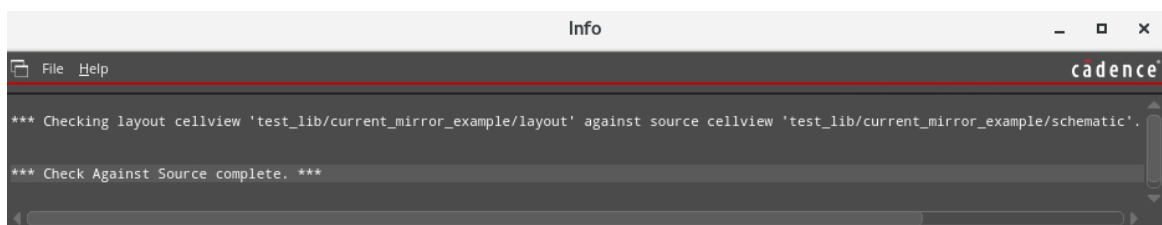


- a. Click **Close** to close the Generate Clones form and remove the highlight from the reference group.

Now you have the complete layout for your current mirror, as shown here.



30. Run **Check Against Source (CAS)** again and observe the *Info* window report, as shown here.



There are no info/warning messages reported now.

31. Close the schematic and layout views. Do **NOT** save.
32. Close the Virtuoso session.
33. In the shell, enter the following command:

```
cd ~
```

This command places you in your home directory.

Summary

In this lab, you

- ◆ Cloned a structure with “exact” connectivity
- ◆ Used groups to align regular layout structures
- ◆ Copied and mapped to reuse blocks which do not have a matching connectivity
- ◆ Worked with groups and cloning



Module 3: Updating Connectivity and Nets

Lab 3-1 Implementing an Engineering Change Order (ECO)

Objective: To update the layout to match the schematic.

In this lab, you work on a completed OpAmp design in which a new schematic is introduced to implement changes to the layout. You use the features of Virtuoso® Layout Suite EXL to implement the changes to the layout.

Lab Setup

Follow the steps below to set up the environment and perform the lab:

1. If you continue from the previous lab, move to the sub-heading: **Starting the session** Otherwise, proceed to the next step.
2. Update the following line in the /VLPT4_IC_25_1/.cshrc_VLPT4 file with the appropriate valid local tool paths for Virtuoso.

```
setenv CDSHOME /ccstools/cdsind1/Software/IC251ISR_lnx86
```

3. Save and close the file.
4. Source the updated class setup file.

```
Source .cshrc_VLPT4
```

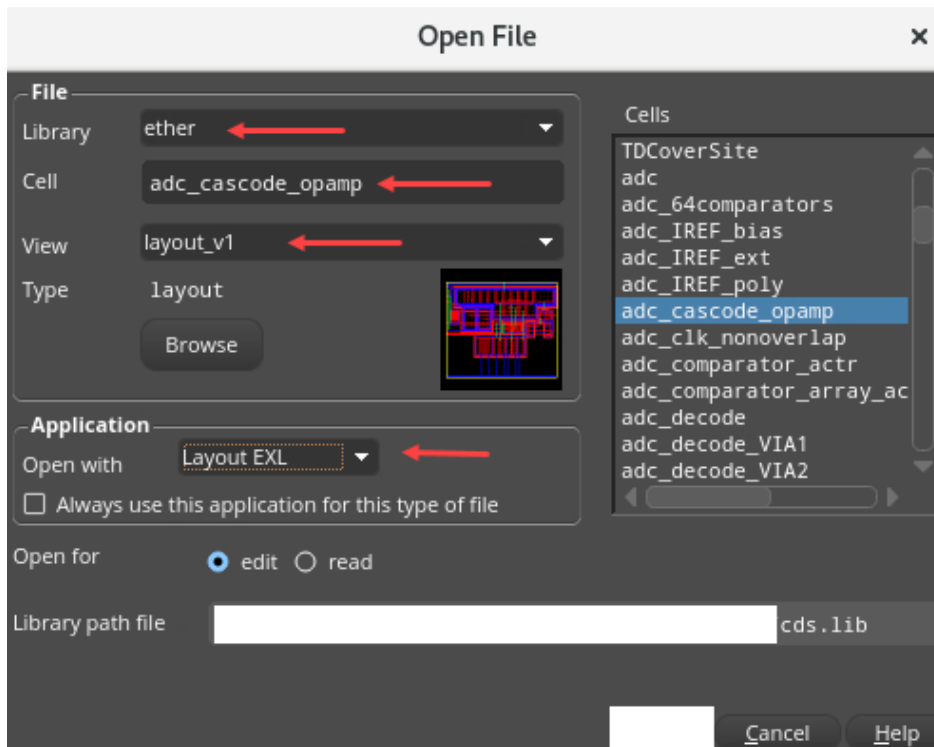
Starting the session

1. In a terminal window, enter the following command from your home (login) directory to get to the lab database directory when you initially logged in:

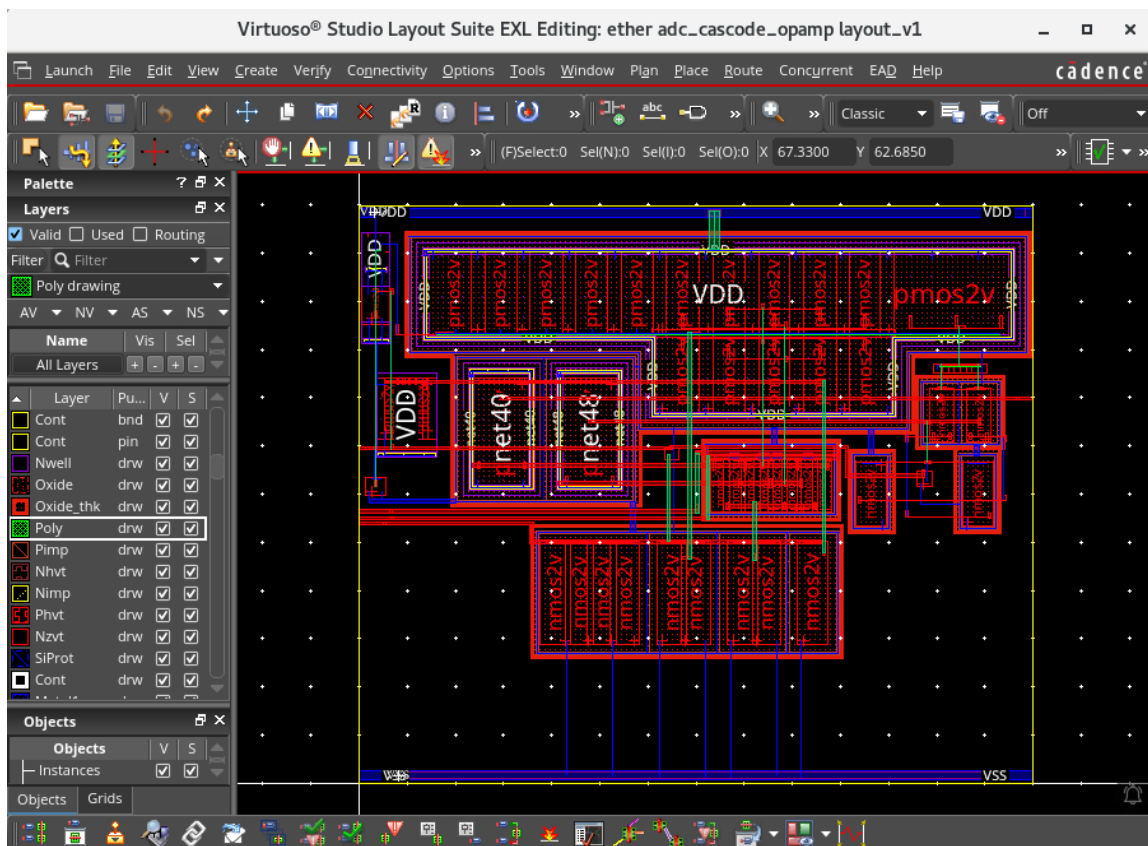
```
cd ./VLPT4_IC_25_1/VLPT4_M03_UpCN
```

2. In the CIW, choose **File – Open** menu command and enter these values in the form.

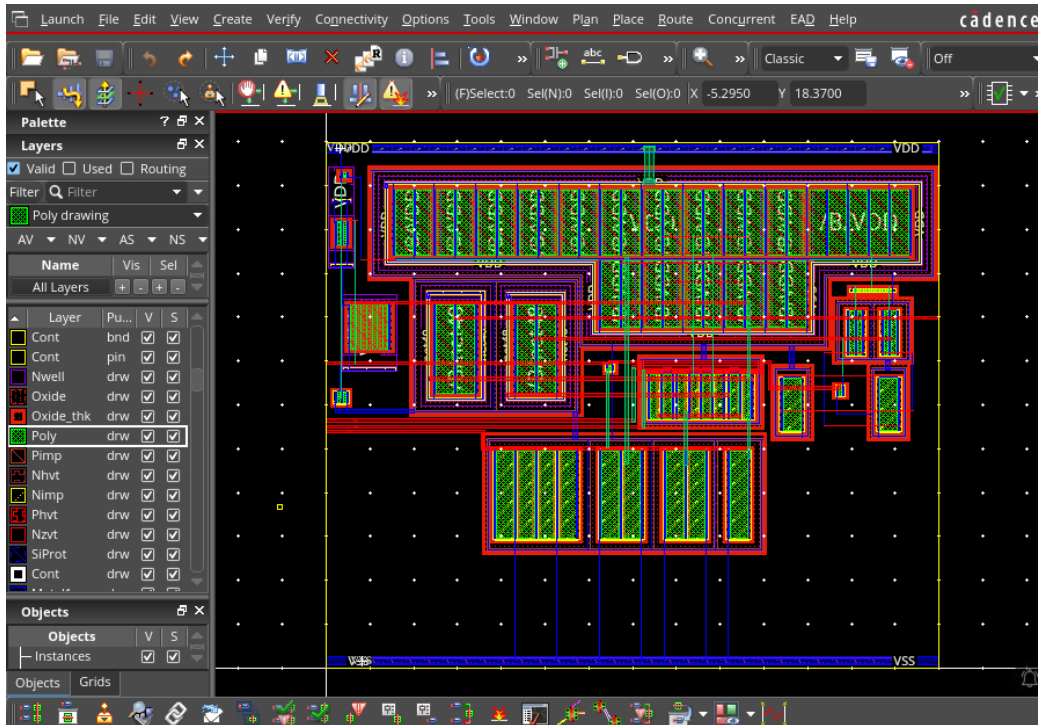
Library	ether
Cell	adc_cascode_opamp
View	layout_v1



The schematic appears in XL mode, and layout views appear in EXL mode. The Virtuoso Layout Suite EXL palette opens with the *ether/adc_cascode_opamp/layout_v1* view.



3. Press **Shift-F** to view all levels of the hierarchy.



Updating the Connectivity Reference

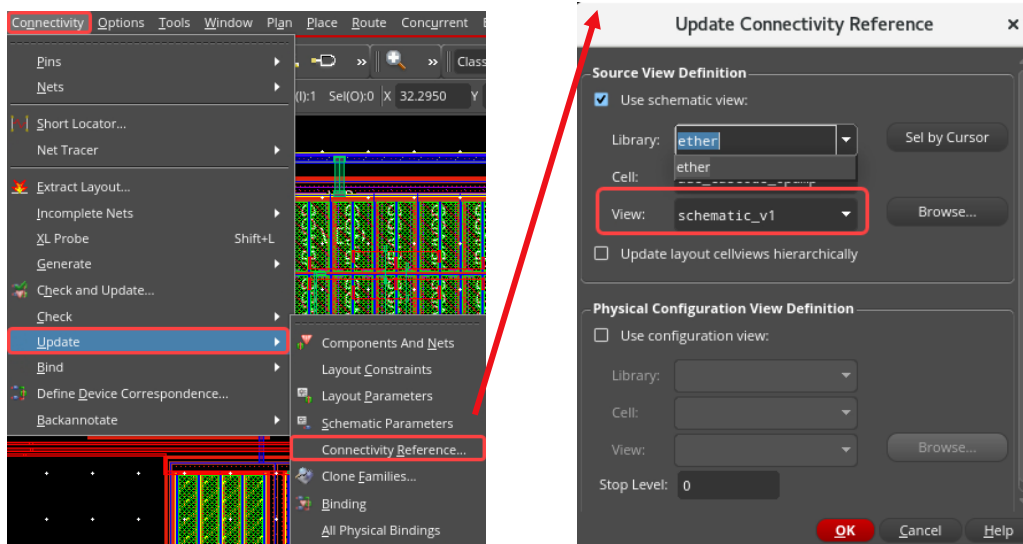
Notice that when you first invoked the layout view, because you are in *Virtuoso EXL*, the associated schematic was invoked as well. Currently, the schematic associated with *layout_v1* is *schematic_v1*.

You are now going to associate a new schematic with *layout_v1* to show the procedures for completing an **ECO**.

The new schematic modifies the shutdown transistors that connect to the pin **SIDDQ** and changes the width of device **NM7**.

1. In the layout view, choose the **Connectivity – Update – Connectivity Reference** menu command.

The Update Connectivity Reference form appears.

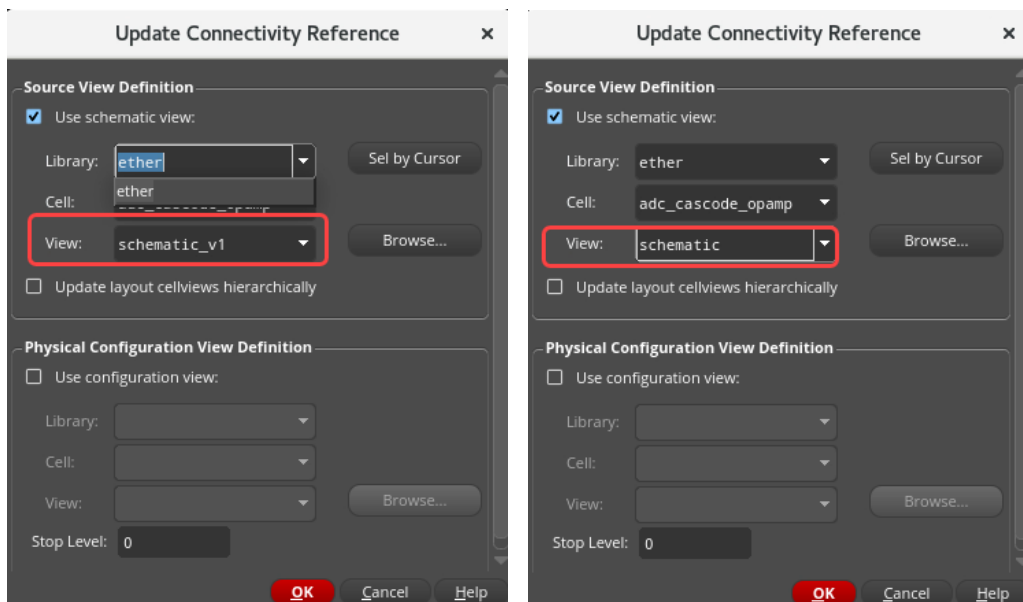


Currently, the schematic view associated with *layout_v1* is *schematic_v1*.

2. In the Update Connectivity Reference form, check the **Use schematic view** option and update the **View** value from *schematic_v1* to *schematic*.
3. Click **OK** in the form.

The *schematic* view opens, which is referenced with the *layout_v1* view.

The new *schematic* view is now the connectivity reference for *layout_v1*.



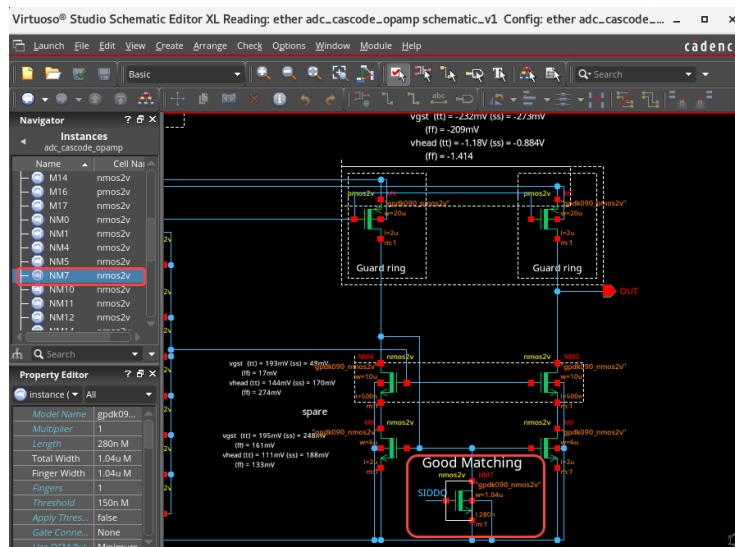
Examining the Differences Between the Original *schematic_v1* and the New *schematic* Reference

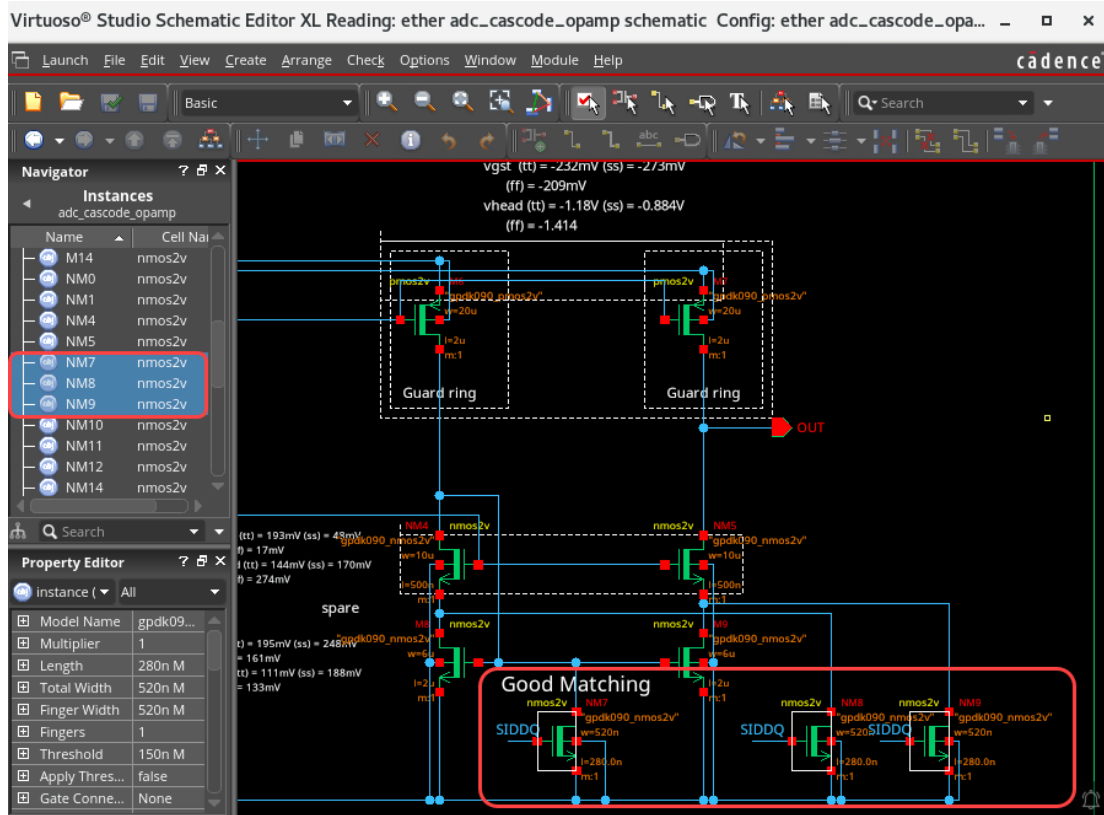
1. Examine the differences between the original *schematic_v1* and the new *schematic* reference.

The changes are made in the lower right of the new connectivity reference, *schematic* view.

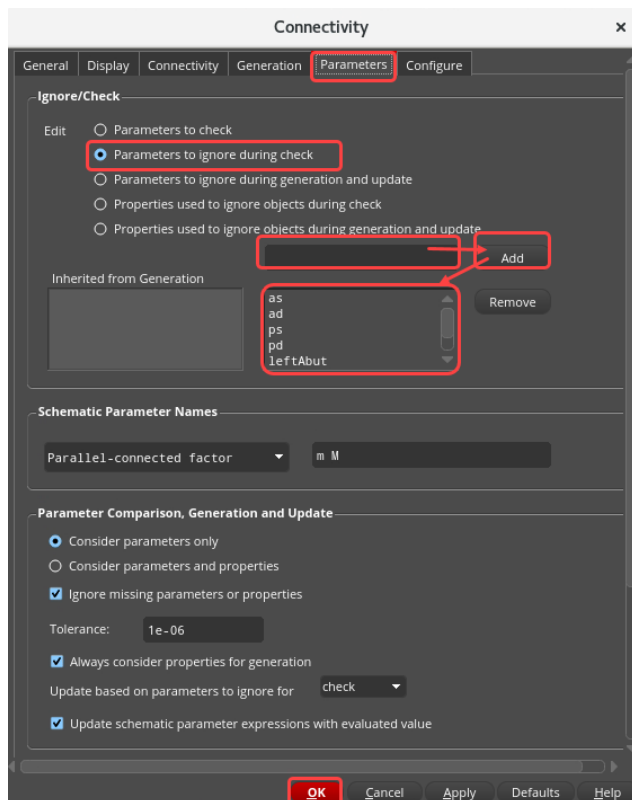
The original *schematic_v1* view remains open.

In the new *schematic* cellview, there are two additional devices, **NM8** and **NM9**, in the lower right of the schematic. The **NM7** device width size has also changed from **1.04μ** to **0.52μ**.





- In the *layout_v1* view, choose the **Options – Connectivity** menu command to open the Layout EXL Options form.



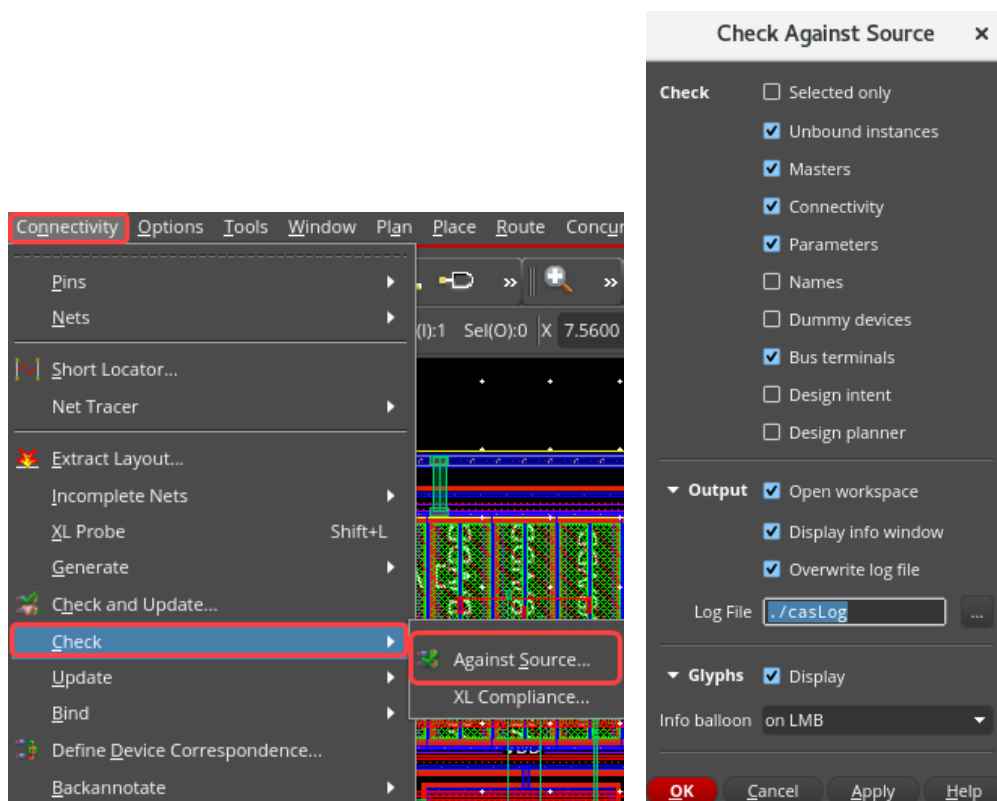
- a. In the **Parameters** tab, select *Parameters to ignore during check* option.
- b. Make sure that the following parameters are set for Layout EXL to ignore during the check between schematic and layout:
 - As ad ps pd leftAbut rightAbut simW

Note: as = area of the source, ad = area of the drain, ps = periphery of the source, pd = periphery of the drain, leftAbut and rightAbut = ignore common contact on abutment, simW = simulation data.
- c. Add these seven parameters separated by a space into the *Ignore* field.
- d. Click the **Add** button.
- e. Click **OK** in the Layout Options EXL form.

Verifying the Design

Now, you will use the *Check Against Source* command to show any differences between the schematic and the layout.

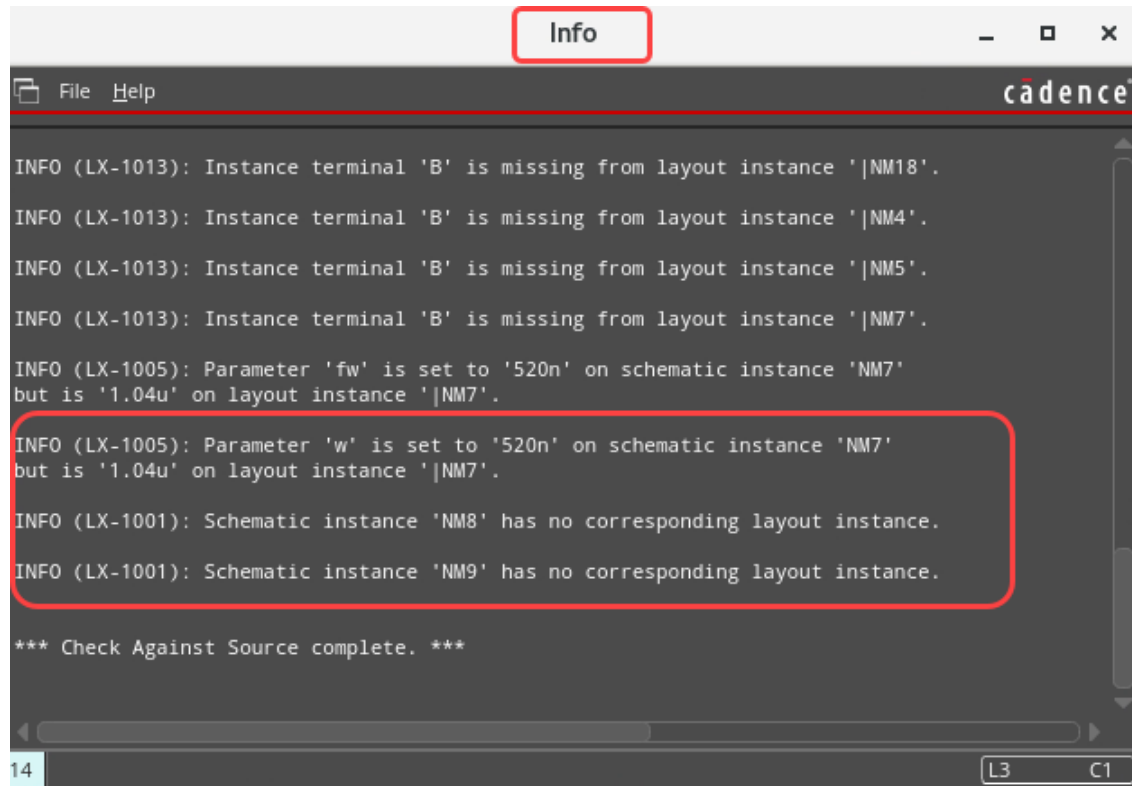
1. In the layout canvas, choose the **Connectivity – Check – Against Source** menu command, or click the **Check Against Source (CAS)** icon in the Layout EXL toolbar to open the Check Against Source form.



2. Match the settings, as shown here, and click **OK**.

The *Info* window opens with information about the differences between the layout and the new schematic. You can check the *casLog* file as well from the current working directory.

3. Scroll to the top and bottom of the report and notice that two instances are missing (**NM8** and **NM9**) in the layout and that the width parameter on the **NM7** device has changed.



4. In the *Info* window, choose **File – Close Window** to close the window.

5. Examine the layout for any flight lines or markers.

Markers in the layout would indicate the location of devices in the layout with no corresponding devices in the schematic.

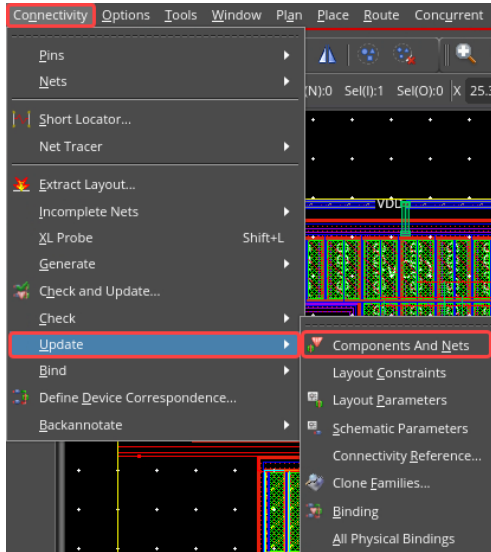
No devices were removed in this layout based on the new schematic, so no markers or flight lines are present in the layout at this time.

Updating the Components and Nets

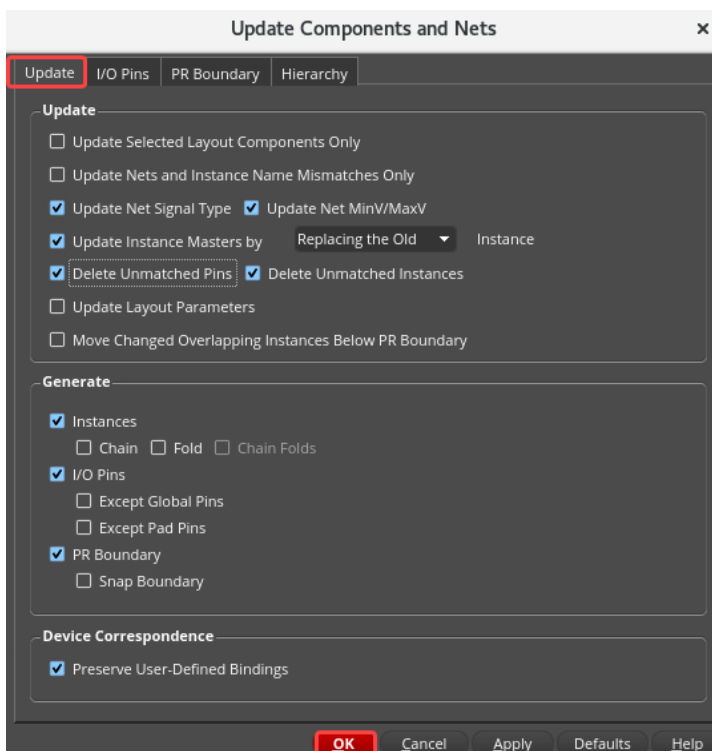
In this part of the lab, you update the layout from the schematic.

1. In the layout canvas, choose the **Connectivity – Update – Components And Nets** menu command, or click the **Update Components and Nets (UCN)** icon in the Layout EXL toolbar.

The Update Components and Nets form appears.

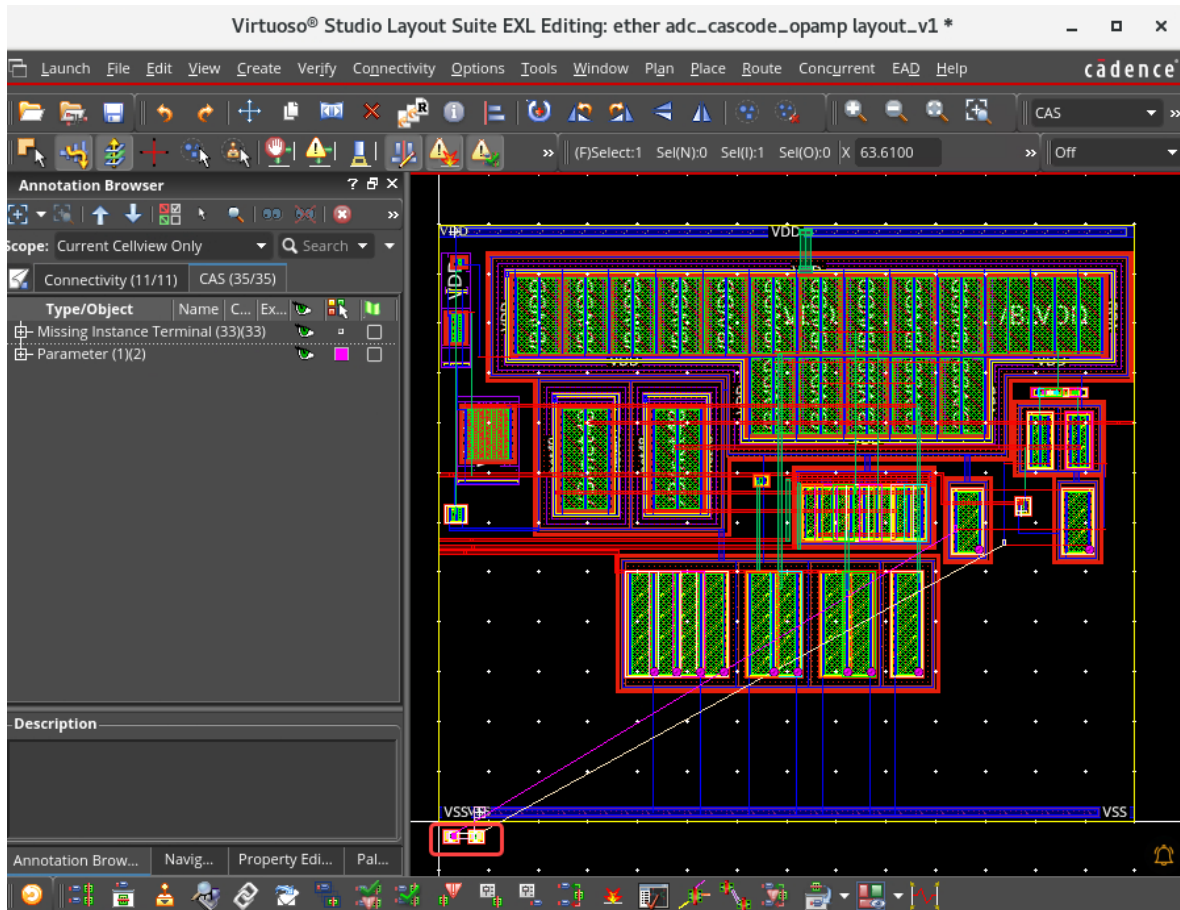


2. Select the options, as shown here.



- Click **OK** in the Update Components and Nets form.

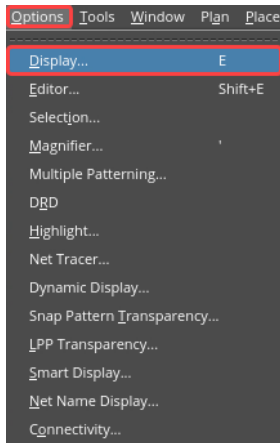
Notice that there are two devices added below the PR Boundary, **NM8** and **NM9**. Also, there are flight lines now that show the connectivity.



Updating the Layout Parameters

In this part of the lab, you update the layout parameters with the new schematic parameters. Because this layout design contains parameterized cells, the devices change in size if the width or length has been changed in the schematic.

1. Choose the **Options – Display** menu command or press the bindkey **E** from the layout window.



The Display Options form appears.

Display Options

Display Controls

- ☒ Open to Stop Level
- ☒ Axes
- ☒ Instance Origins
- ☐ Instance Orientation
- ☒ EIP Surround
- ☒ Pin Names
- ☒ Dot Pins
- ☒ Net Expressions
- ☐ Stretch Handles
- ☐ Via Shapes
- ☐ Transparent Group
- ☐ Group Name Only
- ☐ Traversed instance BBox
- ☐ Traversed Instance Name
- ☐ Transparent Clone
- ☐ Nets
- ☐ Access Edges
- ☐ Instance Pins
- ☒ Instance Pin Names
- ☐ Array Icons
- ☒ Label Origins
- ☒ Use True BBox
- ☐ Cross Cursor
- ☐ Row Name
- ☐ Row Site
- ☒ Dynamic Highlight
- ☒ Dragged Object Ghost
- ☐ Selection Hint
- ☐ Current Level PR Boundary
- ☐ Transparent Modgen

Scroll Percent: 25

Instance Drawing Mode: BBox

Path Display: Borders and Centerlines

Snap Pattern Display: Full

Set LPP Visibility Mode: Do not check validity

Show Name Of: ☐ instance ☒ master ☐ both

Array Display: ☒ Full ☐ Border ☐ Source

Display Levels: Start 0 Stop 31

Grid Controls

Type: ☐ none ☒ dots ☐ lines

☐ Dim Major Dots

Minor Spacing: 1

Major Spacing: 5

X Snap Spacing: 0.005

Y Snap Spacing: 0.005

Resolution

High

☐ Highest Resolution for P&R Objects

Snap Modes

Create: orthogonal

Edit: orthogonal

Display Abstraction Controls

☐ Merge Shapes

Scope: Pcells

☐ Trim Shapes

☐ Slot Shapes

Dimming

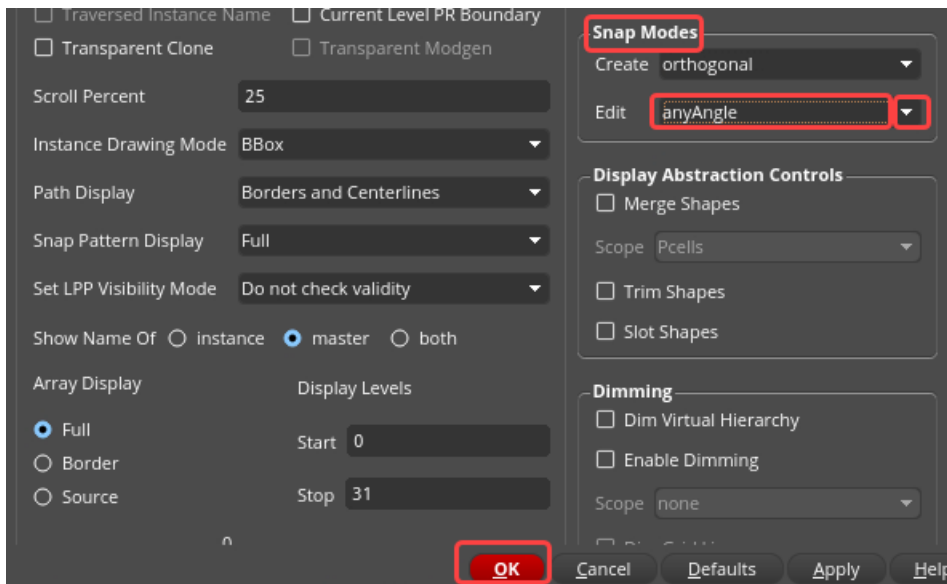
☐ Dim Virtual Hierarchy

☐ Enable Dimming

Scope: none

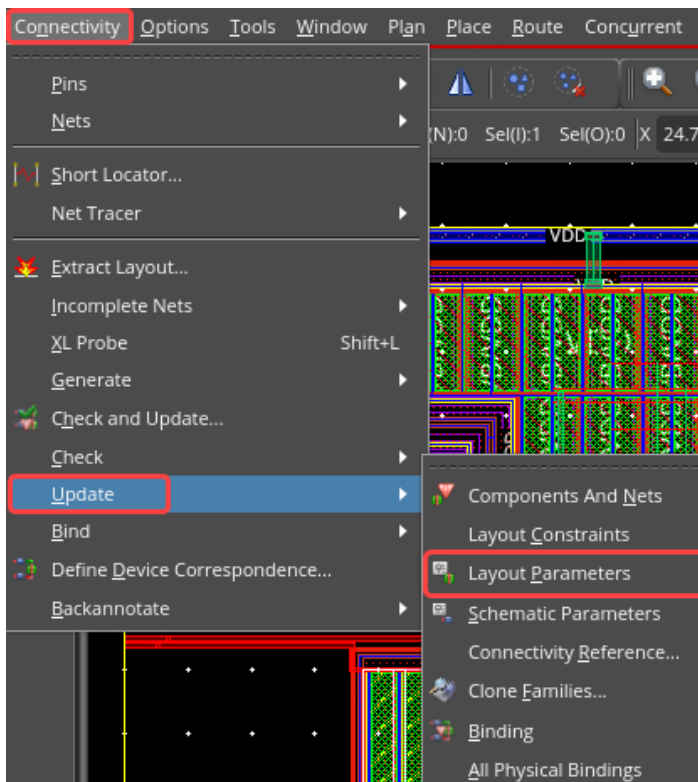
OK Cancel Defaults Apply Help

2. In the **Snap Modes** section, change the *Edit* field from *orthogonal* to **anyAngle**.

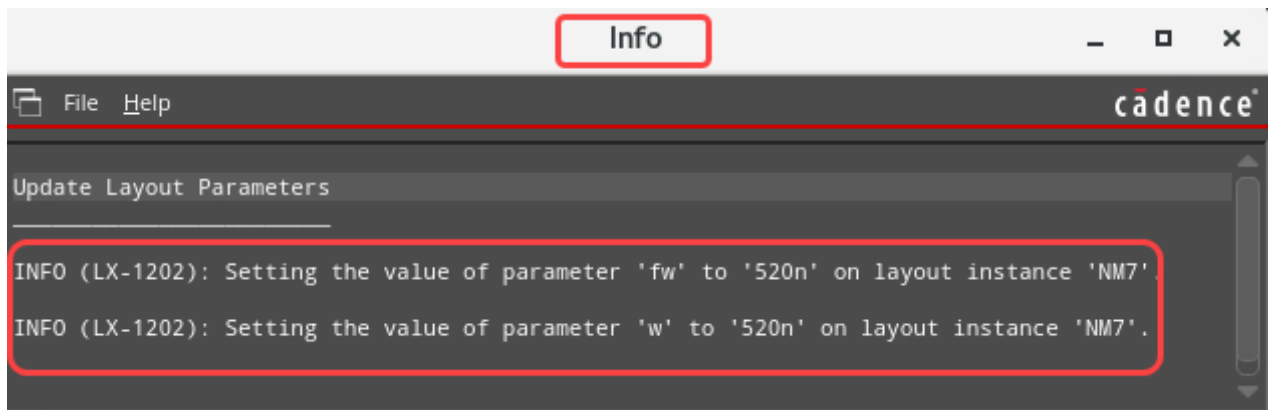


Changing the snap mode makes it easier to move devices. They will move at any angle rather than being locked into an orthogonal direction.

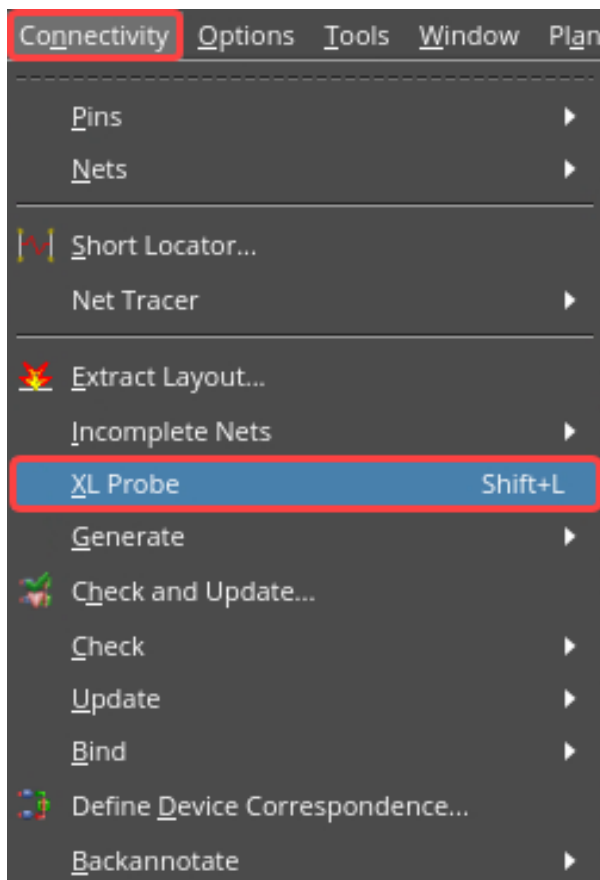
3. Click **OK** in the form.
4. In the layout canvas, choose the **Connectivity – Update – Layout Parameters** menu command or click the **Update Layout Parameters** icon in the Layout EXL toolbar.



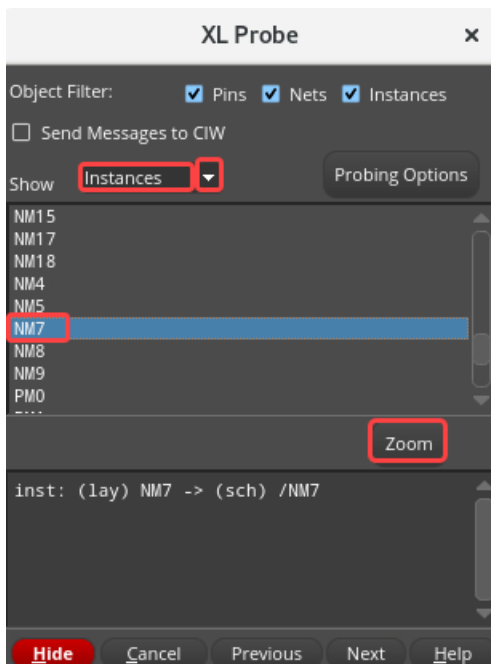
5. An *Info* window appears with information about which devices have been updated.



6. Choose **File – Close Window** in the *Info* window.
7. To probe the devices in the layout, choose the **Connectivity – XL Probe** menu command or press the bindkey **Shift+L** to invoke the XL Probe form.

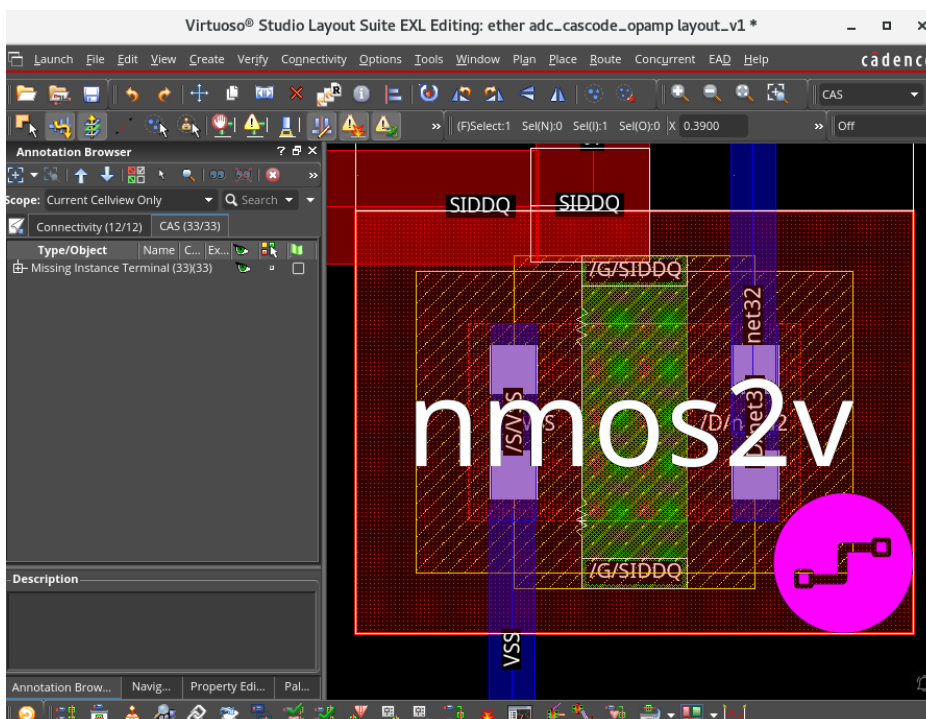


8. Set *Show* criteria to **Instances**, click the device **NM7** and click the **Zoom** button.



This brings you right to the target device.

Notice that the poly connection for the **NM7** device is broken because the device parameters have been changed to a smaller value. (The transistor width was reduced from **1.04μ** to **0.52μ** causing an open gate.)

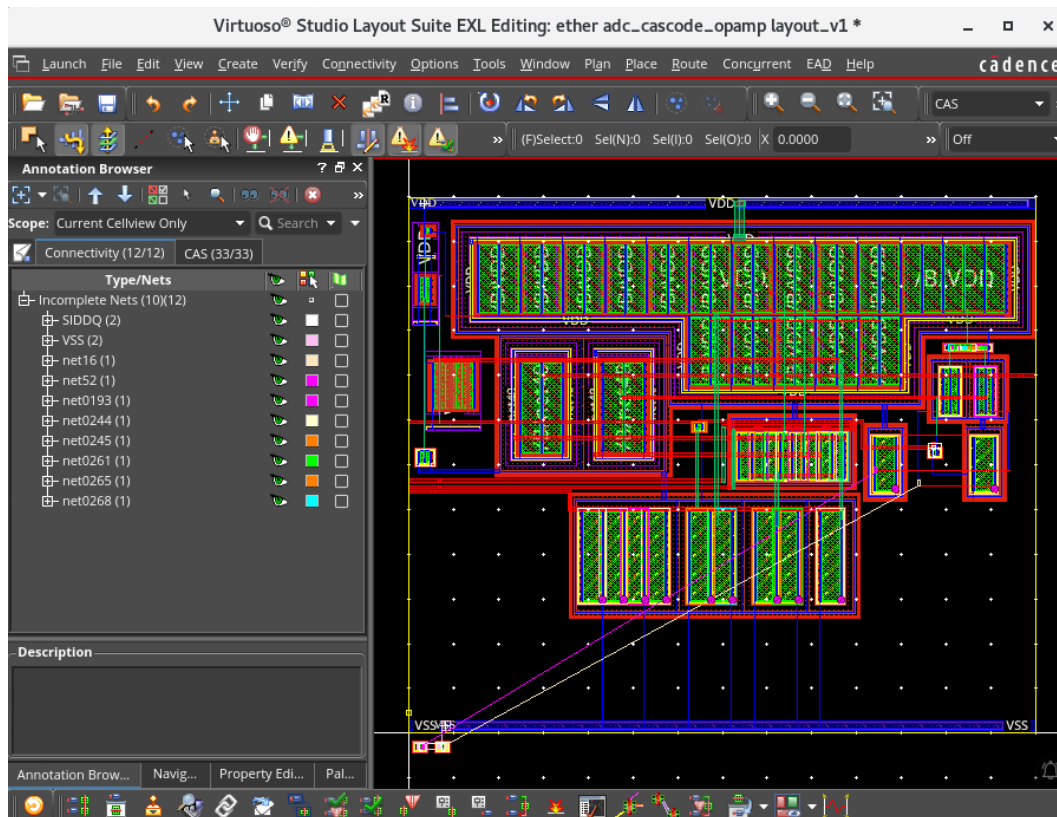


Note: You will update the wiring later in this lab.

- Unhighlight any probed devices by clicking in any blank space in the layout window.
- Click **Cancel** in the *XL Probe* form.

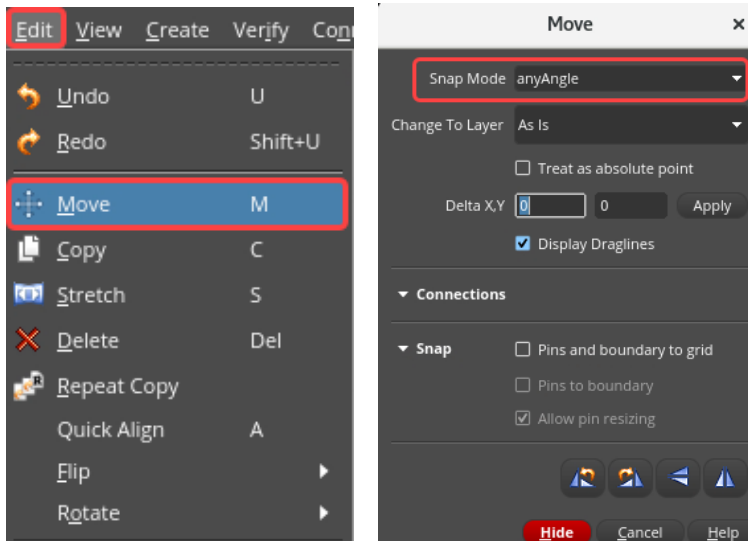
Placing the New Devices

- Activate the **Annotation Browser** to show the unconnects, if not already activated.



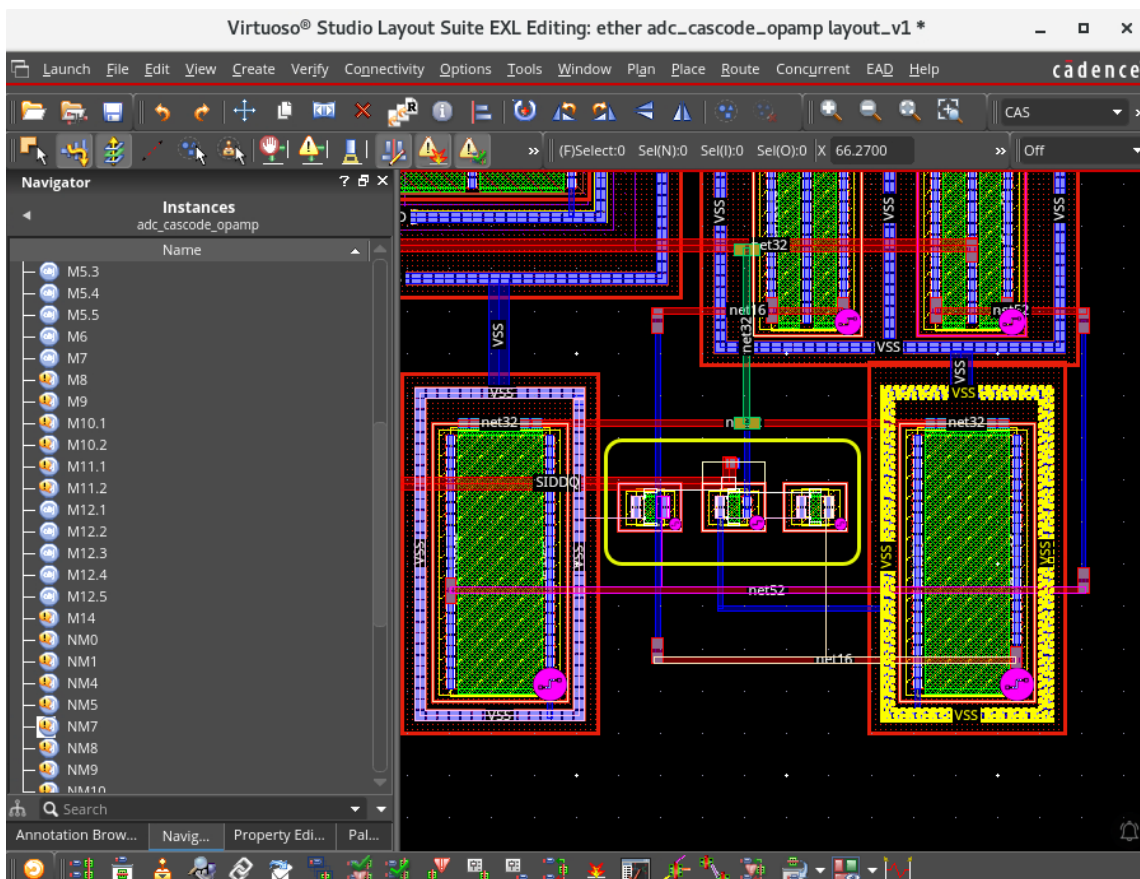
- Identify the area where the two new devices will be placed.
Suggest moving **NM8** to the left and **NM9** to the right of **NM7** for rewiring.

3. In the layout canvas, choose the **Edit – Move** menu command or click the **Move** icon in the Edit toolbar or press the bindkey **M** to invoke the Move form.



4. Use the **Move** command to move each of the two devices from below the *prBoundary* into the lower right design area approximately, as noted here.

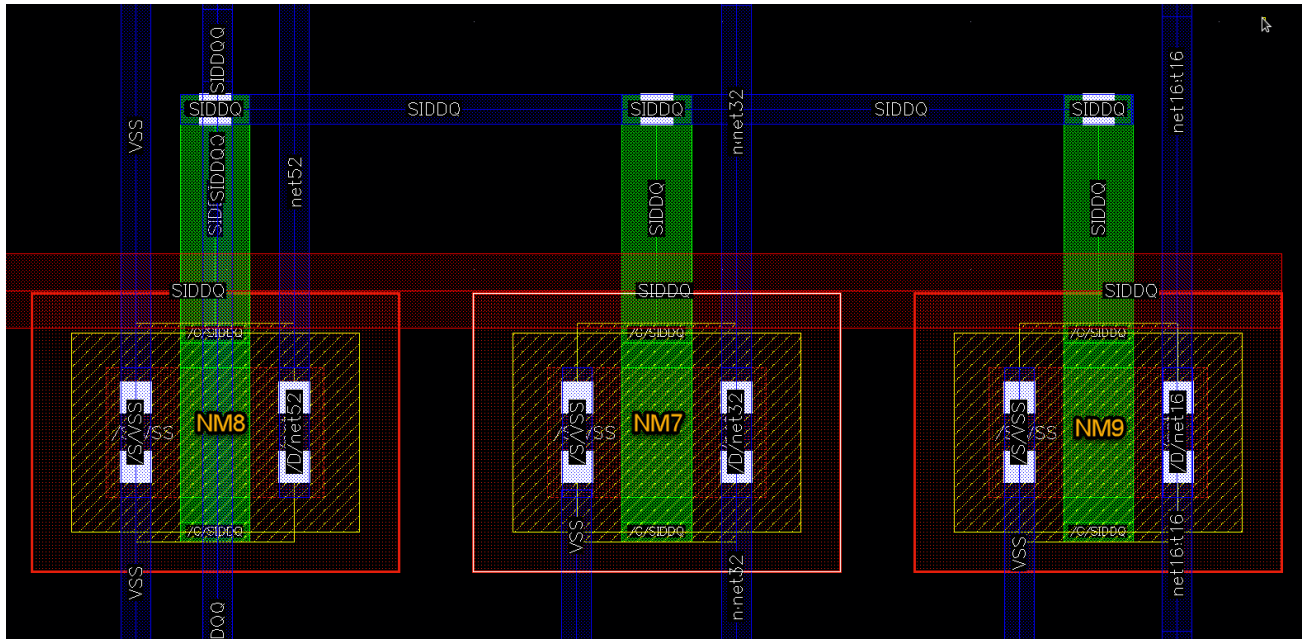
The design looks something like this one.



Wiring the New Components

Before you complete the wiring for the new devices, you need to delete any previous wiring or stray vias that might conflict with the new routing.

At this point, it is up to you to decide how best to clean up and complete the routing at this location. A suggested routing solution is shown here.



1. Close all the windows. Do **NOT** save.

You have successfully updated layout parameters and components.

2. Keep the Virtuoso session running for the next lab.

Summary

In this lab, you

- ◆ Implemented an Engineering Change Order (ECO)
- ◆ Updated the layout to match the schematic



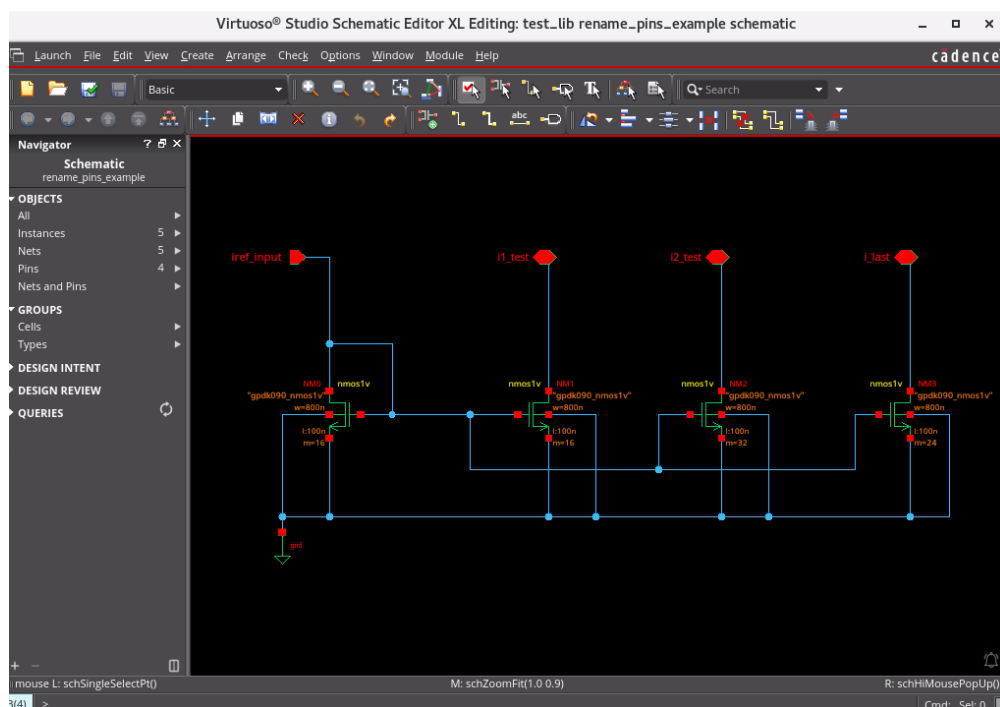
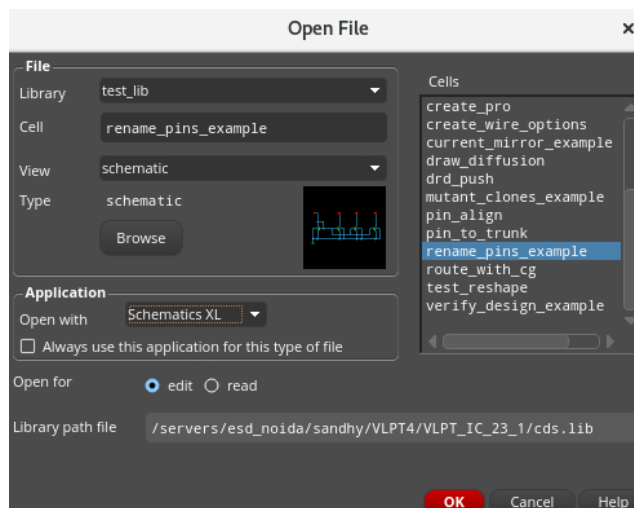
Lab 3-2 Updating the Net and the Pin Names

Objective: To implement the schematic changes in the layout.

In this lab, you identify the differences in the net names between schematic and layout and update the pin names in the layout using the *Update Components and Nets (UCN)* command.

1. In the CIW, choose **File – Open** menu command and open the following schematic view in XL mode.

Library	Cell	View
test_lib	rename_pins_example	schematic

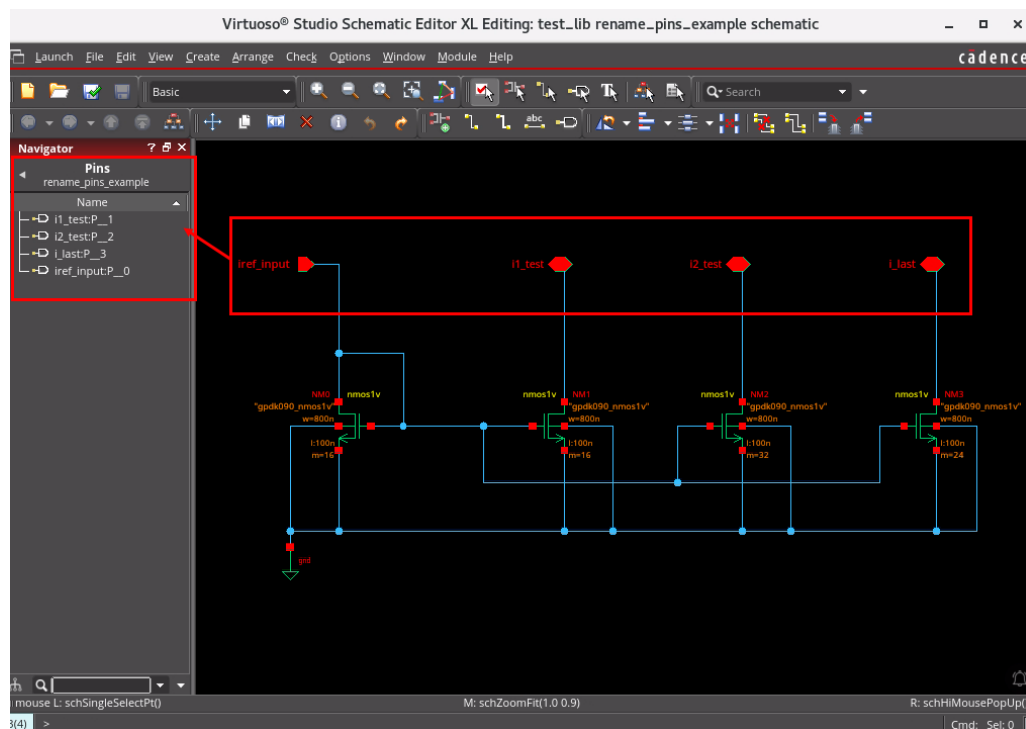


The *test_lib/rename_pins_example/schematic* cellview opens for editing.

In this schematic, the names of the pins are different from the layout:

- Pin *iref* has been renamed to *iref_input*
- Pin *i1* has been renamed to *i1_test*
- Pin *i2* has been renamed to *i2_test*
- Pin *i3* has been renamed to *i_last*

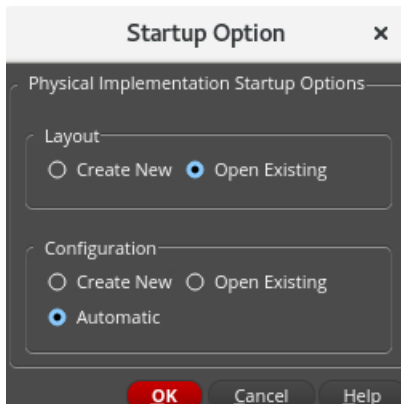
2. In the Navigator assistant, activate **Pins** in the *OBJECTS* category.



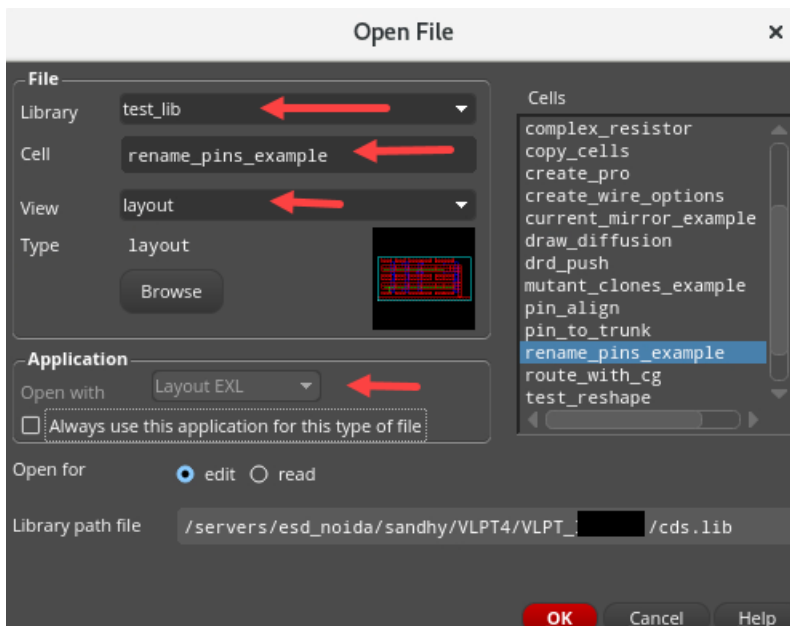
The purpose of this lab is to update the layout to reflect the pin names changes using the *Update Components and Nets (UCN)* command.

3. In the schematic, choose **Launch – Layout EXL** to bring up the *Startup Option* menu.

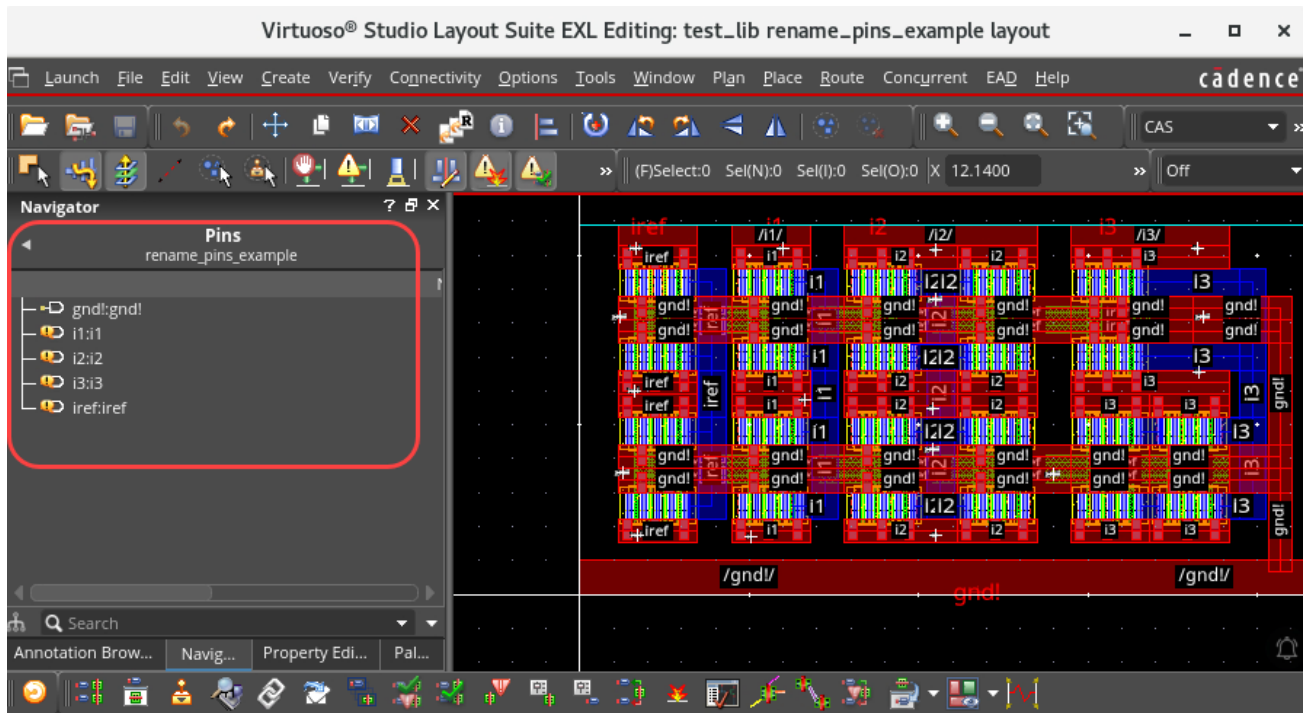
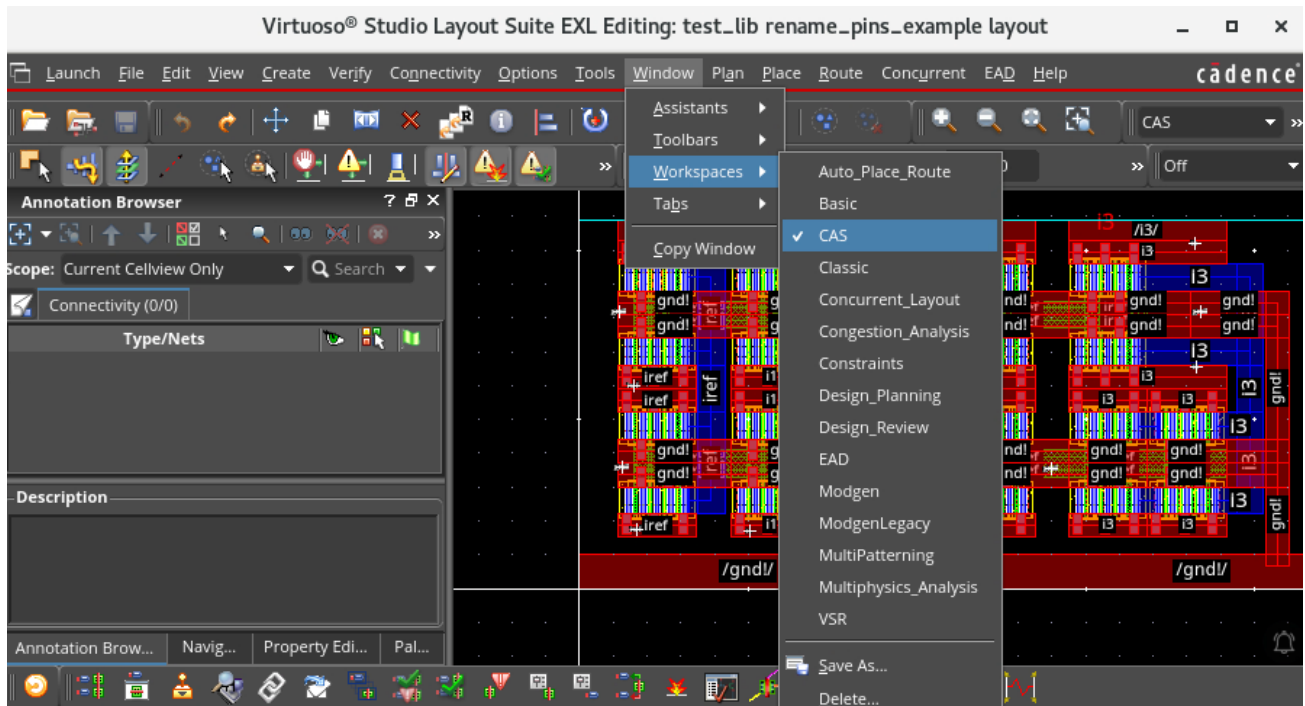
4. Select the options, as shown here, to open an existing layout view and click **OK**.



5. Fill out the *Open File* form, as shown here, and click **OK**.



The Virtuoso Layout Suite EXL palette opens. Choose the workplace as CAS as shown below

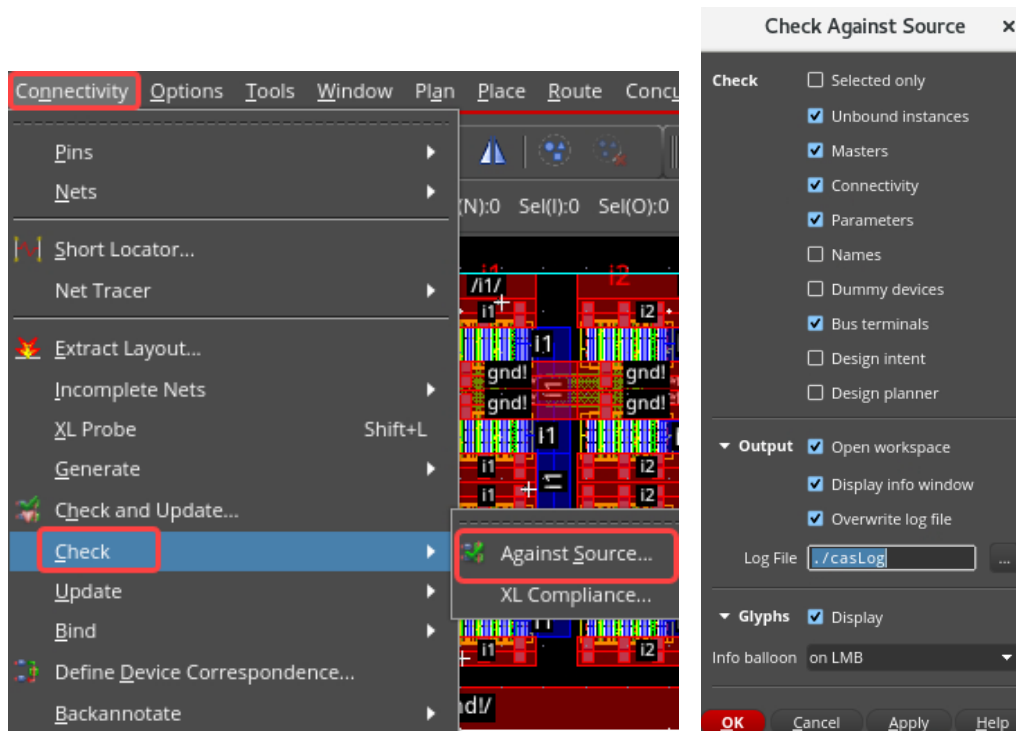


The names of the pins in the layout are still using the **OLD** names and do not reflect the latest schematic changes.

6. To verify this, you will check the layout cellview against the schematic source cellview.

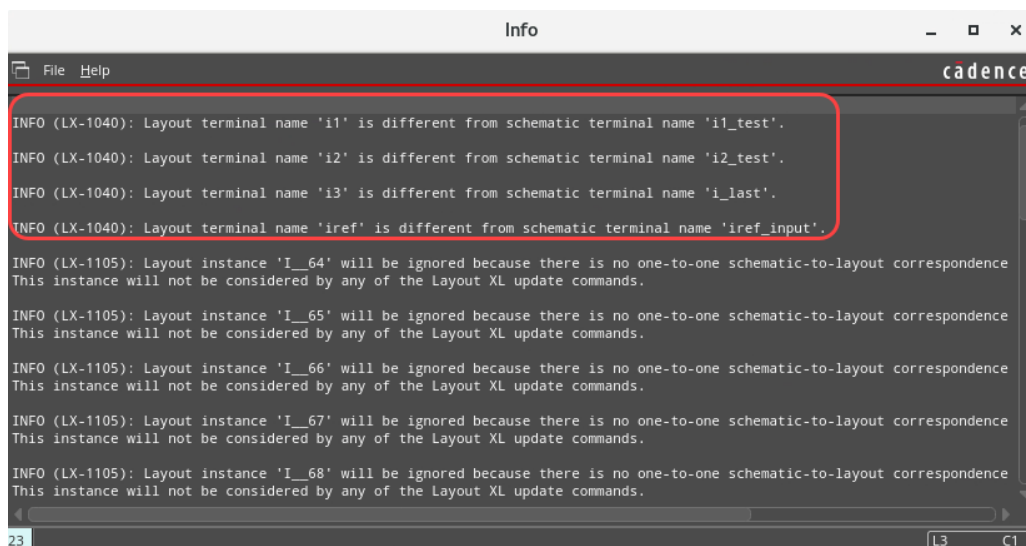
You will use the *Check Against Source* command to show any differences between the schematic and the layout.

- In the layout canvas, choose the **Connectivity – Check – Against Source** menu command, or click the **Check Against Source (CAS)** icon in the Layout EXL toolbar to open the Check Against Source form.
- Match the settings, as shown here.

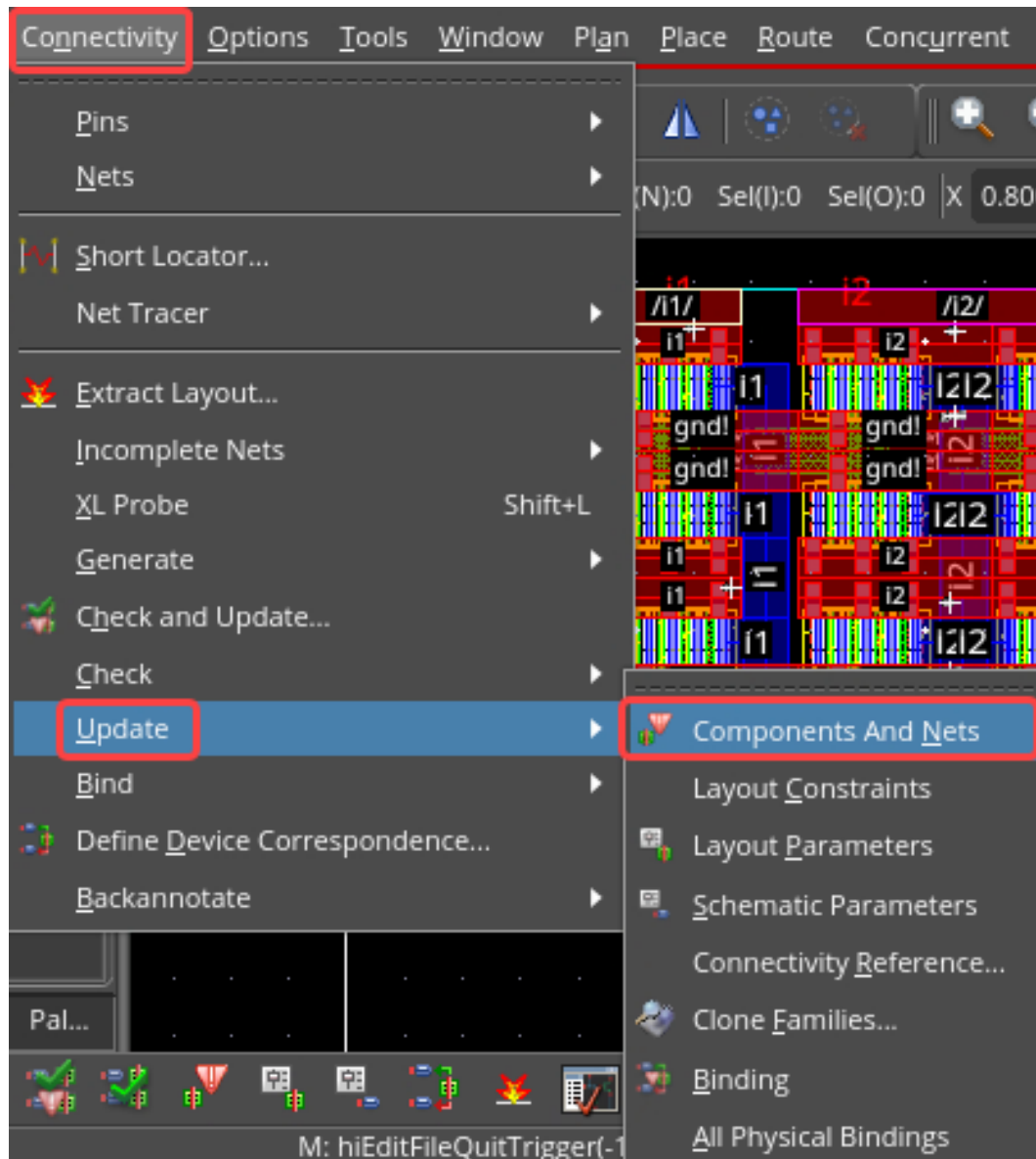


- Click **OK** in the CAS form.

The following *Info* report is displayed. You can check the *casLog* file as well from the current working directory.

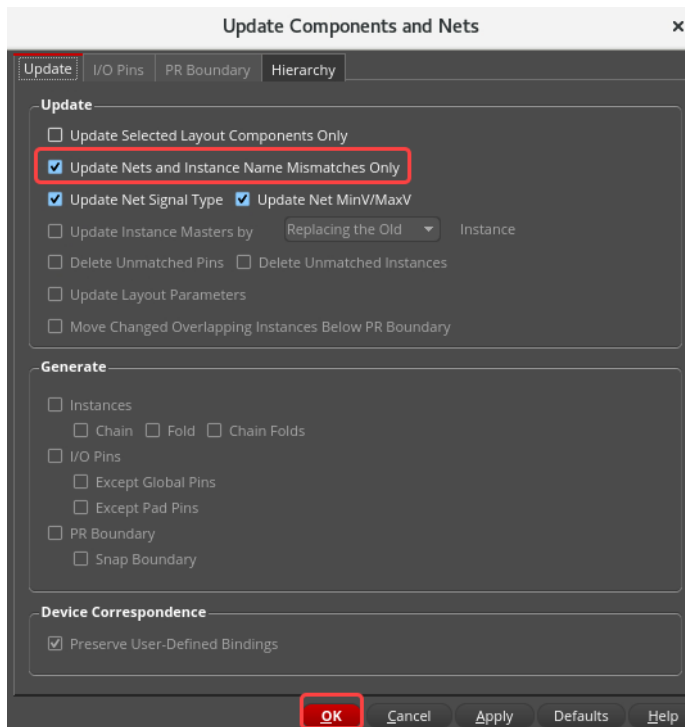


- d. Review the *Info* window for the reported issues to check the terminal names differences reported and then close the *Info* window.
7. Now, you update the pin names in the layout.
- a. In the layout canvas, choose the **Connectivity – Update – Components And Nets** menu command, or click the **Update Components and Nets (UCN)** icon in the Layout EXL toolbar.



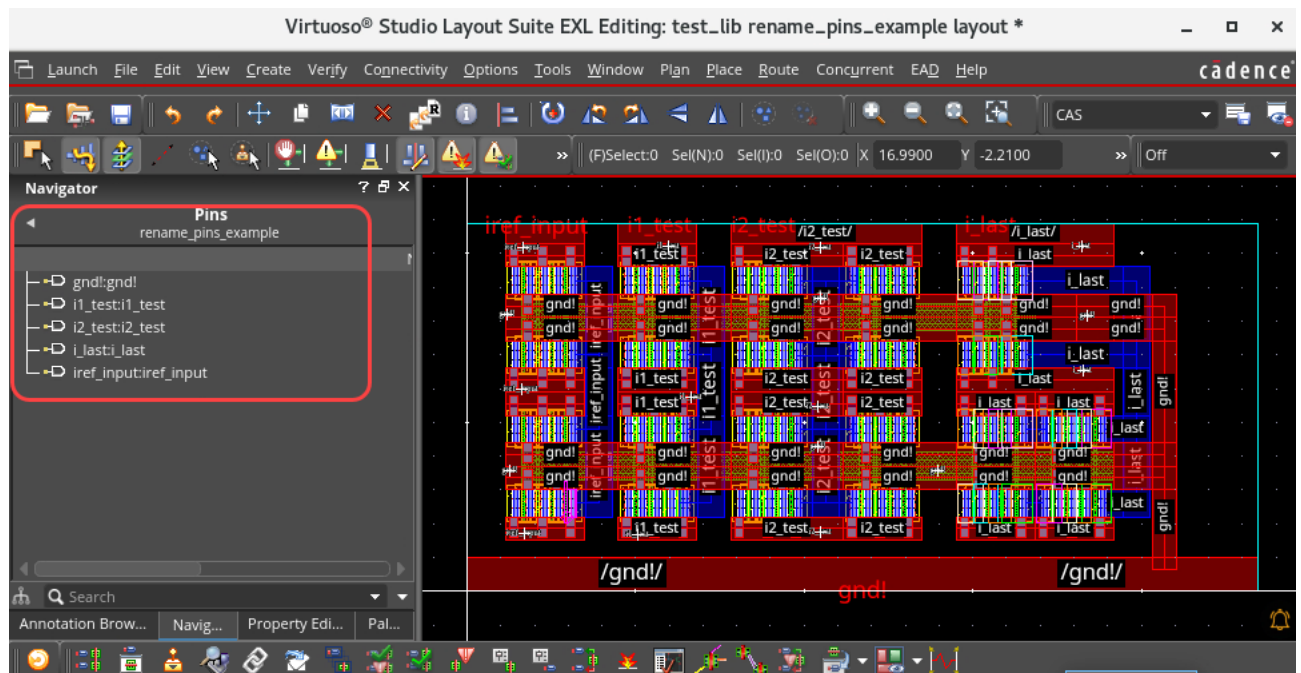
The Update Components and Nets form appears.

- b. Because the “net-names” are the only thing that you need to update in the layout, then you need to activate the **Update Nets and Instance Name Mismatches Only** option. (This will gray out several option fields.)



- c. Click **OK** in the form.

This updates the pin names in the layout according to the new names in the schematic view. Your results are similar to the ones shown in the following graphic.





The net labels in the middle of the layout have been updated as well because these labels are generated using the auto-labeling functionality.

8. Close the schematic and layout views without saving them.
9. Close all the cellviews without saving the changes.
10. Close the Virtuoso platform.

Summary

In this lab, you

- ◆ Identified the differences in the net names between the schematic and the layout
- ◆ Used the Update Components and Nets command to update the pin names in the layout

